

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 411 — Past Questions & Materials
by Renance

Course: GET 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 8
Levels: 400

This study pack was compiled and restructured by Renance for GET 411 students at Federal University of Petroleum Resources, Effurun. It brings together 8 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 8 — GET 411 Engineering Mathematics V

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
 First Semester Examination: 2024/2025 Academic Session
 Course Code & Title: GET 411 - Engineering Mathematics V
 Date: 15/05/2025

INSTRUCTION

- ANSWER QUESTION ONE (1) AND ANY OTHER FOUR (4)
- TIME ALLOWED: 3:00 HOURS

6, 7, 1

QUESTION 1

- a) In problem-solving, identify two major reasons one would seek solutions through numerical methods. [2 mks]
- b) Highlight and demonstrate the main type of error in numerical methods/calculations [3 mks]
- c) Highlight and explain two typical compromises/trade-offs encountered when applying numerical methods to solve problems on a large computational scale [4 mks]
- d) In computer programming, what is meant by a control statement, and list two categories of these statements. [3 mks]
- e) Explain the difference between interpolation and approximation functions in numerical methods. [2mks]
- f) What are the pros and cons of using open-domain methods over closed-domain methods when solving nonlinear systems of equations? [2 mks]
- g) Discuss briefly the advantages and disadvantages of decreasing versus increasing the step size (h) in numerical computations [2 mks]
- h) Suppose that the following differential equation governs the position of a falling object

$$\frac{d^2y}{dx^2} + \frac{c}{m} \frac{dy}{dx} - g = 0$$

Where c (a first-order drag coefficient) = 12.5 kg/s, m (mass) = 70 kg, and g (gravitational acceleration) = 9.81 m/s². With three interior nodes (four (4) subintervals) equally spaced. Employ the finite-difference method to solve this equation for the positions given the boundary conditions, y(0) = 0 and y(12) = 500. [10 mks]

QUESTION 2

- a) Supposing the base-24 numbering system is described by the following symbols, just as in base 16: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, G, H, I, J, K, L, M, N. Hence
- Convert the decimal number 527 base 10 to the base-24 number. [2 mks]
 - Convert the base-24 number 10N into its decimal equivalent. [1 mks]
- b) Explain why computers prefer binary systems in their operations. [1.5 mks]
- c) The CPU consists of key components essential for processing numerical methods. Identify and briefly explain their roles [4 mks]
- d) What is the significance of the two's complement system in binary arithmetic? [2 mks]

QUESTION 3

Table 3 shows the daily measurements of algal biomass (in grams) along a fixed shoreline section of a freshwater lake.

Table 3:

Day x (days)	1	2	3	4	5
Biomass y (g)	1	2	4	8	16

- a) Based on expert judgement, demonstrate which of these three models, "linear, exponential, and quadratic functions," most consistently reflects the data trend. [1.5 mks]
- b) Fit any two of your two chosen models to the data and decide which best approximates the data using RMSE as an error metric. [9 mks]

QUESTION 4

a) Solve the following system of linear equations using either the Gauss-Jordan or Adaptive Successive Over-Relaxation (SOR) method. [7.5 mks]

$$5x + y + z + w = 15; \quad x + 4y + z = 10; \quad x + y + 6z - w = 12; \quad x + y + z + 3w = 8$$

For Adaptive SOR:

Start with $a = b = c = d = 0$ and relaxation weight $w = 1.0$

After each iteration, compute

$$\Delta = \max\{|a^k - a^{k-1}|, |b^k - b^{k-1}|, |c^k - c^{k-1}|, |d^k - d^{k-1}|\}.$$

Then update the relaxation weight w as follows

- if $\Delta < 0.01$, and $w > 0.05$, then $w \leftarrow w - 0.05$
- if $\Delta \geq 0.10$, and $w < 1.95$, then $w \leftarrow w + 0.10$
- Otherwise: keep w unchanged

Continue the iterations until Δ falls below 0.001 and present your results in a table showing the values of a , b , c , d , Δ and w for each iteration.

Hint: The method converges in fewer than 7 iterations when the update rule for w is applied correctly.

b) Construct the divided differences table for the following data [3 mks]

X	1	2	3	4	5	6	7
F(x)	1	8	27	64	125	216	343

QUESTION 5

a) Using the secant method, find the approximate root of $2x^3 + 4x^2 - 4x - 6 = 0$, starting from $x_0 = -3$, $x_1 = -2$, correct up to three decimal places [6.5 mks]

b) Considering the Newton-Raphson method of root finding,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Proof that the error E at each stage is proportional to the square of the error in the previous stage, i.e.

$$E_{n+1} \approx \frac{E_n^2}{2} \frac{f''(\alpha)}{f'(\alpha)}$$

[4 mks]

QUESTION 6

A surveillance missile is launched vertically from ground level during a controlled military test. The motion of the missile is described by the first-order differential equation:

$$\frac{dy}{dx} = -9.8x + 20, \quad y(0) = 0$$

Where:

$y(x)$ is the height of the missile (in meters) at time x seconds

x is the time in seconds since launch

(a) Use either Runge-Kutta 4th Order or Modified Euler's, estimate the height of the missile at $x = 2$, correct to two decimal places, with a step size of $h = 1$. [8 mks]

(b) Solve the differential equation analytically and determine the exact height of the missile at $x = 2$ [3 mks]

(c) Compare both solutions [0.5 mks]

QUESTION 7

a) Evaluate the definite integral $\int_0^6 \frac{1}{1+x} dx$, taking h (interval spacing) $= 1$, using an appropriate numerical integration method [3 mks]

b) A civil engineer is analyzing the area of land between two intersecting roads to determine the space available for construction. The curves representing the roads are given by the following equations:

$$f(x) = x^2; \quad g(x) = 4x$$

Where the area between the two roads is bounded between $x = 0$ and $x = 2$

Use Simpson's Rule, estimate the area between the curves using a step size of $h = 0.25$ [7.5 mks]

.....good luck!

Paper 2 of 8 — GET 411 2024/2025

Level 400 • First Semester • PastQuestion

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First Semester Examination 2024/2025 Academic Session
Course Code & Title: GET 411 - Engineering Mathematics V
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[4 mks]

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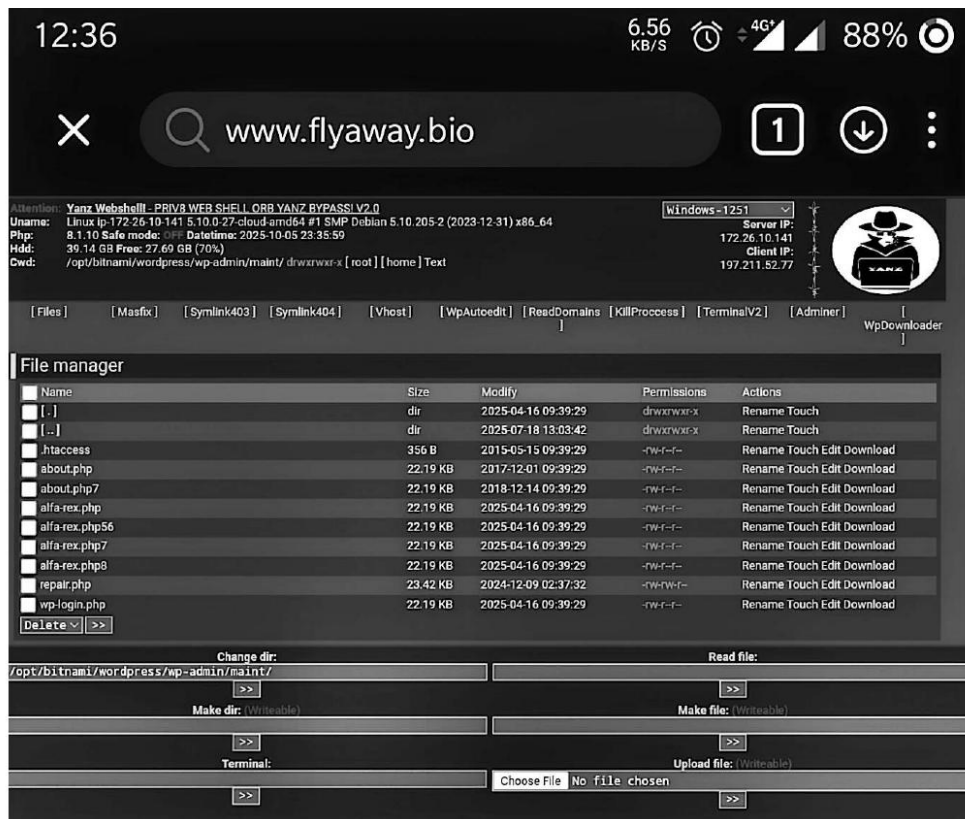
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.....good luck!



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Paper 3 of 8 — GET 411 Engineering mathematics 2024/2025

Level 400 • First Semester • PastQuestion

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Paper 4 of 8 — GET 411 Past Questions

Level 400 • First Semester • PastQuestion

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QUESTION Five ee i - ee a at

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FEDERAL UNIVERSITY OF PETROLE

"TROLEUM REGOU!

DEPARTMENT OF MARINE ENGINEERING nN

First Semester Examination 2014/2015 Academic Session

. CODE AND TITLE: *

.. GET 411 - NUMERICAL METHODS AND COMPUTER APPLICATIONS

DATE: 30TM JUNE, 2015 TIME ALL

' ' ME A '

: LLOWED JHRS

Instruction: Answer Question One (1) and any Other Four | :

N ark i

{a}. The BCD system is the code used by microprocessors. Hente or otherwise;

2 (i) Show how the computer systeth will perform the follow operation

aa (a) 111001001×1001011 (b) $101101101 + 1011$ (c) (eb $\times 9$

ter will express and store (as a Floating point) the result of

- (7 ; marks)

(i). Show how the compu

a,b,andc in question d (i) above.

it Rounding arithmetic, estimate the Relative

tb). vsing 2 3-digit chopping and 3-dig

; 9 evaluate the function,

' Error when thi

' $f@=3$

$\% 4$

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7 ¢ & X, and X: ermine;

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the Lower and Upper

(75 marks)

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~ error for x and y.

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ne > ag 2x4

Ar care

Consider the Equation $3.00x + 0.75 = 2.509 = 0$

$1.60x = 0.80 \sim 0.00Y$ |

s set of equations is

The unique solution to thi

hopping arithmetic and lise

{a) Solve this equation using five digit ϕ

five-digit values of y and X. :

(b).Explain why the answers obtained in (a) differs significantly from t
xandy, $\sim \sim$ ' h es

(c). What 1s the error, the absolute error, relative error ar

TION FIV

Solve the linear system of eq

$xq\ 4) \ 332 - 2x, = 6\ 4^\circ$

$4x, + 5x2 + 2x3 = 3$

$yt\ 3x0 + 4\%3 = 7$

QUESTION SEX

$19 < 90^\circ = (3 - j4 - +2)$

$15 < 0^\circ - 10 < 90^\circ = ($

Solve the equation for the'

ai! $4324 \#3 *$

ye 0,30 and $y = 0.40,$

your result to find

he actual values of

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(10 marks)

(10 marks)

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Paper 5 of 8 — GET 411 PQ 2024

Level 400 • First Semester • PastQuestion

FEDERAL

ALGERIA OF PETROLEUM RESOURCES, EFFURUN

First Semester in ENGINEERING & TECHNOLOGY

Course Code & Examination, 2023/2024 Academic Session

Title: GET 411 - Engineering Mathematics v

Date: 04/05/2024 | 01

INSTRUCTIONS

° A

ANSWER QUESTION ONE AND ANY OTHER FOUR

Question 1 * TIME ALLOWED: 3 HOURS

9) In problem 1

-solve it

; in addition (2) what could have motivated one to sort for solutions through numerical

c) Discuss briefly one (1) common trade-off in implementation

of using numerical methods for large-scale

©) Briefly discuss two real-life problems studied by the following

: + disasters in it

" since we have studied that were caused by inadequate numerical

capabilities in MATLAB

1) in MATLAB Programming, what do you understand by the term "control statement"?

Describe the categories of these statements and their uses. [3 marks]

number systems are more efficient for Computers and Humans carrying numerical

in and out

operations? Hence perform the following tasks.

i) Convert 11101011 base 2 to octal. [3 marks]

ii, given on 8-bit machine, find the 2's complement of -6

is a reserved bit denoting the sign. (2 marks)

3) After the normalization of floating point is carried out at each

stage by the following

$c = 0.5644 \times 10^0$

{5 marks}

noting that the leftmost bit

in the stage, prove the following:

$10! ; b = 0.5566 \times 10^0 ;$

identify two sources of numerical errors.

g) In an experimental exercise the following set of data obtained is observed to follow a trend linear law of the form $F = mL + e$ where

aight line, In this context establish o

describing o st

¥ ipproximates this trend. Lo

| [Force | 12 il Ee ES

| [Leod 50 70 | 100 [120

* h) Given the data below, find the isothermal work done on the gos ° / it is compressed from "n= 22
Liter to vz = 2 Liter using tropezoical rule [3 mks]

2 17 22 27

7

C23 6, bse [410 1350]

stem, the concentration C of @ substonce changes over *li: acco: im

¢ t is time in hours, Given C(1}=1 ond o time step ++ at

4 é

to ++ gdifferential 'equation 7

he Runge-Kutto (fourth order) method on) (99 "33

Total 40 mks

a) oS 4 moving with a velocity vat any tine i

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Usiig Logrange's interpolation formula, estab

the distance travelled in the first

ak)

imate approximately the distanc

) During a heatiny process, t! + f4mp

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Temperature (C%),

lationship between th> jemp

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Paper 6 of 8 — GET 411 PAST QUESTION 2020-2021

Level 400 • First Semester • PastQuestion

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First Semester Examination 2020/2021 Academic Session

Course code and title: GET 411-Numerical Methods and Computer Applications

INSTRUCTION: Answer question one and any other four Time Allowed: 3 hours

Date: 17th November, 2021

QUESTION ONE

A. The BCD system is the code used by microprocessors. Hence, show how the computer will perform the following operations internally.

- $(A3_{16}) + (69.8125_{10})$
- $(245_8) \div (6_{10})$
- $(1AC_{16}) \times (67_{10})$

10 marks

B. Suppose that $S = \frac{1}{3}$ and $T = \frac{5}{7}$ and that a 4 digit chopping is used for the arithmetic calculation involving S and T. compute or estimate the absolute and relative error in the arithmetic and show which one has the maximum error.

5 marks

✓ C. Use Euler method to obtain a numerical solution of the ODE $y' = 1+x+y$ with initial conditions that $y=2, x=1$ for the range $x=1(0.2)2.0$. Determine the particular solution and evaluate the percentage relative error by comparing the result with the actual value.

15 marks

QUESTION TWO

✓ A. If $Ax = \lambda x$ determine Eigen Values and the Corresponding Eigen Vectors in the matrix below:

$$A = \begin{pmatrix} 2 & -5 \\ 1 & -4 \end{pmatrix}$$

5 marks

✓ B. Mesh-current analysis produces the following three equations:

$$20\angle 0^\circ = (5 + 3 - j4)I_1 - (3 - j4)I_2$$

$$10\angle 90^\circ = (3 - j4 + 2)I_2 - (3 - j4)I_1 - 2I_3$$

$$-15\angle 0^\circ - 10\angle 90^\circ = (12 + 2)I_3 - 2I_2$$

Solve the equations for the loop currents I_1, I_2 and I_3 using determinant method, leaving your answer in polar form.

12.5 marks

Handwritten solution for Question Two B:

$$\frac{D_x}{D} - \frac{Y}{D_y} + \frac{Z}{D_z} = \frac{-1}{D}$$

Handwritten calculations for determinants and currents are visible below the formula.

- ✓ B. Carryout the first three(3) iteration of the Gauss-Seidel method for the following system of equation to 4 d.p and obtain the actual values of x, y, z and w.

$$\begin{aligned} 13x + 5y - 3z + w &= 18 \\ 2x + 12y + z - 4w &= 13 \\ 3x - 4y + 10z + w &= 29 \\ 2x + y - 3z + 9w &= 31 \end{aligned}$$

at: $\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

10 marks

QUESTION SEVEN

- ✓ A. The simultaneous equations representing the currents flowing in an unbalanced, three-phase, star-connected, electrical network are as follows:

$$\begin{aligned} I_1 + 2I_2 + 5I_3 + I_4 &= 4 \\ 3I_1 - 4I_2 + 3I_3 - 2I_4 &= 7 \\ 4I_1 + 3I_2 + 2I_3 - I_4 &= 1 \\ I_1 - 2I_2 - 4I_3 - I_4 &= 2 \end{aligned}$$

Using Gaussian elimination method, solve the equations for I_1 , I_2 , I_3 and I_4 .

10 marks

- ✓ B. Solve the linear equation by carrying out the first four(4) iteration of the Gauss-Jacobi iteration to 4 d.p and obtain the actual values of x, y and z.

$$\begin{aligned} 10x - y - 2z &= 6 \\ -x + 11y - z &= 25 \\ 2x - y + 10z &= -11 \end{aligned}$$

At: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

7.5 marks

GOODLUCK

QUESTION THREE

- ✓ A. The relationship between the displacement s , velocity v and acceleration a , of a piston is given by the equations:

$$s + 2.0v + 3.0a + 7.8 = 0.0$$

$$2.0s + 5.0v - a - 1.4 = 0.0$$

$$5.0s - v + 7.0a - 3.5 = 0.0$$

Use matrix method to determine the values of s , v and a .

7.5 marks

- ✓ B. Apply the Euler Cauchy method to solve the ODE $y' = 3(1+x) - y$, $x = 0(0.1)0.5$, $y = 2$

10 marks

QUESTION FOUR

- ✓ A. In a mass-spring-damper system, the acceleration a m/s², velocity u m/s and displacement x m are related by the following simultaneous equation

$$6.2a + 7.9u + 12.6x = 18.0$$

$$7.5a + 4.8u + 4.8x = 6.39$$

$$13.0a + 3.5u - 13x = -17.4$$

Using Triangular Decomposition method, determine the acceleration, velocity and displacement for the system.

10 marks

- ✓ B. If $Ax = b$, determine Eigen Values and the Corresponding Eigen Vectors in the matrix below:

$$A = \begin{pmatrix} 4 & 10 & -8 \\ 1 & 2 & 1 \\ -1 & 2 & 3 \end{pmatrix}$$

7.5 marks

QUESTION FIVE

- ✓ A. Use Runge Kutta method to obtain a numerical solution of the ODE $y' = y - x$ with initial conditions that $y=2$, $x=0$ for the range $x=0(0.2)1.4$. However, if the Exact solution to the ODE is given as $y = x + 1 + e^x$, compute the exact values.

12.5 marks

B. Numerical methods are important or studied for various reasons, briefly highlight at least five of these reasons.

5 marks

QUESTION SIX

- ✓ A. The tensions, T_1 , T_2 and T_3 in a simple framework are given by the equations:

$$5T_1 + 5T_2 + 5T_3 = 7.0$$

$$T_1 + 2T_2 + 4T_3 = 2.4$$

$$4T_1 + 2T_2 = 4.0$$

Determine T_1 , T_2 and T_3 using Gauss-Jordan Reduction method

7.5 marks

400 LEVEL COURSE CONTENT**GET 411: ENGINEERING MATHEMATICS V****3 UNITS**

Review of the number systems and error analysis; numerical schemes, error analysis, computer algorithms and programs for the solution of problems including interpolation by polynomials, nonlinear equations, systems of linear equations, determinants and matrices, and eigenvalue problems; approximations including data fitting, orthogonal polynomials, least squares, splines and fast Fourier transform; differentiation and integration; difference equation; differential equations by Runge-Kutta and other methods; boundary value problems in ODE; introduction to the finite-difference method for partial differential equations.



Paper 7 of 8 — GET 411 ENGINEERING MATHEMATICS V

Level 400 • First Semester • Handout

SOLUTION TO NON LINEAR

(computer algorithm)

An expression of the form $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$, where $a_0, a_1, a_2, \dots, a_n$ are constants and n is a positive integer, is called an algebraic polynomial of degree n if $a_0 \neq 0$. The equation $f(x) = 0$ is called an algebraic equation if $f(x)$ is an algebraic polynomial, eg... $x^3 - 4x - 9 = 0$. If $f(x)$ contains functions such as trigonometric, logarithmic, exponential, etc., then $f(x) = 0$ is called a transcendental equation, eg., $2x^3 - \log(x + 3) \tan x + e^x = 0$.

In general, an equation is solved by factorization. But in many cases, the method of factorization fails. In such cases, numerical iteration methods are used. There are some methods to solve the equation $f(x) = 0$ such as

Closed domain or Bracketing method

These methods are robust in that they are guaranteed to obtain a solution since the root is trapped in the closed interval. They can be slow to converge

1. Bisection method
2. Regula Falsi method

Open domain method

These methods do not restrict the root to remain trapped in a closed interval. Consequently, they are not as robust as bracketing methods and can actually diverge. However, they use information about the nonlinear function itself to refine the estimates of the root. Thus they are considerably more efficient than bracketing methods.

3. Newton – Raphson method
4. Secant method

These numerical iteration methods is a mathematical procedure that generates a sequence of improving approximate solutions for a class of problems. A specific way of implementation of an iteration method, including the termination criteria. In the problems of finding the solution of an equation an iteration method uses an initial guess to generate successive approximations to the solution. Since the iteration methods involve repetition of the same process many times, computers can act well for finding solutions of equation numerically.

1

A numerical method to solve equations may be a long process in some cases. If the method leads to value close to the exact solution, then we say that the method is convergent. Otherwise, the method is said to be divergent.

CONCEPT OF STABILITY AND CONVERGENCE

STABILITY: Stability in numerical analysis refers to the trend of error change iterative scheme. It is related to the concept of convergence.

It is stable if initial errors or small errors at any time remain small when iteration progresses. It is unstable if initial errors or small errors at any time get larger and larger, or eventually get unbounded.

CONVERGENCE: There are two different meaning of convergence in numerical analysis:

If the discretized interval is getting finer and finer after discretizing the continuous problems, the solution is convergent to the true solution.

For an iterative scheme, convergence means the iteration will get closer to the true solution when it progresses.

Given a function $f(x)$, which normally is nonlinear, the problem of computing zeros means to find all possible points, say

α

$0 < \alpha$

$1 > \alpha, \alpha$

$\sim \alpha$

Such that

$f(\alpha^m)$

$0) = f(\alpha^1) = \dots = f(\alpha^m) = 0$

However, it is often that we are required to find a single point α

~ 0 in certain interval

say $[a, b]$ such that

$f(\alpha)$

$0) = 0$

General strategy is to design an iterative process of the form

$\alpha_{n+1} = g(\alpha_n)$

With some starting points α_0 . So that the numerical solution as

$\alpha_0 \rightarrow \alpha$

0 as $n \rightarrow \infty$

Thus, instead of finding the exact solution, we find an approximation.

We shall focus on the aforementioned method and also try to explore any innovation

2

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9.2 Bisection Method

Let $f(x) = 0$ be the given equation. Let x_0 and x_1 be two real values of x at P and Q respectively such that $f(x_1)$ is positive and $f(x_0)$ is negative or vice versa (Fig. 9.1).

Then there is one root of the equation $f(x) = 0$ between x_0 and x_1 . Now, this interval $[x_0, x_1]$ is divided into two sub-intervals

$x + x$

$[x_0, x_2]$ and $[x_2, x_1]$, where $x_2 = \frac{x_0 + x_1}{2}$.

2

If $f(x_0)$ and $f(x_2)$ are of opposite signs then the interval $[x_0, x_2]$ is divided into

$x + x_2$

$[x_0, x_3]$ and $[x_3, x_2]$, where $x_3 = \frac{x_0 + x_2}{2}$.

Fig. 9.1

2

However, if $f(x_0)$ and $f(x_2)$ are of the same sign then $f(x_1)$ and $f(x_2)$ will be opposite signs and the interval $[x_1, x_2]$ is divided into

$x + x$

$[x_1, x_3]$ and $[x_3, x_2]$, where $x_3 = \frac{x_1 + x_2}{2}$. This process is continued till the desired

2

accuracy is obtained.

example 1

Find the positive root of $x^3 - 2x - 5 = 0$, correct up to two decimal places.

Solution

Let $f(x) = x^3 - 2x - 5$

$f(1) = -6$ and $f(2) = -1$, $f(3) = 16$

Since $f(2) < 0$ and $f(3) > 0$, the root lies between 2 and 3.

2+3

$x_1 = \frac{2+3}{2} = 2.5$

2

$f(x_1) = f(2.5) = 5.625$

Since $f(2.5) > 0$ and $f(2) < 0$, the root lies between 2.5 and 2.

2.5 + 2

$x_2 = \frac{2.5 + 2}{2} = 2.25$

2

$f(x_2) = f(2.25) = 1.8906$

Since $f(2.25) > 0$ and $f(2) < 0$, the root lies between 2.25 and 2.

3

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9.2 Bisection Method 9.3

2.25 + 2

$x_3 = \frac{2.25 + 2}{2} = 2.125$

2

$f(x_3) = f(2.125) = 0.3457$

Since $f(2.125) > 0$ and $f(2) < 0$, the root lies between 2.125 and 2.

$$2.125 + 2$$

$$x_4 = 2.0625$$

$$2$$

$$f(x_4) = f(2.0625) = -0.3513$$

Since $f(2.0625) < 0$ and $f(2.125) > 0$, the root lies between 2.0625 and 2.125.

$$2.0625 + 2.125$$

$$x_5 = 2.09375$$

$$2$$

$$f(x_5) = f(2.09375) = -0.0089$$

Since $f(2.09375) < 0$ and $f(2.125) > 0$, the root lies between 2.09375 and 2.125.

$$2.09375 + 2.125$$

$$x_6 = 2.109375$$

$$2$$

$$f(x_6) = f(2.109375) = 0.1668$$

Since $f(2.109375) > 0$ and $f(2.09375) < 0$, the root lies between 2.109375 and 2.09375.

$$2.109375 + 2.09375$$

$$x_7 = 2.10156$$

$$2$$

Since x_6 and x_7 are same up to two decimal places, the positive root is 2.10.

example 2

Find a root of $x^3 - 5x + 3 = 0$ by the bisection method correct up to four decimal places.

Solution

$$\text{Let } f(x) = x^3 - 5x + 3$$

$$f(0) = 3 \text{ and } f(1) = -1$$

Since $f(0) > 0$ and $f(1) < 0$, the root lies between 0 and 1.

$$0 + 1$$

$$x_1 = 0.5$$

$$2$$

$$f(x_1) = f(0.5) = 0.625$$

Since $f(0.5) > 0$ and $f(1) < 0$, the root lies between 0.5 and 1.

$$0.5 + 1$$

$$x_2 = 0.75$$

$$2$$

$$f(x_2) = f(0.75) = -0.3281$$

$$4$$

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$$9.4$$

Since $f(0.75) < 0$ and $f(0.5) > 0$, the root lies between 0.75 and 0.5.

$$0.75 + 0.5$$

$$x_3 = 0.625$$

2

$$f(x_3) = f(0.625) = 0.1191$$

Since $f(0.625) > 0$ and $f(0.75) < 0$, the root lies between 0.625 and 0.75.

$$0.625 + 0.75$$

$$x_4 = 0.6875$$

2

$$f(x_4) = f(0.6875) = -0.1125$$

Since $f(0.6875) < 0$ and $f(0.625) > 0$, the root lies between 0.6875 and 0.625.

$$0.6875 + 0.625$$

$$x_5 = 0.65625$$

2

$$f(x_5) = f(0.65625) = 0.00137$$

Since $f(0.65625) > 0$ and $f(0.6875) < 0$, the root lies between 0.65625 and 0.6875.

$$0.65625 + 0.6875$$

$$x_6 = 0.67188$$

2

$$f(x_6) = f(0.67188) = -0.0561$$

Since $f(0.67188) < 0$ and $f(0.65625) > 0$, the root lies between 0.67188 and 0.65625.

$$0.67188 + 0.65625$$

$$x_7 = 0.66407$$

2

$$f(x_7) = f(0.66407) = -0.02750$$

Since $f(0.66407) < 0$ and $f(0.65625) > 0$, the root lies between 0.66407 and 0.65625.

$$0.66407 + 0.65625$$

$$x_8 = 0.66016$$

2

$$f(x_8) = f(0.66016) = -0.01309$$

Since $f(0.66016) < 0$ and $f(0.65625) > 0$, the root lies between 0.66016 and 0.65625.

$$0.66016 + 0.65625$$

$$x_9 = 0.65821$$

2

$$f(x_9) = f(0.65821) = -0.00589$$

Since $f(0.65821) < 0$ and $f(0.65625) > 0$, the root lies between 0.65821 and 0.65625.

$$0.65821 + 0.65625$$

$$x_{10} = 0.65723$$

2

$$f(x_{10}) = f(0.65723) = -0.0023$$

Since $f(0.65723) < 0$ and $f(0.65625) > 0$, the root lies between 0.65723 and 0.65625.

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9.2 Bisection Method 9.5

$0.65723 + 0.65625$

$x_{11} = = 0.65674$

2

$f(x_{11}) = f(0.65674) = -0.00044$

Since $f(0.65674) < 0$ and $f(0.65625) > 0$, the root lies between 0.65674 and 0.65625.

$0.65674 + 0.65625$

$x_{12} = = 0.6565$

2

$f(x_{12}) = f(0.6565) = 0.00044$

Since $f(0.6565) > 0$ and $f(0.65674) < 0$, the root lies between 0.6565 and 0.65674.

$0.6565 + 0.65674$

$x_{13} = = 0.6566$

2

$f(x_{13}) = f(0.6566) = 0.00075$

Since $f(0.6566) > 0$ and $f(0.65674) < 0$, the root lies between 0.6566 and 0.65674.

$0.6566 + 0.65674$

$x_{14} = = 0.65667$

2

Since x_{13} and x_{14} are same up to four decimal places, the root is 0.6566.

example 3

Perform the $\square\square$ ve iterations of the bisection method to obtain a root of the equation $f(x) = x^3 - x - 1 = 0$.

Solution

Let $f(x) = x^3 - x - 1$

$f(1) = -1$ and $f(2) = 5$

Since $f(1) < 0$ and $f(2) > 0$, the root lies between 1 and 2.

$1 + 2$

$x_1 = = 1.5$

2

$f(x_1) = f(1.5) = 0.875$

Since $f(1.5) > 0$ and $f(1) < 0$, the root lies between 1.5 and 1.

$1.5 + 1$

$x_2 = = 1.25$

2

$f(x_2) = f(1.25) = -0.2968$

Since $f(1.25) < 0$ and $f(1.5) > 0$, the root lies between 1.25 and 1.5.

$1.25 + 1.5$

$x_3 = = 1.375$

2

$$f(x_3) = f(1.375) = 0.2246$$

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9.6

Since $f(1.375) > 0$ and $f(1.25) < 0$, the root lies between 1.375 and 1.25.

$$1.375 + 1.25$$

$$x_4 = 1.3125$$

2

$$f(x_4) = f(1.3125) = -0.0515$$

Since $f(1.3125) < 0$ and $f(1.375) > 0$, the root lies between 1.3125 and 1.375.

$$1.3125 + 1.375$$

$$x_5 = 1.3438$$

2

Hence, the root is 1.3438 up to 5 iterations.

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example 7

Perform three iterations of the bisection method to obtain the root of the equation $2 \sin x - x = 0$, correct up to three decimal places.

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9.10

Solution

$$\text{Let } f(x) = 2 \sin x - x$$

$$f(1) = 0.6829 \text{ and } f(2) = -0.1814$$

Since $f(1) > 0$ and $f(2) < 0$, the root lies between 1 and 2.

$$1 + 2$$

$$x_1 = 1.5$$

2

$$f(x_1) = f(1.5) = 0.4949$$

Since $f(1.5) > 0$ and $f(2) < 0$, the root lies between 1.5 and 2.

$$1.5 + 2$$

$$x_2 = 1.75$$

2

$$f(x_2) = f(1.75) = 0.2179$$

Since $f(1.75) > 0$ and $f(2) < 0$, the root lies between 1.75 and 2.

$$1.75 + 2$$

$$x_3 = 1.875$$

2

Hence, the root is 1.875 up to three iterations.

9

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9.14

Since $f(0.625) < 0$ and $f(0.5) > 0$, the root lies between 0.625 and 0.5.

$$0.625 + 0.5$$

$$x_4 = 0.5625$$

2

$$f(x_4) = f(0.5625) = -0.1413$$

Since $f(0.5625) < 0$ and $f(0.5) > 0$, the root lies between 0.5625 and 0.5.

$$0.5625 + 0.5$$

$$x_5 = 0.53125$$

2

Hence, the root is 0.53125 in four steps.

exerclse

Find a positive root of the following equations correct to four decimal

places using the bisection method:

1. x

3

$$-4x - 9 = 0$$

[Ans.: 2.7065]

2. x

3

$$+3x - 1 = 0$$

[Ans.: 0.3222]

5. $3x = 1 + \sin x$

[Ans.: 0.3918]

9. $x \log$

10

$x = 1.2$ lying between 2 and 3

[Ans.: 2.7406]

10

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9.3 Regula Falsi Method 9.15

9.3 regula Falsi Method

This method resembles the bisection method. In this method, two points x_0 and x_1 are chosen such that $f(x_0)$ and $f(x_1)$ are of opposite signs, i.e., the graph of $y = f(x)$

crosses the x-axis between these points. Hence, a root lies between x_0 and x_1 and $f(x_0)f(x_1) < 0$ (Fig.9.2).

The equation of the chord joining the points

$P[x_0, f(x_0)]$ and $Q[x_1, f(x_1)]$ is

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0) = f(x_0)$$

In this method, the curve PQ is replaced by the chord PQ and the point of intersection of the chord with the x-axis is taken as an approximation to the root.

If x_2 is the point of intersection of the x-axis and the line joining $P[x_0, f(x_0)]$ and $Q[x_1, f(x_1)]$ then x_2 is closer to the root than x_0 and x_1 .

Fig. 9.2

Using the slope formula,

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_2) - f(x_0)}{x_2 - x_0}$$

$$m = m$$

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_2) - f(x_0)}{x_2 - x_0}$$

$$x_2 - x_0 = \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = x_0 + \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = x_0 - \frac{f(x_0)}{\frac{f(x_1) - f(x_0)}{x_1 - x_0}}$$

$$x_2 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

which is an approximation to the root.

If $f(x_0)$ and $f(x_2)$ are of opposite signs, the root lies between x_0 and x_2 , and the next approximation x_3 is obtained as

$$x_3 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_3 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_3 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

If the root lies between x_1 and x_2 , the next approximation x_3 is obtained as

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

This process is repeated till the root is obtained to the desired accuracy. This iteration process is known as the method of false position or regula falsi method.

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9.16

example 1

Find a positive root of $x^3 - 4x + 1$ correct up to three decimal places.

Solution

Let $f(x) = x^3 - 4x + 1$

$f(0) = 1$ and $f(1) = -2$

Since $f(0) > 0$ and $f(1) < 0$, the root lies between 0 and 1.

Let $x_0 = 0, x_1 = 1$

$x_1 - x_0$

$x_2 = x_0 - f(x_0)$

$f(x_1) - f(x_0)$

$1 - 0$

$= 0 - (1)$

$-2 - 1$

$= 0.3333$

$f(x_2) = f(0.3333) = -0.2962$

Since $f(0.3333) < 0$ and $f(0) > 0$, the root lies between 0.3333 and 0, i.e., x_2 and x_0 .

$x_2 - x_0$

$x_3 = x_0 - f(x_0)$

$f(x_2) - f(x_0)$

$0.3333 - 0$

$= 0 - (1)$

$-0.2962 - 1$

$= 0.2571$

$f(x_3) = f(0.2571) = -0.0114$

Since $f(0.2571) < 0$ and $f(0) > 0$, the root lies between 0.2571 and 0, i.e., x_3 and x_0 .

$x_3 - x_0$

$x_4 = x_0 - f(x_0)$

$f(x_3) - f(x_0)$

$0.2571 - 0$

$= 0 - (1)$

$-0.0114 - 1$

$= 0.2542$

$f(x_4) = f(0.2542) = -0.0004$

Since $f(0.2542) < 0$ and $f(0) > 0$, the root lies between 0.2542 and 0, i.e., x_4 and x_0 .

$x_4 - x_0$

$x_5 = x_0 - f(x_0)$

$f(x_4) - f(x_0)$

$0.2542 - 0$

$= 0 - (1)$

$-0.0004 - 1$

$$= 0.2541$$

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9.3 Regula Falsi Method 9.17

Since x_4 and x_5 are same up to three decimal places, a positive root is 0.254.

example 2

Find the root of the equation $2x - \log_{10}x = 7$, which lies between 3.5 and 4, correct up to 4 places of decimal.

Solution

$$\text{Let } f(x) = 2x - \log_{10}x - 7$$

$$f(3.5) = -0.54407 \text{ and } f(4) = 0.39794$$

Since $f(3.5) < 0$ and $f(4) > 0$, the root lies between 3.5 and 4.

$$\text{Let } x_0 = 3.5, x_1 = 4$$

$$x_1 - x_0$$

$$x_2 = x_0 - f(x_0)$$

$$f(x_1) - f(x_0)$$

$$4 - 3.5$$

$$= 3.5 - (-0.54407)$$

$$0.39794 + 0.54407$$

$$= 3.78878$$

$$f(x_2) = f(3.78878) = -0.00094$$

Since $f(3.78878) < 0$ and $f(4) > 0$, the root lies between 3.78878 and 4, i.e., x_2 and x_1 .

$$x_1 - x_2$$

$$x_3 = x_2 - f(x_2)$$

$$f(x_1) - f(x_2)$$

$$4 - 3.78878$$

$$= 3.78878 - (-0.00094)$$

$$0.39794 + 0.00094$$

$$= 3.78928$$

$$f(x_3) = f(3.78928) = 0.000003$$

Since $f(3.78928) > 0$ and $f(3.78878) < 0$, the root lies between 3.78928 and 3.78878, i.e., x_3 and x_2 .

$$x_3 - x_2$$

$$x_4 = x_2 - f(x_2)$$

$$f(x_3) - f(x_2)$$

$$3.78928 - 3.78878$$

$$= 3.78878 - (-0.00094)$$

$$0.000003 + 0.00094$$

$$= 3.78928$$

Since x_3 and x_4 are same up to 4 decimal places, the root is 3.78928.

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9.18

example 3

Find a real root of the equation $x \log_{10} x = 1.2$ by the regula falsi method.

Solution

Let $f(x) = x \log_{10} x - 1.2$

$f(2) = -0.5979$ and $f(3) = 0.2314$

Since $f(2) < 0$ and $f(3) > 0$, the root lies between 2 and 3.

Let $x_0 = 2, x_1 = 3$

$x_1 - x_0$

$x_2 = x_0 - f(x_0)$

$f(x_1) - f(x_0)$

3-2

$= 2 - (-0.5979)$

$0.2314 + 0.5979$

$= 2.721$

$f(x_2) = f(2.721) = -0.0171$

Since $f(2.721) < 0$ and $f(3) > 0$, the root lies between 2.721 and 3, i.e., x_2 and x_1 .

$x_1 - x_2$

$x_3 = x_2 - f(x_2)$

$f(x_1) - f(x_2)$

3 - 2.721

$= 2.721 - (-0.0171)$

$0.2314 + 0.0171$

$= 2.7402$

$f(x_3) = f(2.7402) = -0.0004$

Since $f(2.7402) < 0$ and $f(3) > 0$, the root lies between 2.7402 and 3, i.e., x_3 and x_1 .

$x_3 - x_1$

$x_4 = x_1 - f(x_1)$

$f(x_3) - f(x_1)$

2.7042 - 3

$= 3 - (0.2314)$

$-0.0004 - 0.2314$

$= 2.7406$

Since x_3 and x_4 are same up to three decimal places, a real root is 2.740.

example 4

Solve the equation $x \tan x = -1$, starting with $x_0 = 2.5$ and $x_1 = 3$, correct up to three decimal places.

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9.3 Regula Falsi Method 9.19

Solution

Let $f(x) = x \tan x + 1$ $f(2.5) = -0.8676$ and $f(3) = 0.5724$ Since $f(2.5) < 0$ and $f(3) > 0$, the root lies between 2.5 and 3.Let $x_0 = 2.5$, $x_1 = 3$ $x_1 - x_0$ $x_2 = x_0 - f(x_0)$ $f(x_1) - f(x_0)$ $3 - 2.5$ $= 2.5 - (-0.8676)$ $0.5724 + 0.8676$ $= 2.8013$ $f(x_2) = f(2.8013) = 0.0082$ Since $f(2.8013) > 0$ and $f(2.5) < 0$, the root lies between 2.8013 and 2.5, i.e., x_2 and x_0 . $x_2 - x_0$ $x_3 = x_0 - f(x_0)$ $f(x_2) - f(x_0)$ $2.8013 - 2.5$ $= 2.5 - (-0.8676)$ $0.0082 + 0.8676$ $= 2.7985$ $f(x_3) = f(2.7985) = 0.0003$ Since $f(2.7985) > 0$ and $f(2.5) < 0$, the root lies between 2.7985 and 2.5, i.e., x_3 and x_0 . $x_3 - x_0$ $x_4 = x_0 - f(x_0)$ $f(x_3) - f(x_0)$ $2.7985 - 2.5$ $= 2.5 - (-0.8676)$ $0.0003 + 0.8676$ $= 2.7984$ Since x_3 and x_4 are same up to three decimal places, the root is 2.798.

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exerclse

Find a real root of the following equations correct to three decimal places

using the regula falsi method:

1. x

3

$+ x^{-1} = 0$

[Ans.: 0.682]

4. x^e

3

$= 3$

[Ans.: 1.050]

9.4 newton—raphson Method

Let $f(x) = 0$ be the given equation and x_0 be an approximate root of the equation. If

$x_1 = x_0 + h$ be the exact root then $f(x_1) = 0$.

i.e., $f(x_0 + h) = 0$

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9.4 Newton—Raphson Method 9.23

h^2

$f(x_0) + hf'(x_0) + \frac{f''(x_0)}{2!}h^2 + \dots = 0$ [By Taylor's series]

2!

Since h is small, neglecting h^2 and higher powers of h ,

$f(x_0) + hf'(x_0) = 0$

$f(x_0)$

$h = -$

$\frac{f(x_0)}{f'(x_0)}$

$f(x_0)$

$x_1 = x_0 + h = x_0 -$

$\frac{f(x_0)}{f'(x_0)}$

Similarly, starting with x_1 , a still better approximation x_2 is obtained.

$f(x_1)$

$x_2 = x_1 -$

$\frac{f(x_1)}{f'(x_1)}$

$f(x_n)$

In general, $x_{n+1} = x_n -$

$\frac{f(x_n)}{f'(x_n)}$

This equation is known as the Newton-Raphson formula or Newton's iteration formula.

9.4.1 geometrical Interpretation

Let x_0 be a point near the root a of the

equation $f(x) = 0$ (Fig. 9.3). The equation of

the tangent at $P_0[x_0, f(x_0)]$ is

$$y - f(x_0) = f'(x_0)(x - x_0)$$

This line cuts the x-axis at x_1 .

$$f(x_0)$$

$$x_1 = x_0 -$$

$$f'(x_0)$$

Fig. 9.3

which is a first approximation to the root a .

If P_1 is the point corresponding to x_1 on the curve then the tangent at P_1 will cut the x-axis at x_2 which is nearer to a and is the second approximation to the root. Repeating this process, the root a is approached quite rapidly. Thus, this method consists of replacing the part of the curve between the point P_0 and the x-axis by means of the tangent to the curve at P_0 .

9.4.2 convergence of the newton—raphson Method

By the Newton-Raphson method,

$$f(x_n)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \dots (9.1)$$

$$f'(x_n)$$

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9.24

The Newton-Raphson method converges if $|f'(x)| < 1$.

$$f(x)$$

$$f(x) = x -$$

$$f'(x)$$

$$\frac{f(x)}{f'(x)} = \frac{f(x)}{f'(x)} \sim \frac{f(x)}{f'(x)}$$

$$f'(x) = 1 - \frac{f(x)}{f'(x)}$$

$$\frac{f(x)}{f'(x)}$$

$$\frac{f(x)}{f'(x)} = \frac{f(x)}{f'(x)}$$

$$\frac{f(x)}{f'(x)} = \frac{f(x)}{f'(x)}$$

$$f(x) f'(x)$$

$$f'(x) =$$

$$[f'(x)]^2$$

Hence, the Newton-Raphson method converges if

$$f(x) f'(x)$$

$$< 1$$

$$[f'(x)]^2$$

$$f(x) f'(x) < [f'(x)]^2$$

$$2$$

$$\dots (9.2)$$

If a is the actual root of $f(x) = 0$, a small interval should be selected in which $f(x)$, $f'(x)$ and $f''(x)$ are all continuous and the condition given by Eq. (9.2) is satisfied.

Hence, the Newton-Raphson method always converges provided the initial approximation x_0 is taken very close to the actual root a .

9.4.3 rate of convergence of the newton—raphson

Method

Let a be exact root of $f(x) = 0$ and let x_n, x_{n+1} be two successive approximations to the actual root. If ϵ_n and ϵ_{n+1} are the corresponding errors then

$$x_n = a + \epsilon_n$$

$$x_{n+1} = a + \epsilon_{n+1}$$

Substituting in Eq. (9.1),

$$f(a + \epsilon_n)$$

$$a + \epsilon_{n+1} = a + \epsilon_n -$$

$$f'(a + \epsilon_n)$$

$$f(a + \epsilon_n)$$

$$\epsilon_{n+1} = \epsilon_n -$$

$$f'(a + \epsilon_n)$$

$$\epsilon_{2n}$$

$$f(a) + \epsilon_n f'(a) + \frac{f''(a)}{2} \epsilon_n^2 + \dots$$

$$= \epsilon_n - 2! [\text{By Taylor's series}]$$

$$f'(a) + \epsilon_n f''(a) + \dots$$

$$\epsilon_{2n}$$

$$\epsilon_n f'(a) + f''(a) \epsilon_n^2 + \dots$$

$$= \epsilon_n - 2 [\because f(a) = 0]$$

$$f'(a) + \epsilon_n f''(a)$$

$$18$$

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9.4 Newton—Raphson Method 9.25

Neglecting the derivatives of order higher than two,

$$\epsilon_{2n}$$

$$\epsilon_n f'(a) + f''(a) \epsilon_n^2$$

$$\epsilon_{n+1} = \epsilon_n - 2$$

$$f'(a) + \epsilon_n f''(a)$$

$$1 \approx \epsilon_{2n} f''(a) \sim$$

$$= \frac{1}{2}$$

$$2 \approx f'(a) + \epsilon_n f''(a) \approx$$

$$\epsilon f''(a) \sim$$

$$\epsilon_{2n} \approx f'(a)$$

$$= \frac{1}{2}$$

$$2 \approx f''(a)$$

$$1 + \epsilon_n$$

$$\hat{f}(a) \approx$$

$$\epsilon_{2n} f'(a)$$

$$\approx \dots (9.3)$$

$$2 f'(a)$$

Equation (9.3) shows that the error at each stage is proportional to the square of the error in the previous stage. Hence, the Newton-Raphson method has a quadratic convergence and the convergence is of the order 2.

example 1

Find the root of the equation $x^3 + x - 1 = 0$, correct up to four decimal places.

Solution

$$\text{Let } f(x) = x^3 + x - 1$$

$$f(0) = -1 \text{ and } f(1) = 1$$

Since $f(0) < 0$ and $f(1) > 0$, the root lies between 0 and 1.

$$\text{Let } x_0 = 1$$

$$f'(x) = 3x^2 + 1$$

By the Newton-Raphson method,

$$f(x_n)$$

$$x_{n+1} = x_n -$$

$$f'(x_n)$$

$$f(x_0) = f(1) = 1$$

$$f'(x_0) = f'(1) = 4$$

$$f(x_0)$$

$$x_1 = x_0 -$$

$$f'(x_0)$$

$$1$$

$$= 1 -$$

$$4$$

$$= 0.75$$

$$19$$

$$f(x_1) = f(0.75) = 0.171875$$

$$f'(x_1) = f'(0.75) = 2.6875$$

$$f(x_1)$$

$$x_2 = x_1 -$$

$$f'(x_1)$$

$$0.171875$$

$$= 0.75 -$$

$$2.6875$$

$$= 0.68605$$

$$f(x_2) = f(0.68605) = 0.00894$$

$$f'(x_2) = f'(0.68605) = 2.41198$$

$$f(x_2)$$

$$x_3 = x_2 -$$

$$f'(x_2)$$

$$0.00894$$

$$= 0.68605 -$$

$$2.41198$$

$$= 0.68234$$

$$f(x_3) = f(0.68234) = 0.000028$$

$$f'(x_3) = f'(0.68234) = 2.39676$$

$$f(x_3)$$

$$x_4 = x_3 -$$

$$f'(x_3)$$

$$0.000028$$

$$= 0.68234 -$$

$$2.39676$$

$$= 0.68233$$

Since x_3 and x_4 are same up to four decimal places, the root is 0.6823.

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9.28

example 3

Find the root of $x^4 - x - 10 = 0$, correct up to three decimal places.

Solution

$$\text{Let } f(x) = x^4 - x - 10$$

$$f(1) = -10, \text{ and } f(2) = 4$$

Since $f(1) < 0$ and $f(2) > 0$, the root lies between 1 and 2.

$$\text{Let } x_0 = 2$$

$$f'(x) = 4x^3 - 1$$

By the Newton-Raphson method,

$$f(x_n)$$

$$x_{n+1} = x_n -$$

$$f'(x_n)$$

$$f(x_0) = f(2) = 4$$

$$f'(x_0) = f'(2) = 31$$

$$f(x_0)$$

$$x_1 = x_0 -$$

$$f'(x_0)$$

4

= 2-

31

= 1.871

$f(x_1) = f(1.871) = 0.3835$

$f'(x_1) = f'(1.871) = 25.1988$

$f(x_1)$

$x_2 = x_1 -$

$f'(x_1)$

0.3835

= 1.871 -

25.1988

= 1.8558

$f(x_2) = f(1.8558) = 5.2922 \times 10^{-3}$

$f'(x_2) = f'(1.8558) = 24.5655$

$f(x_2)$

$x_3 = x_2 -$

$f'(x_2)$

5.2922×10^{-3}

= 1.8558 -

24.5655

= 1.8556

Since x_2 and x_3 are same up to three decimal places, the root is 1.855.

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9.34

By the Newton-Raphson method,

$f(x_n)$

$x_{n+1} = x_n -$

$f'(x_n)$

$f(x_0) = f(1) = 0.4597$

$f'(x_0) = f'(1) = 1.8415$

$f(x_0)$

$x_1 = x_0 -$

$f'(x_0)$

0.4597

= 1-

1.8415

= 0.7504

$f(x_1) = f(0.7504) = 0.019$

$f'(x_1) = f'(0.7504) = 1.6819$

$$f(x_1)$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.7504 - \frac{1.6819}{0.7391}$$

$$f(x_2) = f(0.7391) = 0.00002$$

$$f'(x_2) = f'(0.7391) = 1.6736$$

$$f(x_2)$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.7391 - \frac{0.00002}{1.6736}$$

$$= 0.7391$$

Since x_2 and x_3 are same up to three decimal places, the root is 0.739.

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example 10

Find an iterative formula for x^N , where N is a positive number and hence, evaluate (i) 3^{11} , and (ii) 3^{58}

Solution

Let $x = x^N$

$x^N - N = 0$

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9.4 Newton—Raphson Method 9.37

Let $f(x) = x^N - N$

$f'(x) = Nx^{N-1}$

By the Newton—Raphson method,

$f(x_n)$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$f(x_n)$

$x_n^N - N$

$= x_n^N - N$

Nx_n^{N-1}

$(N-1)x_n^{N-1} + N$

Nx_n^{N-1}

This is the iterative formula for $k = 3$ and N .

(i) When $N = 11$ and $k = 3$,

$$f(x) = x^3 - 11$$

$$f(2) = -3 \text{ and } f(3) = 16$$

Since $f(2) < 0$ and $f(3) > 0$, the root lies between 2 and 3.

$$\text{Let } x_0 = 3$$

$$2x_n^3 + 11$$

$$x_{n+1} =$$

$$3x_n^2$$

$$2x_0^3 + 11$$

$$x_1 = 2.4074$$

$$3x_0^2$$

$$2x_1^3 + 11$$

$$x_2 = 2.2376$$

$$3x_1^2$$

$$2x_2^3 + 11$$

$$x_3 = 2.2240$$

$$3x_2^2$$

$$2x_3^3 + 11$$

$$x_4 = 2.2240$$

$$3x_3^2$$

Since x_3 and x_4 are same up to four decimal places,

$$x = 2.2240$$

$$11 = 2.2240$$

(ii) When $N = 58$ and $k = 3$,

$$f(x) = x^3 - 58$$

$$f(3) = -31 \text{ and } f(4) = 6$$

Since $f(3) < 0$ and $f(4) > 0$, the root lies between 3 and 4.

$$24$$

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$$9.38$$

$$\text{Let } x_0 = 4$$

$$2x_n^3 + 58$$

$$x_{n+1} =$$

$$3x_n^2$$

$$2x_0^3 + 58$$

$$x_1 = 3.875$$

$$3x_0^2$$

$$2x_1^3 + 58$$

$$x_2 = 3.8709$$

$$3 \times 12$$

$$2 \times 23 + 58$$

$$x_3 = 3.8709$$

$$3 \times 22$$

Since x_2 and x_3 are same up to four decimal places,

$$3$$

$$58 = 3.8709$$

exercIse

I. Find the roots of the following equations using the newton-raphson method:

$$3$$

$$1. \ x - x - 1 = 0$$

$$[\text{Ans.: } 1.3247]$$

$$3 \ 2$$

$$2. \ x^3 + 2x^2 + 50x + 7 = 0$$

$$[\text{Ans.: } -0.1407]$$

$$2$$

$$9. \ \cos x = x$$

$$[\text{Ans.: } 0.8241]$$

$$25$$

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$$9.5 \text{ Secant Method } 9.39$$

II. Find the values of the following:

$$1. \ 35$$

$$[\text{Ans.: } 5.916]$$

$$3$$

$$2. \ 24$$

$$[\text{Ans.: } 2.884]$$

$$1$$

$$3.$$

$$14$$

$$[\text{Ans.: } 0.2673]$$

$$9.5 \text{ secAnt Method}$$

The Newton-Raphson method requires the evaluation of two functions (the function and its derivative) per iteration. For complicated expressions, the method takes a large amount of time. Hence, it is desirable to have a method that converges as fast as the Newton-Raphson method but involves only evaluation of the function.

Let $f(x) = 0$ be the given equation. Let x_0 and x_1 be the approximate roots of the equation $f(x) = 0$ and $f(x_0)$ and $f(x_1)$ are their function values respectively. If x_2 is the point of intersection of the x-axis and the line joining points $P[x_0, f(x_0)]$ and $Q[x_1, f(x_1)]$ then x_2 is closer to the root a than x_0 and x_1 (Fig. 9.4).

Fig. 9.4

Using the slope formula,

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{0 - f(x_1)}{x_2 - x_1}$$

$$m = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$= \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$x_1 - x_0$$

$$x_2 - x_1 = -\frac{f(x_1)}{f(x_1) - f(x_0)}$$

$$f(x_1) - f(x_0)$$

$$x_1 - x_0$$

$$x_2 = x_1 - \frac{f(x_1)}{f(x_1) - f(x_0)}$$

$$f(x_1) - f(x_0)$$

Using x_1 and x_2 , the process is repeated to obtain x_3 .

In general,

$$x_n - x_{n-1}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f(x_n) - f(x_{n-1})}, n \geq 1$$

$$f(x_n) - f(x_{n-1})$$

This method is similar to the regula falsi method. This method starts with two initial approximations x_0 and x_1 and calculates x_2 by the same formula as in the regula falsi

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9.40

method but proceeds to the next iteration without considering any root bracketing, i.e., the condition $f(x_0)f(x_1) < 0$.

convergence of the secant Method

By the Secant method,

$$x_n - x_{n-1}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f(x_n) - f(x_{n-1})} \dots (9.4)$$

$$f(x_n) - f(x_{n-1})$$

Let a be the exact root of $f(x) = 0$ and let x_n, x_{n+1} be two successive approximations to the actual root.

If $\epsilon_n, \epsilon_{n-1}$, are the corresponding error then

$$x_n = a + \epsilon_n$$

$$x_{n-1} = a + \Delta x_{n-1}$$

$$x_{n+1} = a + \Delta x_{n+1}$$

Substituting in Eq. (9.4),

$$\Delta x_n - \Delta x_{n-1}$$

$$a + \Delta x_{n+1} = a + \Delta x_n - f(a + \Delta x_n)$$

$$f(a + \Delta x_n) - f(a + \Delta x_{n-1})$$

$$\Delta x_n - \Delta x_{n-1}$$

$$\Delta x_{n+1} = \Delta x_n - f(a + \Delta x_n)$$

$$f(a + \Delta x_n) - f(a + \Delta x_{n-1})$$

$$\approx \Delta x^2$$

$$(\Delta x_n - \Delta x_{n-1}) \left[f(a) + \Delta x_n f'(a) + \frac{n}{2} f''(a) + \dots \right]$$

$$= \Delta x_n - \hat{\Delta x}^2$$

$$\frac{1}{2} \Delta x^2$$

$$(\Delta x_n - \Delta x_{n-1}) f'(a) + (\Delta x_n - \Delta x_{n-1}) f''(a) + \dots \quad [\text{By Taylor's series}]$$

$$\frac{1}{2}$$

$$\approx \Delta x^2$$

$$\therefore f(a) = 0$$

$$(\Delta x_n - \Delta x_{n-1}) \left[\Delta x_n f'(a) + 2 \frac{n}{2} f''(a) \right] \approx$$

$$= \Delta x_n - \hat{\Delta x}^2 \quad \text{and neglecting}$$

$$\frac{1}{2}$$

$$(\Delta x_n - \Delta x_{n-1}) f'(a) + 2 \Delta x_n - \Delta x_{n-1} f''(a) \approx$$

$$\frac{1}{2}$$

higher order

derivatives of $f(a)$

$$\approx \Delta x$$

$$(\Delta x_n - \Delta x_{n-1}) \Delta x_n f'(a) \approx 1 + 2 \frac{n}{2} f''(a) \approx$$

$$= \Delta x_n - \hat{\Delta x}^2$$

$$\approx (\Delta x_n + \Delta x_{n-1}) f''(a) \approx$$

$$(\Delta x_n - \Delta x_{n-1}) f'(a) + \frac{1}{2} + 2$$

$$f(a) \approx$$

$$\hat{\Delta x}$$

$$\Delta x_n (1 + c \Delta x_n)$$

$$= \Delta x_n -$$

$$1 + c (\Delta x_n + \Delta x_{n-1})$$

$$1 f''(a)$$

where $c = \dots$ (9.5)

$$2 f(a)$$

$$27$$

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9.5 Secant Method 9.41

$$\begin{aligned}
 & \epsilon_n + c \epsilon_{2n} + c \epsilon_n \epsilon_{n-1} - \epsilon_n - c \epsilon_{2n} \\
 & \epsilon_{n+1} = \\
 & 1 + c (\epsilon_n + \epsilon_{n-1}) \\
 & c \epsilon_n \epsilon_{n-1} \\
 & 1 + c (\epsilon_n + \epsilon_{n-1}) \\
 & \approx c \epsilon_n \epsilon_{n-1} \hat{=} 1 + c (\epsilon_n + \epsilon_{n-1}) \approx 1^{\sim} \dots (9.6) \\
 & \hat{=}^{\circ}
 \end{aligned}$$

Equation (9.5) is a nonlinear difference equation which can be solved by letting

$$\epsilon_{n+1} = A \epsilon_n^p \text{ or } \epsilon_n = A \epsilon_{n-1}^p .$$

$$1 \quad 1$$

$$\epsilon_{n-1} = \epsilon_n^p A^p$$

Substituting in Eq. (9.6),

$$1 \quad 1$$

$$A \epsilon_n^p = c \epsilon_n \epsilon_n^p A^p$$

$$\hat{=} 1^{\wedge} 1$$

$$- \hat{=} 1^+ \sim 1^+$$

$$\epsilon_n^p = c A \hat{=} p^-$$

$$\epsilon_n^p$$

Equating the power of ϵ_n on both the sides,

$$1$$

$$p = 1^+$$

$$p$$

$$p^2 - p - 1 = 0$$

$$1$$

$$p = (1 \pm 5)$$

$$2$$

Taking the positive sign only,

$$p = 1.618$$

$$\epsilon_{n+1} = A \epsilon_{1.618}$$

$$n$$

Hence, the rate of convergence of the secant method is 1.618 which is lesser than the Newton-Raphson method. The secant method evaluates the function only once in each iteration, whereas the Newton-Raphson method evaluates two functions $f(x)$ and $f'(x)$ in each iteration. Hence, the secant method is more efficient than the Newton-Raphson method.

example 1

Find the approximate root of $x^3 - 2x - 1 = 0$, starting from $x_0 = 1.5$,

$x_1 = 2$, correct upto three decimal places.

Solution

$$\text{Let } f(x) = x^3 - 2x - 1$$

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9.42

$x_0 = 1.5, x_1 = 2$

$f(x_0) = f(1.5) = -0.625$ and $f(x_1) = f(2) = 3$

By the secant method,

$x_n - x_{n-1}$

$x_{n+1} = x_n - f(x_n)$

$f(x_n) - f(x_{n-1})$

$x_1 - x_0$

$x_2 = x_1 - f(x_1)$

$f(x_1) - f(x_0)$

$2 - 1.5$

$= 2 - (3)$

$3 + 0.625$

$= 1.5862$

$f(x_2) = f(1.5862) = -0.1815$

$x_2 - x_1$

$x_3 = x_2 - f(x_2)$

$f(x_2) - f(x_1)$

$1.5862 - 2$

$= 1.5862 - (-0.1815)$

$-0.1815 - 3$

$= 1.6098$

$f(x_3) = f(1.6098) = -0.0479$

$x_3 - x_2$

$x_4 = x_3 - f(x_3)$

$f(x_3) - f(x_2)$

$1.6098 - 1.5862$

$= 1.6098 - (-0.0479)$

$-0.0479 + 0.1815$

$= 1.6183$

$f(x_4) = f(1.6183) = 0.0016$

$x_4 - x_3$

$x_5 = x_4 - f(x_4)$

$f(x_4) - f(x_3)$

$1.6183 - 1.6098$

$= 1.6183 - (0.0016)$

$0.0016 + 0.0479$

$= 1.6181$

Since x_4 and x_5 are same up to three decimal places, the root is 1.618.

example 2

Find the approximate root of the equation $x^3 + x^2 - 3x - 3 = 0$, correct up to 4 decimal places.

Solution

Let $f(x) = x^3 + x^2 - 3x - 3 = 0$

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9.5 Secant Method 9.43

Let $x_0 = 1, x_1 = 2$

$f(x_0) = f(1) = -4$ and $f(x_1) = f(2) = 3$

By the secant method,

$x_n - x_{n-1}$

$x_{n+1} = x_n - f(x_n)$

$f(x_n) - f(x_{n-1})$

$x_1 - x_0$

$x_2 = x_1 - f(x_1)$

$f(x_1) - f(x_0)$

$2 - 1$

$= 2 - (3)$

$3 + 4$

$= 1.57143$

$f(x_2) = f(1.57143) = -1.36442$

$x_2 - x_1$

$x_3 = x_2 - f(x_2)$

$f(x_2) - f(x_1)$

$1.57143 - 2$

$= 1.57143 - (-1.36442)$

$-1.36442 - 3$

$= 1.70541$

$f(x_3) = f(1.70541) = -0.24775$

$x_3 - x_2$

$x_4 = x_3 - f(x_3)$

$f(x_3) - f(x_2)$

$1.70541 - 1.57143$

$= 1.70541 - (-0.24775)$

$-0.24775 + 1.36442$

$= 1.73514$

$f(x_4) = f(1.73514) = 0.0293$

$x_4 - x_3$

$x_5 = x_4 - f(x_4)$

$$f(x_4) - f(x_3)$$

$$1.73514 - 1.70541$$

$$= 1.73514 - (0.0293)$$

$$0.0293 + 0.24775$$

$$= 1.732$$

$$f(x_5) = f(1.732) = -0.00048$$

$$x_5 - x_4$$

$$x_6 = x_5 - f(x_5)$$

$$f(x_5) - f(x_4)$$

$$1.732 - 1.73514$$

$$= 1.732 - (-0.00048)$$

$$-0.00048 - 0.0293$$

$$= 1.73205$$

$$f(x_6) = f(1.73205) = -0.000008$$

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9.44

$$x_6 - x_5$$

$$x_7 = x_6 - f(x_6)$$

$$f(x_6) - f(x_5)$$

$$1.73205 - 1.732$$

$$= 1.73205 - (-0.000008)$$

$$-0.000008 + 0.00048$$

$$= 1.73205$$

Since x_6 and x_7 are same up to five decimal places, the root is 1.73205.

example 3

Find the root of $x \log_{10} x - 1.9 = 0$, correct up to three decimal places with $x_0 = 3$ and $x_1 = 4$.

Solution

Let $f(x) = x \log_{10} x - 1.9$

$x_0 = 3, x_1 = 4$

$f(x_0) = f(3) = -0.4686$ and $f(x_1) = f(4) = 0.5082$

By the secant method,

$x_n - x_{n-1}$

$x_{n+1} = x_n - f(x_n)$

$f(x_n) - f(x_{n-1})$

$x_1 - x_0$

$x_2 = x_1 - f(x_1)$

$f(x_1) - f(x_0)$

4-3

$$\begin{aligned}
 &= 4 - (0.5082) \\
 &0.5082 + 0.4686 \\
 &= 3.4797 \\
 f(x_2) &= f(3.4797) = -0.0156 \\
 x_2 - x_1 \\
 x_3 &= x_2 - f(x_2) \\
 f(x_2) - f(x_1) \\
 3.4797 - 4 \\
 &= 3.4797 - (-0.0156) \\
 &-0.0156 - 0.5082 \\
 &= 3.4952 \\
 f(x_3) &= f(3.4952) = -0.0005 \\
 x_3 - x_2 \\
 x_4 &= x_3 - f(x_3) \\
 f(x_3) - f(x_2) \\
 3.4952 - 3.4797 \\
 &= 3.4952 - (-0.0005) \\
 &-0.0005 + 0.0156 \\
 &= 3.4957
 \end{aligned}$$

Since x_3 and x_4 are same up to three decimal places, the root is 3.495.

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9.5 Secant Method 9.45

example 4

Find the positive solution of $x - 2 \sin x = 0$, correct up to three decimal places starting from $x_0 = 2$ and $x_1 = 1.9$.

Solution

Let $f(x) = x - 2 \sin x$

$x_0 = 2, x_1 = 1.9$

$f(x_0) = f(2) = 0.1814$ and $f(x_1) = f(1.9) = 0.0074$

By the secant method,

$$\begin{aligned}
 x_n - x_{n-1} \\
 x_{n+1} &= x_n - f(x_n) \\
 f(x_n) - f(x_{n-1}) \\
 x_1 - x_0 \\
 x_2 &= x_1 - f(x_1) \\
 f(x_1) - f(x_0) \\
 1.9 - 2 \\
 &= 1.9 - (0.0074) \\
 0.0074 - 0.1814
 \end{aligned}$$

$$= 1.8957$$

$$f(x_2) = f(1.8957) = 0.00034$$

$$x_2 - x_1$$

$$x_3 = x_2 - f(x_2)$$

$$f(x_2) - f(x_1)$$

$$1.8957 - 1.9$$

$$= 1.8957 - (0.00034)$$

$$0.00034 - 0.0074$$

$$= 1.8955$$

Since x_2 and x_3 are same up to three decimal places, the positive root is 1.895.

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9.48

$$x_5 - x_4$$

$$x_6 = x_5 - f(x_5)$$

$$f(x_5) - f(x_4)$$

$$0.5169 - 0.5317$$

$$= 0.5169 - (0.0026)$$

$$0.0026 + 0.0429$$

$$= 0.5177$$

$$f(x_6) = f(0.5177) = 0.0002$$

$$x_6 - x_5$$

$$x_7 = x_6 - f(x_6)$$

$$f(x_6) - f(x_5)$$

$$0.5177 - 0.5169$$

$$= 0.5177 - (0.0002)$$

$$0.0002 - 0.0026$$

$$= 0.5178$$

Since x_6 and x_7 are same up to three decimal places, the root is 0.517.

exerclse

Find a real root of the following equations correct upto three decimal places using the secant method:

3 2

$$1. x^3 - 2x^2 + 3x - 4 = 0$$

[Ans.: 1.650]

3 2

$$2. x^3 + 3x^2 - 3 = 0$$

[Ans.: 0.879]

x

$$3. e - 4x = 0$$

[Ans.: 0.357]

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Points to Remember 9.49

points to remember

Bisection Method

In this method, two points x_0 and x_1 are chosen such that $f(x_0)$ and $f(x_1)$ are of opposite signs. The first approximation to the root is

$$x_0 + x_1$$

$$x_2 =$$

$$2$$

If $f(x_0)$ and $f(x_2)$ are of opposite signs, the root lies between x_0 and x_2 and the next approximation x_3 is obtained as

$$x_0 + x_2$$

$$x_3 =$$

$$2$$

This process is repeated till the root is obtained to the desired accuracy.

Regula Falsi Method

In this method, two points x_0 and x_1 are chosen such that $f(x_0)$ and $f(x_1)$ are of opposite signs.

$$x_1 - x_0$$

$$x_2 = x_0 - f(x_0)$$

$$f(x_1) - f(x_0)$$

which is an approximation to the root.

If $f(x_0)$ and $f(x_2)$ are of opposite signs, the root lies between x_0 and x_2 , and the next approximation x_3 is obtained as

$$x_2 - x_0$$

$$x_3 = x_0 - f(x_0)$$

$$f(x_2) - f(x_0)$$

If the root lies between x_1 and x_2 , the next approximation x_3 is obtained as

$$x_1 - x_2$$

$$x_3 = x_2 - f(x_2)$$

$$f(x_1) - f(x_2)$$

This process is repeated till the root is obtained to the desired accuracy.

Newton-Raphson Method

$$f(x_n)$$

$$x_{n+1} = x_n -$$

$$f'(x_n)$$

The Newton-Raphson method has a quadratic convergence and the convergence is

of the order 2.

Secant Method

$x_n - x_{n-1}$

$x_{n+1} = x_n - f(x_n)$

$f(x_n) - f(x_{n-1})$

The rate of convergence of the secant method is 1.618.

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6

Finite Differences and

Interpolation

chapter outline

6.1 Introduction

6.2 Finite Differences

6.3 Different Operators and their Relations

6.4 Interpolation

6.5 Newton's Forward Interpolation Formula

6.6 Newton's Backward Interpolation Formula

6.7 Central Difference Interpolation

6.11 Interpolation with Unequal Intervals

6.12 Lagrange's Interpolation Formula

6.13 Divided Differences

6.14 Newton's Divided Difference Formula

6.15 Inverse Interpolation

6.1 Introduction

Interpolation is the process of reading between the lines of a table. It is the process of computing intermediate values of a function from a given set of tabular values of the function. Extrapolation is used to denote the process of finding the values outside the given interval.

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In the interpolation process, the given set of tabular values are used to find an expression

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for $f(x)$ and then using it to find its required value for a given value of x . But it is

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difficult to find an exact form of $f(x)$ using the limited values in the table. Hence, $f(x)$

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is replaced by another function $f(x)$, which matches with $f(x)$ at the discrete values in

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the table. This function $f(x)$ is known as the interpolating function.

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6.2 Finite Differences and Interpolation

When the interpolating function is a polynomial function, the process is known as polynomial interpolation. Polynomial interpolations are preferred because of the following reasons:

- (i) They are simple forms of functions which can be easily manipulated.
- (ii) Polynomials are free from singularities whereas rational functions or other types have singularities.

6.2 Finite Differences

Let the function $y = f(x)$ be tabulated for the equally spaced values $y_0 = f(x_0)$, $y_1 = f(x_0 + h)$, $y_2 = f(x_0 + 2h)$, ..., $y_n = f(x_0 + nh)$, as

$x \quad x_0 \quad x_0 + h \quad x_0 + 2h \quad \dots \quad x_0 + nh \quad \dots$

$y = f(x) \quad y_0 \quad y_1 \quad y_2 \quad \dots \quad y_n \quad \dots$

To determine the values of $f(x)$ for some intermediate values of x , the following three types of differences can be used.

6.2.1 Forward differences

If $y_0, y_1, y_2, \dots, y_n$ denote a set of values of y then $y_1 - y_0, y_2 - y_1, \dots, y_n - y_{n-1}$ are called the first forward differences of y and are denoted by $Dy_0, Dy_1, \dots, Dy_{n-1}$ respectively.

$$Dy_0 = y_1 - y_0$$

$$Dy_1 = y_2 - y_1$$

□ □

$$Dy_{n-1} = y_n - y_{n-1}$$

where D is called the forward difference operator. The differences of the first forward differences are called second forward differences and are denoted by $D^2y_0, D^2y_1, \dots, D^2y_{n-1}$. Similarly, third forward differences, fourth forward differences, etc., can be defined.

$$D^2y_0 = D(Dy_0)$$

$$= D(y_1 - y_0)$$

$$= Dy_1 - Dy_0$$

$$= y_2 - y_1 - (y_1 - y_0)$$

$$= y_2 - 2y_1 + y_0$$

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$$D^3y_0 = D^2y_1 - D^2y_0$$

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$$= (y_3 - 2y_2 + y_1) - (y_2 - 2y_1 + y_0)$$

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$$= y_3 - 3y_2 + 3y_1 - y_0$$

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6.2 Finite Differences 6.3

$$D_4y_0 = D_3y_1 - D_3y_0$$

$$= (y_4 - 3y_3 + 3y_2 - y_1) - (y_3 - 3y_2 + 3y_1 - y_0)$$

$$= y_4 - 4y_3 + 6y_2 - 4y_1 + y_0$$

Since the coefficients occurring on the right-hand side are the binomial coefficients, the general form is

$$D_n y_0 = y_n - nC_1 y_{n-1} + nC_2 y_{n-2} - \dots + (-1)^n y_0$$

Forward Difference Table

x y Dy D²y D³y D⁴y

x₀ y₀

$$Dy_0 = y_1 - y_0$$

$$x_0 + h = x_1 \quad y_1 \quad D_2y_0 = Dy_1 - Dy_0$$

$$Dy_1 = y_2 - y_1 \quad D_3y_0 = D_2y_1 - D_2y_0$$

$$x_0 + 2h = x_2 \quad y_2 \quad D_2y_1 = Dy_2 - Dy_1 \quad D_4y_0 = D_3y_1 - D_3y_0$$

$$Dy_2 = y_3 - y_2 \quad D_3y_1 = D_2y_2 - D_2y_1$$

$$x_0 + 3h = x_3 \quad y_3 \quad D_2y_2 = Dy_3 - Dy_2$$

$$Dy_3 = y_4 - y_3$$

$$x_0 + 4h = x_4 \quad y_4$$

In a difference table, x is called the argument, and y is called the function or entry.

note When (n + 1) values are given, the highest-order difference is n, e.g., when 5 values are given, the highest-order difference is 4.

6.2.2 Backward differences

If $y_0, y_1, y_2, \dots, y_n$ denote a set of values of y then $y_1 - y_0, y_2 - y_1, \dots, y_n - y_{n-1}$ are called the first backward differences of y and are denoted by $-y_1, -y_2, \dots, -y_n$, respectively.

$$-y_1 = y_1 - y_0$$

$$-y_2 = y_2 - y_1$$

□ □ □

$$-y_n = y_n - y_{n-1}$$

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where $-$ is called the backward difference operator. Similarly, backward differences graded

of higher order can be defined.

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$$-2y_2 = -(-y_2)$$

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6.4 Finite Differences and Interpolation

$$= -(y_2 - y_1)$$

$$= -y_2 - -y_1$$

$$= y_2 - y_1 - (y_1 - y_0)$$

$$= y_2 - 2y_1 + y_0$$

$$-3y_3 = -2y_3 - -2y_2$$

$$= y_3 - 3y_2 + 3y_1 - y_0 \text{ etc.}$$

Backward Difference Table

$$x \quad y \quad -y \quad -2y \quad -3y \quad -4y$$

$$x_0 \quad y_0$$

$$-y_1 = y_1 - y_0$$

$$x_1 \quad y_1 \quad -2y_2 = -y_2 - -y_1$$

$$-y_2 = y_2 - y_1 \quad -3y_3 = -2y_3 - -2y_2$$

$$x_2 \quad y_2 \quad -2y_3 = -y_3 - -y_2 \quad -4y_4 = -3y_4 - -3y_3$$

$$-y_3 = y_3 - y_2 \quad -3y_4 = -2y_4 - -2y_3$$

$$x_3 \quad y_3 \quad -2y_4 = -y_4 - -y_3$$

$$-y_4 = y_4 - y_3$$

$$x_4 \quad y_4$$

6.2.3 central differences

If $y_0, y_1, y_2, \dots, y_n$ denote a set of values of y then $y_1 - y_0, y_2 - y_1, \dots, y_n - y_{n-1}$ are called the central differences of y and are denoted by $\Delta y_1, \Delta y_3, \dots, \Delta y_{n-1}$ respectively.

n-

$$\Delta y_1$$

$$\Delta y_1 = y_1 - y_0$$

$$\Delta y_3$$

$$\Delta y_3 = y_2 - y_1$$

$$\Delta y_{n-1}$$

$$\Delta y_{n-1}$$

$$\Delta y_{n-1} = y_n - y_{n-1}$$

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$$\Delta y_{n-1}$$

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where d is called the central difference operator. Similarly, higher-order central differences can be defined.

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6.2 Finite Differences 6.5

$d^2 y_1 = d^2 y_3 - d^2 y_1$

$d^2 y_1 = d^2 y_3 - d^2 y_1$

$d^2 y_1 = d^2 y_3 - d^2 y_1$

$d^2 y_1 = d^2 y_3 - d^2 y_1$

$d^2 y_1 = d^2 y_3 - d^2 y_1$

$d^2 y_1 = d^2 y_3 - d^2 y_1$ etc.

3

2

Central Difference Table

$x \ y \ d \ d^2 \ d^3 \ d^4$

$x_0 \ y_0$

$d y_1 = y_1 - y_0$

2

$d^2 y_1 = d y_3 - d y_1$

$x_1 \ y_1$

2 2

$d y_3 = y_2 - y_1 \ d^3 y_3 = d^2 y_2 - d^2 y_1$

2 2

$d^2 y_2 = d y_5 - d y_3 \ d^4 y_2 = d^3 y_5 - d^3 y_3$

$x_2 \ y_2$

2 2 2 2

$d y_5 = y_3 - y_2 \ d^3 y_5 = d^2 y_3 - d^2 y_2$

2 2

$d^2 y_3 = d y_7 - d y_5$

$x_3 \ y_3$

2 2

$d y_7 = y_4 - y_3$

2

$x_4 y_4$

From the central difference table, it is clear that the central differences on the same horizontal line have the same suffix x . Also, the differences of odd orders are known only for half values of the suffix x and those of even orders, for only integral values of the suffix x .

note It is clear from the three difference tables that it is only the notations which change and not the differences.

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$$y_1 - y_0 = \Delta y_0 = -y_1 = \Delta y_1$$

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6.6 Finite Differences and Interpolation

6.3 Different operators and their relations

1. Forward difference operator The forward difference operator is denoted by Δ and is defined as

$$\Delta f(x) = f(x+h) - f(x)$$

$$\text{or } \Delta y_r = y_{r+1} - y_r$$

where h is known as the interval of differencing.

2. Backward difference operator The backward difference operator is denoted by ∇ and is defined as

$$\nabla f(x) = f(x) - f(x-h)$$

$$\text{or } \nabla y_r = y_r - y_{r-1}$$

3. central difference operator The central difference operator is denoted by δ and is defined as

$$\delta h = \delta h$$

$$\delta f(x) = f(x+\frac{h}{2}) - f(x-\frac{h}{2})$$

$$\delta^2 y_r = y_{r+1} - y_{r-1}$$

$$\text{or } \delta^2 y_r = y_{r+1} - y_{r-1}$$

$$\delta^2 y_r = y_{r+1} - y_{r-1}$$

$$\delta^2 y_r = y_{r+1} - y_{r-1}$$

4. shift operator The shift operator is denoted by E and is defined as

$$E[f(x)] = f(x+h)$$

$$\text{or } E y_r = y_{r+1}$$

Similarly, $E^{-1} [f(x)] = f(x - h)$

or $E^{-1} y_r = y_{r-1}$

5. averaging operator The averaging operator is denoted by μ and is defined as

$$\hat{E} h^+ \hat{E} h^+$$

$$f \hat{A} x + \sim + f \hat{A} x - \sim$$

$$\hat{E} 2^- \hat{E} 2^-$$

$$\mu f(x) =$$

$$2$$

6. differential operator The differential operator is denoted by D and is defined as

d

$$D f(x) = \frac{f(x) - f(x-h)}{h}$$

$d x$

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6.3 Different Operators and their Relations 6.7

6.3.1 relations between operators

1. relation between D and E

$$D f(x) = \frac{f(x+h) - f(x)}{h}$$

$$= \frac{E f(x) - f(x)}{h} [\because E f(x) = f(x+h)]$$

$$= (E - 1) f(x)$$

$$\backslash D \int E - 1 \text{ or } E \int 1 + D$$

2. relation between Δ and E

$$\Delta f(x) = f(x) - f(x-h)$$

$$= f(x) - E^{-1} f(x) [\because E^{-1} f(x) = f(x-h)]$$

$$= (1 - E^{-1}) f(x)$$

$$\backslash \Delta \int 1 - E^{-1}$$

3. relation between d and E

$$\hat{E} h^+ \hat{E} h^+$$

$$d f(x) = \frac{f(x+h) - f(x-h)}{2h}$$

$$\hat{E} 2^- \hat{E} 2^-$$

$$1 1$$

$$= E f(x) - E f(x)$$

2 2

$\hat{E} 1 - ^$

1

$= \ddot{E} E 2 - E 2 ^ f (x)$

1 1

$\backslash d \int E - E 2$

2

4. relation between m and E

$\hat{E} h ^ \hat{E} h ^$

$f \acute{A} x + \sim + f \acute{A} x - \sim$

$\ddot{E} 2 ^ \ddot{E} 2 ^$

$m f (x) =$

2

1 1

$E 2 f (x) + E 2 f (x)$

2

1 1

$E 2 + E 2$

$\backslash m \int$

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6.8 Finite Differences and Interpolation

5. relation between D and E

$E f (x) = f (x + h)$

h^2

$= f (x) + h f '(x) + f '' (x) + L$ [By Taylor's series]

$2!$

$h^2 2$

$= f (x) + h D f (x) +$

$D f (x) + L$

$2!$

$\hat{E} h^2 2 ^$

$= \acute{A} 1 + h D + D + L \sim f (x)$

$\tilde{E} 2!^{-}$

$= ehD f (x)$

$\setminus E \int ehD$

Also, $hD \int \log E \int \log(1 + D)$

$1 hD$

--

Corollary $E 2 \int e 2$

Proof

1

$- \hat{E} h^{\wedge}$

$E 2 f (x) = f \hat{A} x - \sim$

$\tilde{E} 2^{-}$

2

$\hat{E} h^{\wedge}$

$h \hat{A} \tilde{E} 2 \sim^{-}$

$= f (x) - f' (x) + f'' (x) - L$

$2 2!$

$\tilde{E} \hat{E} h^{\wedge}$

$2 \sim$

$\hat{A} \sim \cdot$

$\tilde{E} 2^{-}$

$= \hat{A} (1 - D + D 2 - L \cdot \cdot f (x))$

h

$\hat{A} 2 2!^{\circ}$

hD

$= e 2 f (x)$

$1 hD$

--

$\setminus E 2 \int e 2$

6. relation between m and d

$1 -1$

$E 2 + E 2$

$m \int$

2

2

$\hat{E} 12 -1^{\wedge}$

$E + E 2$

$m \int \hat{A} 2 \sim$

Sg Smi by

$\hat{A} \tilde{E} 2 \sim^{-}$

Dg th

gr ed

$E + 2 + E^{-1}$

En pil

4

m

Co

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6.3 Different Operators and their Relations 6.9

$4 + (E - E^{-1})^2$

4

2

d

$\int 1 +$

4

d^2

$m \int 1 +$

4

example 1

Prove that $(1 + D)(1 - \frac{1}{E}) = 1$

Solution

$E \int 1 + D$

$\frac{1}{E} \int 1 - E^{-1}$

$E^{-1} \int 1 - \frac{1}{E}$

$(1 + D)(1 - \frac{1}{E}) \int E E^{-1} = 1$

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Dg th

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6.3 Different Operators and their Relations 6.11

example 5

Evaluate (a) $D(x^2 + \sin x)$, and (b) $D^2 \cos 3x$, the interval of differencing being h .

Solution

$$(i) D(x^2 + \sin x) = [(x+h)^2 + \sin(x+h)] - (x^2 + \sin x)$$

$$= h^2 + 2hx + \sin(x+h) - \sin x$$

$$\hat{E} h^h$$

$$= h^2 + 2hx + 2 \cos x + \sim \sin$$

$$\hat{E} 2^{-2}$$

$$(ii) D^2 \cos 3x = D(D \cos 3x)$$

$$= D[\cos 3(x+h) - \cos 3x]$$

$$= D \cos 3(x+h) - D \cos 3x$$

$$= \cos 3[(x+h)+h] - \cos 3(x+h) - \cos 3(x+h) + \cos 3x$$

$$= \cos 3(x+2h) - 2 \cos 3(x+h) + \cos 3x$$

$$= \cos 3(x+2h) + \cos 3x - 2 \cos 3(x+h)$$

Sg Smi by

Dg th

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En pil

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6.3 Different Operators and their Relations 6.13

example 9

Find the missing term in the following table:

x 1 2 3 4 5

y 7 - 13 21 37

Solution

Difference Table

x y Dy D²y D³y D⁴y

1 7

y₁ - 7

2 y₁ 20 - 2y₁

13 - y₁ 3y₁ - 25

$$3 \cdot 13 y_1 - 5 \cdot 38 - 4 y_1$$

$$8 \cdot 13 - y_1$$

$$4 \cdot 21 \cdot 8$$

$$16$$

$$5 \cdot 37$$

Since only four entries are given, the fourth-order difference will be zero.

$$D^4 y_0 = 0$$

$$38 - 4 y_1 = 0$$

$$y_1 = 9.5$$

example 10

Obtain the estimate of missing terms in the following table:

$$x \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8$$

$$y \quad 2 \quad 4 \quad 8 \quad -32 \quad -128 \quad 256$$

Solution

Sg Smi by

$$\text{Let } f(4) = a, f(6) = b$$

Dg th

gr ed

En pil

m

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6.14 Finite Differences and Interpolation

Difference Table

$$x \quad y \quad D_y \quad D^2_y \quad D^3_y \quad D^4_y \quad D^5_y \quad D^6_y$$

$$1 \quad 2$$

$$2$$

$$2 \quad 4 \quad 2$$

$$4 \quad a \quad -14$$

$$3 \quad 8 \quad a \quad -12 \quad 66 \quad -4a$$

$$a - 8 \quad 52 \quad -3a \quad 10a + b \quad -222$$

$$4 \quad a \quad 40 \quad -2a \quad b + 6a \quad -156 \quad 706 \quad -20a \quad -6b$$

$$32 \quad -a \quad b + 3a \quad -104 \quad 484 \quad -10a \quad -5b$$

$$15b + 15a$$

$$5 \quad 32 \quad b + a \quad -64 \quad 328 \quad -4a \quad -4b$$

$$-1196$$

$$224 \quad -3b$$

$$b \quad -32 \quad 10b + 5a \quad -712$$

-a

$$6b + 160 - 2b + a = 384$$

$$128 - 3b = 160$$

$$7128b$$

$$128$$

$$8256$$

Since only six values are given,

$$D_6 y_0 = 0$$

$$20a + 6b = 706 \dots (1)$$

$$15a + 15b = 1196 \dots (2)$$

Solving Eqs (1) and (2),

$$a = 16.26$$

$$b = 63.48$$

example 11

The following table gives the value of y which is a polynomial of degree

Sg Smi by

□□ve. It is known that y = f(3) is in error. Correct the error.

Dg th

gr ed

$$x \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

En pil

$$y \quad 1 \quad 2 \quad 33 \quad 254 \quad 1025 \quad 3126 \quad 7777$$

m

Co

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6.3 Different Operators and their Relations 6.15

Solution

Let f(3) = a

Since y is a polynomial of degree 5,

$$D_6 y_0 = 0$$

$$(E - 1)^6 y_0 = 0$$

$$(E^6 - 6E^5 + 15E^4 - 20E^3 + 15E^2 - 6E + 1) y_0 = 0$$

$$y_6 - 6y_5 + 15y_4 - 20y_3 + 15y_2 - 6y_1 + y_0 = 0$$

$$7777 - 6(3126) + 15(1025) - 20a + 15(33) - 6(2) + 1 = 0$$

$$-20a = -4880$$

$$a = 244$$

$$\text{Error} = 254 - 244 = 10$$

example 12

If u_x is a function for which the n th difference is constant and $u_1 + u_7 = -784$, $u_2 + u_6 = 686$, $u_3 + u_5 = 1088$, find u_4 .

Solution

Since the n th difference is constant,

$$\Delta^6 u_1 = 0$$

$$(E - 1)^6 u_1 = 0$$

$$(E^6 - 6E^5 + 15E^4 - 20E^3 + 15E^2 - 6E + 1) u_1 = 0$$

$$u_7 - 6u_6 + 15u_5 - 20u_4 + 15u_3 - 6u_2 + u_1 = 0$$

$$(u_7 + u_1) - 6(u_6 + u_2) + 15(u_5 + u_3) - 20u_4 = 0$$

$$-784 - 6(686) + 15(1088) - 20u_4 = 0$$

$$20u_4 = 11420$$

$$u_4 = 571$$

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6.5 Newton's Forward Interpolation Formula 6.19

6.4 Interpolation

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \dots, y_n$ corresponding to the values $x_0, x_1, x_2, \dots, x_n$ of x . The process of finding the value of y corresponding to any value of $x = x_i$ between x_0 and x_n is called interpolation. Thus, interpolation is a technique of finding the value of a function for any intermediate value of the independent variable. The process of computing the value of the function outside the range of given values of the variable is called extrapolation. The study of interpolation is based on the concept of finite differences which were discussed in the preceding section.

6.5 newton's Forward Interpolation Formula

Sg Smi by

Dg th

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \dots$ corresponding to the values $x_0, x_1, x_2, \dots$ of x . Suppose it is required to evaluate $f(x)$ for $x = x_0 + rh$, where r is any

real number.

En pil

m

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6.20 Finite Differences and Interpolation

$$y_r = f(x_0 + rh)$$

$$= E^r f(x_0)$$

$$= (1 + D)^r f(x_0)$$

$$= (1 + D)^r y_0$$

$$= \left[1 + rD + \frac{r(r-1)}{2!} D^2 + \frac{r(r-1)(r-2)}{3!} D^3 + \dots \right] y_0$$

$$= \left[1 + rD + \frac{r(r-1)}{2!} D^2 + \frac{r(r-1)(r-2)}{3!} D^3 + \dots \right] y_0$$

$$\hat{=} \left[1 + rD + \frac{r(r-1)}{2!} D^2 + \frac{r(r-1)(r-2)}{3!} D^3 + \dots \right] y_0$$

[Using Binomial theorem]

$$r(r-1) \frac{r(r-1)(r-2)}{2! 3!}$$

$$= y_0 + rDy_0 + \frac{r(r-1)}{2!} D^2 y_0 + \frac{r(r-1)(r-2)}{3!} D^3 y_0 + \dots \quad (6.1)$$

$$2! 3!$$

Equation (6.1) is known as Newton's forward interpolation formula.

note This formula is used for evaluating the value of y near the initial tabulated value of x , i.e., near x_0 .

example 1

Compute $\cosh(0.56)$ using Newton's forward difference formula from the following table:

x 0.5 0.6 0.7 0.8

$f(x)$ 1.127626 1.185465 1.255169 1.337435

Solution

Let $x = 0.56$, $x_0 = 0.5$, $h = 0.1$

$x - x_0 = 0.56 - 0.5$

$r = \frac{x - x_0}{h} = \frac{0.56 - 0.5}{0.1} = 0.6$

$h = 0.1$

Difference Table

x $f(x)$ $Df(x)$ $D^2f(x)$ $D^3f(x)$

0.5 1.127626

0.057839

0.6 1.185465 0.011865

0.069704 0.000697

0.7 1.255169 0.012562

Sg Smi by

Dg th

0.082266

gr ed

0.8 1.337435

En pil

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6.5 Newton's Forward Interpolation Formula 6.21

By Newton's forward difference formula,

$r(r-1) \frac{r(r-1)(r-2)}{3!} f(x_0 + rh) = f(x_0) + r Df(x_0) + \frac{r(r-1)}{2!} D^2f(x_0) + \frac{r(r-1)(r-2)}{3!} D^3f(x_0) + \dots$

$f(x_0 + rh) = f(x_0) + r Df(x_0) + \frac{r(r-1)}{2!} D^2f(x_0) + \frac{r(r-1)(r-2)}{3!} D^3f(x_0) + \dots$

0.6(0.6 - 1)

$\cosh(0.56) = 1.127626 + 0.6(0.057839) + \frac{0.6(0.6-1)}{2!}(0.011865)$

2!

0.6(0.6 - 1)(0.6 - 2)

+ (0.000697)

3!

$= 1.127626 + 0.034703 - 0.001424 + 0.000039$

$= 1.160944$

example 2

Find the value of $\sin 52^\circ$ using Newton's forward interpolation formula from the following table:

q° 45° 50° 55° 60°

$\sin q^\circ$ 0.7071 0.7660 0.8192 0.8660

Solution

Let $x = 52^\circ$, $x_0 = 45^\circ$, $h = 5^\circ$

$x - x_0 = 52^\circ - 45^\circ$

$r = \frac{x - x_0}{h} = \frac{52^\circ - 45^\circ}{5^\circ} = 1.4$

$h = 5^\circ$

Difference Table

$x = q^\circ$ $y = \sin q^\circ$ Dy D^2y D^3y

45° 0.7071

0.0589

$50^\circ \ 0.7660 - 0.0057$

$0.0532 - 0.0007$

$55^\circ \ 0.8192 - 0.0064$

0.0468

$60^\circ \ 0.8660$

By Newton's forward interpolation formula,

Sg Smi by

Dg th

$r(r-1) \frac{2}{2!} r(r-1)(r-2) \frac{3}{3!}$

$y(x) = y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \dots$

gr ed

$2! \ 3!$

En pil

m

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6.22 Finite Differences and Interpolation

$1.4(1.4 - 1)$

$\sin 52^\circ = 0.7071 + 1.4(0.0589) + \frac{1.4(1.4-1)(1.4-2)}{2!}(-0.0057)$

$2!$

$1.4(1.4 - 1)(1.4 - 2)$

$+ \frac{1.4(1.4-1)(1.4-2)(1.4-3)}{3!}(-0.0007)$

$3!$

$= 0.7071 + 0.0825 - 0.0016 + 0.00004$

$= 0.7880$

example 3

Using Newton's forward interpolation formula, find the value of $f(1.6)$.

$x \ 1.4 \ 1.8 \ 2.2$

$f(x) \ 3.49 \ 4.82 \ 5.96 \ 6.5$

Solution

Let $x = 1.6$, $x_0 = 1$, $h = 0.4$

$x - x_0 \ 1.6 - 1$

$r = \frac{1.6 - 1}{0.4} = 1.5$

$h \ 0.4$

Difference Table

$x \ f(x) \ \Delta f(x) \ \Delta^2 f(x) \ \Delta^3 f(x)$

$1 \ 3.49$

1.33

1.4 4.82 - 0.19

1.14 - 0.41

1.8 5.96 - 0.6

0.54

2.2 6.5

By Newton's forward interpolation formula,

$r(r-1) \frac{2}{2!} r(r-1)(r-2) \frac{3}{3!}$

$f(x_0 + rh) = f(x_0) + r Df(x_0) + \frac{r(r-1)}{2!} D^2f(x_0) + \frac{r(r-1)(r-2)}{3!} D^3f(x_0) + \dots$

$2! 3!$

$1.5(1.5-1) \frac{1.5(1.5-1)(1.5-2)}{3!}$

$f(1.6) = 3.49 + 1.5(1.33) + (-0.19) + (-0.41)$

$2! 3!$

Sg Smi by

Dg th

$= 3.49 + 1.995 - 0.0713 + 0.0256$

gr ed

$= 5.4393$

En pil

m

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example 6

Using Newton's forward interpolation formula, find the value of $f(218)$.

x 100 150 200 250 300 350 400

$f(x)$ 10.63 13.03 15.04 16.81 18.42 19.90 21.27

Solution

Let $x = 218$, $x_0 = 100$, $h = 50$

Sg Smi by

$x - x_0 = 218 - 100$

Dg th

$r = \frac{x - x_0}{h} = \frac{218 - 100}{50} = 2.36$

gr ed

$h = 50$

En pil

m

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6.5 Newton's Forward Interpolation Formula 6.25

Difference Table

x $f(x)$ $Df(x)$ $D^2f(x)$ $D^3f(x)$ $D^4f(x)$ $D^5f(x)$ $D^6f(x)$

100 10.63

2.4

150 13.03 - 0.39

2.01 0.15

200 15.04 - 0.24 - 0.07

1.77 0.08 0.02

250 16.81 - 0.16 - 0.05 0.02

1.61 0.03 0.04

300 18.42 - 0.13 - 0.01

1.48 0.02

350 19.90 - 0.11

1.37

400 21.27

By Newton's forward interpolation formula,

$r(r-1)(r-2)(r-3)(r-4)(r-5)(r-6)$

$f(x) = f(x_0) + r Df(x_0) + \frac{r(r-1)}{2!} D^2f(x_0) + \frac{r(r-1)(r-2)}{3!} D^3f(x_0) + \frac{r(r-1)(r-2)(r-3)}{4!} D^4f(x_0) + \frac{r(r-1)(r-2)(r-3)(r-4)}{5!} D^5f(x_0) + \frac{r(r-1)(r-2)(r-3)(r-4)(r-5)}{6!} D^6f(x_0) + L$

$2! 3!$

$r(r-1)(r-2)(r-3)(r-4)(r-5)(r-6)$

$+ Df(x_0) + D^2f(x_0) + D^3f(x_0) + D^4f(x_0) + D^5f(x_0) + D^6f(x_0) + L$

$4! 5!$

$r(r-1)(r-2)(r-3)(r-4)(r-5)(r-6)$

$+ Df(x_0) + L$

$6!$

$2.36(2.36-1)(2.36-2)(2.36-3)(2.36-4)(2.36-5)(2.36-6)$

$f(218) = 10.63 + 2.36(2.4) + (-0.39) + (0.15)$

$2! 3!$

$2.36(2.36-1)(2.36-2)(2.36-3)(2.36-4)(2.36-5)(2.36-6)$

$+ (-0.07)$

$4!$

$2.36(2.36-1)(2.36-2)(2.36-3)(2.36-4)(2.36-5)(2.36-6)$

$+ (0.02)$

$5!$

$2.36(2.36-1)(2.36-2)(2.36-3)(2.36-4)(2.36-5)(2.36-6)$

$+ (0.02)$

$6!$

$$= 10.63 + 5.664 - 0.6259 + 0.0289 + 0.0022 + 0.0002 - 0.00009$$

Sg Smi by

$$= 15.6993$$

Dg th

gr ed

En pil

m

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6.26 Finite Differences and Interpolation

example 7

From the following table, estimate the number of students who obtained marks between 40 and 45:

Marks 30-40 40-50 50-60 60-70 70-80

Number of students 31 42 51 35 31

Solution

Cumulative Frequency Table

Marks less than (x) 40 50 60 70 80

Number of students (y) 31 73 124 159 190

Since $x = 45$ is nearer to the beginning of the table, Newton's forward interpolation formula is used.

Let $x = 45$, $x_0 = 40$, $h = 10$

$x - x_0 = 45 - 40$

$r = \frac{x - x_0}{h} = \frac{45 - 40}{10} = 0.5$

$h = 10$

Difference Table

$x \quad y \quad \Delta y \quad \Delta^2 y \quad \Delta^3 y \quad \Delta^4 y$

40 31

42

50 73 9

51 -25

60 124 -16 37

35 12

70 159 -4

31

80 190

By Newton's forward interpolation formula,

$$r(r-1) \frac{\Delta^2 y}{2!} + r(r-1)(r-2) \frac{\Delta^3 y}{3!} + \dots$$

$$y(x) = y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \dots$$

Sg Smi by

Dg th

2! 3!

$$r(r-1)(r-2)(r-3) \Delta^4 y_0$$

gr ed

$$+ \Delta^2 y_0 + L$$

En pil

4!

m

Co

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6.5 Newton's Forward Interpolation Formula 6.27

$$0.5(0.5-1)0.5(0.5-1)(0.5-2)$$

$$y(45) = 31 + 0.5(42) + \frac{0.5(0.5-1)}{2!}(-9) + \frac{0.5(0.5-1)(0.5-2)}{3!}(-25)$$

2! 3!

$$0.5(0.5-1)(0.5-2)(0.5-3)$$

$$+ (37)$$

4!

$$= 31 + 21 - 1.1250 - 1.5625 - 1.4453$$

$$= 47.8672$$

$$\approx 48$$

The number of students with marks less than 45 = 48

The number of students with marks less than 40 = 31

Hence, the number of students obtaining marks between 40 and 45 = 48 - 31 = 17

example 8

Determine the polynomial by Newton's forward difference formula from the following table:

x	0	1	2	3	4	5
y	-10	-8	-8	-4	10	40

Solution

Let $x_0 = 0$, $h = 1$

$$x - x_0 = x - 0$$

$$r = \frac{x - x_0}{h} = x$$

h 1

Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	-10				

2

1 -8 -2

0 6

2 -8 4 0

4 6

3 -4 10 0

14 6

4 10 16

Sg Smi by

30

Dg th

5 40

gr ed

En pil

m

Co

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6.28 Finite Differences and Interpolation

By Newton's forward difference formula,

$$r(r-1)2r(r-1)(r-2)3r(r-1)(r-2)(r-3)4$$

$$y(x) = y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \frac{r(r-1)(r-2)(r-3)}{4!} \Delta^4 y_0$$

$$2! 3! 4!$$

$$x(x-1)x(x-1)(x-2)$$

$$= -10 + x(2) + (-2) + (6) + 0$$

$$2! 3!$$

$$= -10 + 2x - x(x-1) + x(x-1)(x-2)$$

$$= -10 + 2x - x^2 + x + x(x^2 - 3x + 2)$$

$$= -10 + 2x - x^2 + x + x^3 - 3x^2 + 2x$$

$$= x^3 - 4x^2 + 5x - 10$$

Sg Smi by

Dg th

gr ed

En pil

m

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example 10

Construct Newton's forward interpolation polynomial for the following data:

x 4 6 8 10

y 1 3 8 16

Solution

Let $x_0 = 4$, $h = 2$

$x - x_0$ $x - 4$

$r =$

h^2

Difference Table

x y Δy $\Delta^2 y$ $\Delta^3 y$

4 1

2

6 3 3

5 0

8 8 3

8

S_g S_{mi} by

10 16

D_g t_h

gr ed

By Newton's forward interpolation formula,

En pil

$r(r-1)2r(r-1)(r-2)3$

$y(x) = y_0 + r\Delta y_0 + D y_0 + D y_0 + L$

m

$2! 3!$

Co

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6.30 Chapter 6 Finite Differences and Interpolation

$\hat{E} x - 4^{\wedge} \hat{E} x - 4^{\wedge}$

$\hat{A}\hat{E} \sim \hat{A} - 1^{\sim}$

$\hat{E} x - 4^{\wedge} 2^{\sim} \hat{E} 2^{\sim}$

$= 1 + \hat{A} \sim (2) + (3) + 0$

$\hat{E} 2^{\sim} 2!$

$(x - 4)(x - 6)$

$= 1 + (x - 4) + (3)$

8

$$(x^2 - 10x + 24)$$

$$= x^2 - 3x + (3)$$

$$8$$

$$3x^2 - 15x$$

$$= x^2 - 3x - 9$$

$$84$$

$$3x^2 - 11x$$

$$= -6$$

$$84$$

$$3(25) - 11(5) = 13$$

$$y(5) = -6 =$$

$$848$$

6.6 newton's Backward Interpolation Formula

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \dots$ corresponding to the values $x_0, x_1, x_2, \dots$ of x . Suppose it is required to evaluate $f(x)$ for $x = x_0 + rh$, where r is any real number.

$$y_r = f(x_0 + rh)$$

$$= E^r f(x_0)$$

$$= (E - 1)^{-r} y_n$$

$$= (1 - \nabla)^{-r} y_n$$

$$= \left[1 + r \nabla + \frac{r(r+1)}{2!} \nabla^2 + \frac{r(r+1)(r+2)}{3!} \nabla^3 + \dots \right] y_n$$

$$= \left[1 + r \nabla + \frac{r(r+1)}{2!} \nabla^2 + \frac{r(r+1)(r+2)}{3!} \nabla^3 + \dots \right] y_n$$

$$\hat{=} 2! 3! \dots$$

[Using Binomial theorem]

$$r(r+1) \frac{r(r+1)(r+2)}{3!}$$

$$= y_n + r \nabla y_n + \frac{r(r+1)}{2!} \nabla^2 y_n + \frac{r(r+1)(r+2)}{3!} \nabla^3 y_n + \dots \quad (6.2)$$

$$2! 3!$$

Equation (6.2) is known as Newton's backward interpolation formula.

note This formula is used for evaluating the value of y near to the end of tabulated value of x , i.e., near x_n .

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6.6 Newton's Backward Interpolation Formula 6.31

example 1

Consider the following tabular values:

x 50 100 150 200 250

y 618 724 805 906 1032

Determine $y(300)$ using Newton's backward interpolation formula.

Solution

Let $x = 300$, $x_n = 250$, $h = 50$

$x - x_n = 300 - 250$

$r = \frac{x - x_n}{h} = \frac{300 - 250}{50} = 1$

$h = 50$

Difference Table

x y Δy $\Delta^2 y$ $\Delta^3 y$ $\Delta^4 y$

50 618

106

100 724 -25

81 45

150 805 20 - 40

101 5

200 906 25

126

250 1032

By Newton's backward interpolation formula,

$$y(x) = y_n + r \Delta y_n + \frac{r(r+1)}{2!} \Delta^2 y_n + \frac{r(r+1)(r+2)}{3!} \Delta^3 y_n + \frac{r(r+1)(r+2)(r+3)}{4!} \Delta^4 y_n + \dots$$

$y(300) = 1032 + 1(126) + \frac{1(1+1)}{2!}(25) + \frac{1(1+1)(1+2)}{3!}(-40) + \frac{1(1+1)(1+2)(1+3)}{4!}(5)$

$= 1032 + 126 + 25 - 40 + 5$

$= 1148$

$y(300) = 1148$

$1(2) 1(2)(3) 1(2)(3)(4)$

$y(300) = 1032 + 1(126) + (25) + (5) + (-40)$

$2! 3! 4!$

$= 1032 + 126 + 25 + 5 - 40$

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$= 1148$

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6.6 Newton's Backward Interpolation Formula 6.33

example 3

From the following table, find P when $t = 142^\circ\text{C}$ and 175°C using appropriate Newton's interpolation formula.

Temperature $t^\circ\text{C}$ 140 150 160 170 180

Pressure P 3685 4845 6302 8076 10225

Solution

Since $x = 142$ is nearer to the beginning of the table, Newton's forward interpolation formula is used.

Let $x = 142$, $x_0 = 140$, $h = 10$

$x - x_0 = 142 - 140$

$r = \frac{x - x_0}{h} = 0.2$

$h = 10$

Difference Table

$x = t^\circ \quad y = P \quad \Delta y \quad \Delta^2 y \quad \Delta^3 y \quad \Delta^4 y$

140 3685

1169

150 4854 279

1448 47

160 6302 326 2

1774 49

170 8076 375

2149

180 10225

By Newton's forward interpolation formula,

$r(r-1) \frac{\Delta^2 y_0}{2!} + r(r-1)(r-2) \frac{\Delta^3 y_0}{3!} + \dots$

$y(x) = y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \dots$

$2! 3!$

$r(r-1)(r-2)(r-3) \frac{\Delta^4 y_0}{4!} + \dots$

$+ \Delta^4 y_0 + \dots$

$4!$

$0.2(0.2-1) \frac{0.2(0.2-1)(0.2-2)}{3!} \Delta^3 y_0 + \dots$

$P(142) = y(142) = 3685 + 0.2(1169) + \frac{0.2(0.2-1)}{2!} (279) + \frac{0.2(0.2-1)(0.2-2)}{3!} (47) + \dots$

$2! 3!$

$0.2(0.2-1)(0.2-2)(0.2-3) \frac{\Delta^4 y_0}{4!} + \dots$

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$$= 3685 + 233.8 - 22.32 + 2.256 - 0.0672$$

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$$= 3898.6688$$

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6.34 Finite Differences and Interpolation

Since $x = 175$ is near to the end of the table, Newton's backward interpolation formula is used.

$$x = 175, x_n = 180, h = 10$$

$$x - x_n = 175 - 180$$

$$r = \frac{x - x_n}{h} = \frac{-5}{10} = -0.5$$

$$h = 10$$

By Newton's backward interpolation formula,

$$y(x) = y_n + r \frac{\Delta y_{n-1}}{1!} + \frac{r(r+1)}{2!} \Delta^2 y_{n-2} + \frac{r(r+1)(r+2)}{3!} \Delta^3 y_{n-3} + \dots$$

$$y(175) = y_{180} + r \frac{\Delta y_{170}}{1!} + \frac{r(r+1)}{2!} \Delta^2 y_{160} + \frac{r(r+1)(r+2)}{3!} \Delta^3 y_{150} + \dots$$

$$2! 3!$$

$$r(r+1)(r+2)(r+3) \frac{\Delta^4 y_{140}}{4!} + \dots$$

$$+ \dots y_n + L$$

$$4!$$

$$(-0.5)(-0.5+1)$$

$$P(175) = y(175) = 10225 + (-0.5)(2149) + (375)$$

$$2!$$

$$(-0.5)(-0.5+1)(-0.5+2)$$

$$+ (49)$$

$$3!$$

$$(-0.5)(-0.5+1)(-0.5+2)(-0.5+3)$$

$$+ (2)$$

$$4!$$

$$= 10225 - 1074.5 - 46.875 - 3.0625 - 0.0781$$

$$= 9100.4844$$

example 4

The population of a town is given below. Estimate the population for the year 1895 and 1930 using suitable interpolation.

Year x 1891 1901 1911 1921 1931

Population y 46 66 81 93 101

(in thousand)

Solution

Since $x = 1895$ is near to the beginning of the table, Newton's forward interpolation

formula is used.

Let $x = 1895$, $x_0 = 1891$, $h = 10$

$x - x_0 = 1895 - 1891$

$r = 0.4$

$h = 10$

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6.6 Newton's Backward Interpolation Formula 6.35

Difference Table

$x \quad y \quad Dy \quad D^2y \quad D^3y \quad D^4y$

1891 46

20

1901 66 -5

15 2

1911 81 -3 -3

12 -1

1921 93 -4

8

1931 101

By Newton's forward interpolation formula,

$r(r-1)(r-2)(r-3)$

$y(x) = y_0 + rDy_0 + \frac{r(r-1)}{2!}D^2y_0 + \frac{r(r-1)(r-2)}{3!}D^3y_0 + \frac{r(r-1)(r-2)(r-3)}{4!}D^4y_0 + \dots$

$r(r-1)(r-2)(r-3)$

$+ D^4y_0 + \dots$

$4!$

$0.4(0.4-1)(0.4-2)(0.4-3)$

$y(1895) = 46 + 0.4(20) + (-5) + \frac{(2)(-3)(-4)}{3!}(-1)$

$2! 3!$

$0.4(0.4-1)(0.4-2)(0.4-3)$

$+ (-3)$

$4!$

$= 46 + 8 + 0.6 + 0.128 + 0.1248$

$= 54.8528$ thousands

Since $x = 1930$ is near to the end of the table, Newton's backward interpolation formula is used.

Let $x = 1930$, $x_n = 1931$, $h = 10$

$x - x_n = 1930 - 1931$

$r = \frac{x - x_n}{h} = \frac{-1}{10} = -0.1$

$h = 10$

By Newton's forward interpolation formula,

$y(x) = y_n + r \Delta y_n + \frac{r(r+1)}{2!} \Delta^2 y_n + \frac{r(r+1)(r+2)}{3!} \Delta^3 y_n + \dots$

$y(1930) = 101 + (-0.1) \Delta y_n + \frac{(-0.1)(-0.1+1)}{2!} \Delta^2 y_n + \frac{(-0.1)(-0.1+1)(-0.1+2)}{3!} \Delta^3 y_n + \dots$

$2! 3!$

$r(r+1)(r+2)(r+3) \Delta^4 y_n + \dots$

$+ \dots + y_n + L$

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6.36 Finite Differences and Interpolation

$(-0.1)(-0.1+1)(-0.1)(-0.1+1)(-0.1+2)$

$y(1930) = 101 + (-0.1)(8) + (-4) + (-1)$

$2! 3!$

$(-0.1)(-0.1+1)(-0.1+2)(-0.1+3)$

$+ (-3)$

4!

$= 101 - 0.8 + 0.18 + 0.0285 + 0.062$

$= 100.4705$ thousands

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6.11 Interpolation with unequal Intervals

If the values of x are unequally spaced then interpolation formulae for equally spaced points cannot be used. It is, therefore, desirable to develop interpolation formulae for unequally spaced values of x . There are two such formulae for unequally spaced values

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of x .

(i) Lagrange's interpolation formula

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(ii) Newton's interpolation formula with divided difference

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6.56 Finite Differences and Interpolation

6.12 laGranGe's Interpolation Formula

Let $y = f(x)$ be a function which take the values $y_0, y_1, y_2, \dots, y_n$ corresponding to $x = x_0, x_1, x_2, \dots, x_n$. Since there are $(n + 1)$ values of x and y , $f(x)$ can be represented by a polynomial in x of degree n .

$$y = f(x) = a_0 (x - x_1)(x - x_2) \dots (x - x_n) + a_1 (x - x_0)(x - x_2) \dots (x - x_n) + \dots + a_n (x - x_0)(x - x_1) \dots (x - x_{n-1}) \dots (6.19)$$

where $a_0, a_1, a_2, \dots, a_n$ are constants.

Putting $x = x_0, y = y_0$ in Eq. (6.19),

$$y_0 = a_0 (x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)$$

y_0

$a_0 =$

$$(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)$$

Similarly, putting $x = x_1, y = y_1$ in Eq. (6.19),

y_1

$a_1 =$

$$(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)$$

Proceeding in the same way,

y_n

$a_n =$

$$(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})$$

Substituting the values of $a_0, a_1, a_2, \dots, a_n$ in Eq. (6.19),

$$(x - x_1)(x - x_2) \dots (x - x_n) (x - x_0)(x - x_2) \dots (x - x_n)$$

$$f(x) = y_0 + y_1 + \dots$$

$$(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n) (x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)$$

$$(x - x_0)(x - x_1) \dots (x - x_{n-1})$$

$$+ y_n \dots (6.20)$$

$$(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})$$

Equation (6.20) is known as Lagrange's interpolation formula.

note This formula can also be used to split the given function into partial fractions.

Dividing both sides of Eq. (6.20) by $(x - x_0)(x - x_1) \dots (x - x_n)$,

$$f(x) y_0 \hat{= 1}^{\wedge}$$

$$(x - x_0)(x - x_1) \dots (x - x_n) (x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n) \hat{= x - x_0}^{\sim}$$

$$y_1 \hat{= 1}^{\wedge}$$

$$+ + \dots$$

$$(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n) \hat{= x - x_1}^{\sim}$$

$$y_n \hat{= 1}^{\wedge}$$

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$$(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1}) \hat{= x - x_n}^{\sim}$$

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6.12 Lagrange's Interpolation Formula 6.57

example 1

Compute $f(9.2)$ by using Lagrange's interpolation method from the following data:

$$x \quad 9 \quad 9.5 \quad 11$$

$$f(x) \quad 2.1972 \quad 2.2513 \quad 2.3979$$

Solution

By Lagrange's interpolation formula,

$$(x - x_1)(x - x_2) (x - x_0)(x - x_2)$$

$$f(x) = f(x_0) + f(x_1)$$

$$(x_0 - x_1)(x_0 - x_2) (x_1 - x_0)(x_1 - x_2)$$

$$(x - x_0)(x - x_1)$$

$$+ f(x_2)$$

$$(x_2 - x_0)(x_2 - x_1)$$

$$(9.2 - 9.5)(9.2 - 11) (9.2 - 9)(9.2 - 11)$$

$$f(9.2) = (2.1972) + (2.2513)$$

$$(9 - 9.5)(9 - 11) (9.5 - 9)(9.5 - 11)$$

$$(9.2 - 9)(9.2 - 9.5)$$

$$\begin{aligned}
 &+ (2.3979) \\
 &(11 - 9)(11 - 9.5) \\
 &= 1.1865 + 1.0806 - 0.048 \\
 &= 2.2191
 \end{aligned}$$

example 2

Find the value of y when x = 10 from the following table:

x 5 6 9 11

y 12 13 14 16

Solution

By Lagrange's interpolation formula,

$$\begin{aligned}
 &(x - x_1)(x - x_2)(x - x_3)(x - x_0)(x - x_2)(x - x_3) \\
 &y(x) = y_0 + y_1 \\
 &(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(x_1 - x_0)(x_1 - x_2)(x_1 - x_3) \\
 &(x - x_0)(x - x_1)(x - x_3)(x - x_0)(x - x_1)(x - x_2) \\
 &+ y_2 + y_3 \\
 &(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)(x_3 - x_0)(x_3 - x_1)(x_3 - x_2) \\
 &(10 - 6)(10 - 9)(10 - 11)(10 - 5)(10 - 9)(10 - 11) \\
 &y(10) = (12) + (13) \\
 &(5 - 6)(5 - 9)(5 - 11)(6 - 5)(6 - 9)(6 - 11)
 \end{aligned}$$

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$$\begin{aligned}
 &(10 - 5)(10 - 6)(10 - 11)(10 - 5)(10 - 6)(10 - 9) \\
 &+ (14) + (16)
 \end{aligned}$$

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$$(9 - 5)(9 - 6)(9 - 11)(11 - 5)(11 - 6)(11 - 9)$$

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$$= 2 - 4.3333 + 11.6666 + 5.3333$$

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$$= 14.6666$$

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example 9

Using Lagrange's interpolation formula, find the interpolating polynomial for the following table:

x 0 1 2 5

f(x) 2 3 12 147

Solution

By Lagrange's interpolation formula,

$$(x - x_1)(x - x_2)(x - x_3)(x - x_0)(x - x_2)(x - x_3)$$

$$f(x) = f(x_0) + f(x_1)$$

$$(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)$$

$$(x - x_0)(x - x_1)(x - x_3)(x - x_0)(x - x_1)(x - x_2)$$

$$+ f(x_2) + f(x_3)$$

$$(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)$$

$$(x - 1)(x - 2)(x - 5)(x - 0)(x - 2)(x - 5)$$

$$= (2) + (3)$$

$$(0 - 1)(0 - 2)(0 - 5)(1 - 0)(1 - 2)(1 - 5)$$

$$(x - 0)(x - 1)(x - 5)(x - 0)(x - 1)(x - 2)$$

$$+ (12) + (147)$$

$$(2 - 0)(2 - 1)(2 - 5)(5 - 0)(5 - 1)(5 - 2)$$

$$(x - 1)(x^2 - 7x + 10)3x(x^2 - 7x + 10)2x(x^2 - 6x + 5)49x(x^2 - 3x + 2)$$

$$= + + +$$

$$-5^4 - 120$$

$$-4(x^3 - 8x^2 + 17x - 10) + (15x^3 - 105x^2 + 150x)$$

$$3^2 3^2 \cdot 0$$

$$j''$$

$$0 - 20(2x^3 - 12x^2 + 10x) + (49x^3 - 147x^2 + 98x) \cdot 0$$

$$20$$

$$20x + 20x - 20x + 40$$

$$3^2$$

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$$= x^3 + x^2 - x + 2$$

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6.12 Lagrange's Interpolation Formula 6.67

Solutions

By Lagrange's interpolation formula,

$$(t - t_1)(t - t_2)(t - t_3) \dots (t - t_0)(t - t_2) \dots (t - t_3) \dots$$

$$v = v_0 + v_1$$

$$(t_0 - t_1)(t_0 - t_2) \dots (t_0 - t_3) \dots (t_1 - t_0)(t_1 - t_2) \dots (t_1 - t_3) \dots$$

$$(t - t_0)(t - t_1) \dots (t - t_3) \dots (t - t_0)(t - t_1) \dots (t - t_2) \dots$$

$$+ v_2 + v_3$$

$$(t_2 - t_0)(t_2 - t_1) \dots (t_2 - t_3) \dots (t_3 - t_0)(t_3 - t_1) \dots (t_3 - t_2) \dots$$

$$(t - 1)(t - 3)(t - 4) \dots (t - 0)(t - 3)(t - 4) \dots$$

$$= (21) + (15)$$

$$(0 - 1)(0 - 3)(0 - 4) \dots (1 - 0)(1 - 3)(1 - 4) \dots$$

$$(t - 0)(t - 1)(t - 4) \dots (t - 0)(t - 1)(t - 3) \dots$$

$$+ (12) + (10)$$

$$(3 - 0)(3 - 1)(3 - 4) \dots (4 - 0)(4 - 1)(4 - 3) \dots$$

$$1$$

$$= (-5t^3 + 38t^2 - 105t + 252)$$

$$12$$

If s is the distance travelled in time t,

$$ds = v dt$$

$$v = (-5t^3 + 38t^2 - 105t + 252)$$

$$ds = v dt$$

$$4$$

$$s = \int v dt$$

$$0$$

$$4$$

$$1$$

$$12 \int_0^4$$

$$= (-5t^4 + 38t^3 - 105t^2 + 252t)$$

$$4$$

$$15t^4 + 38t^3 - 105t^2$$

$$= - + - + 252t$$

$$12 + 4 + 3 + 2$$

$$0$$

$$1 \int_0^4 (-5t^3 + 38t^2 - 105t + 252) dt$$

$$= \left[-\frac{5}{4}t^4 + \frac{38}{3}t^3 - \frac{105}{2}t^2 + 252t \right]_0^4$$

$$12 \int_0^4 (-5t^3 + 38t^2 - 105t + 252) dt$$

$$= 54.88$$

Hence, the distance travelled in 4 seconds = 54.88

$$dv = 1$$

$$a = (-15t^2 + 76t - 105)$$

dt 12

At $t = 4$,

1

$$a = (-15 \times 16 + 76 \times 4 - 105) = 3.416$$

12

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6.68 Finite Differences and Interpolation

exercse 6.4

1. From the table given below, find $y(x = 2)$.

x 0 1 3 4

y 5 6 50 105

[ans. : 19]

2. Use Lagrange's formula to find the velocity of the particle $v = f(t)$ at $t = 3.5$ from the following table:

t 0 1 2 3

v 21 15 12 10

[ans. : 8.75]

3. Find $f(27)$ from the following table:

x 14 17 31 35

f(x) 68.7 64.0 44.0 39.1

[ans. : 49.3]

[ans. : 809.997]

6. If $y = 4.3315$, $y = 7.4046$, $y = 5.6713$, $y = 7.1154$, find the curve

0 1 3 5

passing through these points. Hence, find y and y .

2 4

[ans. : 5.1420, 6.3199]

7. If $f(1) = 3$, $f(2) = -5$, $f(-4) = 4$, find the three-point Lagrange's interpolation polynomial that takes the same values.

È 1 ~

Í ans. : 20 (- 39x - 123x + 252)´

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6.70 Finite Differences and Interpolation

6.13 divided differences

In Lagrange's interpolation formula, if another interpolation value is added then the interpolation coefficients are required to be recalculated. To avoid this recalculation, Newton's general interpolation formula is used.

If $(x_0, y_0), (x_1, y_1), (x_2, y_2) \dots$ be given points then the first divided difference for x_0, x_1 is defined by the relation,

$y - y_0$

$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$

$x_1 - x_0$

$y_2 - y_1$

Similarly, $[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1}$, etc.

$x_2 - x_1$

The second divided difference for x_0, x_1, x_2 is defined as

$[x_1, x_2] - [x_0, x_1]$

$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}$

$x_2 - x_0$

The third divided difference for x_0, x_1, x_2, x_3 is defined as

$[x_2, x_3] - [x_1, x_2]$

$[x_0, x_1, x_2, x_3] = \frac{[x_2, x_3] - [x_1, x_2]}{x_3 - x_0}$

$x_3 - x_0$

notes

(i) The divided differences are symmetrical in their arguments, i.e., independent of the order of arguments:

y_0, y_1

$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$

$x_1 - x_0$

$= [x_1, x_0]$

```

y0 y1 y2
[ x0 , x1 , x2 ] = + +
( x0 - x1 )( x0 - x2 ) ( x1 - x0 ) ( x1 - x2 ) ( x2 - x0 ) ( x2 - x1 )
= [ x1 , x2 , x0 ] or [ x2 , x0 , x1 ]

```

(ii) The n th divided differences of a polynomial of the n th degree are constant.

Let the arguments be equally spaced so that $x_1 - x_0 = x_2 - x_1 = \dots = x_n - x_{n-1} = h$

$$\frac{y_1 - y_0}{x_1 - x_0} =$$

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6.14 Newton's Divided Difference Formula 6.71

$$[x_1, x_2] - [x_0, x_1]$$

$$[x_0, x_1, x_2] =$$

$$x_2 - x_0$$

$$= - \frac{2h}{\hbar} \frac{\partial}{\partial h} \ln Z$$

$$1 = D^2 y_0$$

$2! \cdot h \cdot 2$

In general,

$$1$$

$$[x_0, x_1, x_2, \dots, x_n] =$$

D n y0
n! h n

If the tabulated function is an n th degree polynomial, D_{ny0} will be constant. Hence, the n th divided differences will also be constant.

6.14 newton's divided dIFference Formula

Let the function $y = f(x)$ take values $y_0, y_1, y_2, \dots, y_n$ corresponding to $x_0, x_1, x_2, \dots, x_n$ respectively. According to the definition of divided differences,

$y - y_0$

$$[x, x_0] =$$

$$x - x_0$$

$$y = y_0 + (x - x_0)[x, x_0]$$

...(6.21)

$$[x, x_0] - [x_0, x_1]$$

$$[x, x_0, x_1] =$$

$$x - x_1$$

$$[x, x_0] = [x_0, x_1] + (x - x_1)[x, x_0, x_1]$$

Substituting the value of $[x, x_0]$ in Eq. (6.21),

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1] \dots (6.22)$$

$$[x, x_0, x_1] - [x_0, x_1, x_2]$$

Also, $[x, x_0, x_1, x_2] =$

$$x - x_2$$

$$[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2]$$

Substituting the value of $[x, x_0, x_1]$ in Eq. (6.22),

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

$$+ (x - x_0)(x - x_1)(x - x_2)[x, x_0, x_1, x_2]$$

Proceeding in the same manner,

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$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

gr ed

$$+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] + \dots$$

En pil

$$+ (x - x_0)(x - x_1) \dots (x - x_{n-1})[x, x_0, x_1, \dots, x_n] \dots (6.23)$$

m

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6.72 Finite Differences and Interpolation

Equation (6.23) is known as Newton's general interpolation formula with divided differences.

example 1

1

If $f(x) = \dots$, find the divided difference $[a, b]$ and $[a, b, c]$.

x

Solution

Divided Difference Table

First Divided Second Divided

$x f(x)$

Difference Difference

1

a

a

1 1

$b a = -1$

$b-a ab$

1 1

1 - +

$b bc ab = 1$

b

$c-a abc$

1 1

$c b = -1$

$c-b bc$

1

c

c

1

$[a, b] = -$

ab

1

$[a, b, c] =$

abc

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6.14 Newton's Divided Difference Formula 6.73

example 2

Find the second divided difference for the argument $x = 1, 2, 5,$ and 7 for the function $f(x) = x^2$.

Solution

Divided Difference Table

First Divided Second Divided Third Divided

x $f(x)$

Difference Difference Difference

1 1

4 -1

=3

2 -1

7-3

2 4 =1

5 -1

25 - 4

=7 0

5-2

12 - 7

5 25 =1

7-2

49 - 25

= 12

7-5

7 49

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6.74 Finite Differences and Interpolation

example 3

Find the third divided difference with arguments 2, 4, 9, 10 of the function

$f(x) = x^3 - 2x$.

Solution

Divided Difference Table

First Divided Second Divided Third Divided

x $f(x)$

Difference Difference Difference

2 4

$$56 - 4$$

$$= 26$$

$$4-2$$

$$131 - 26$$

$$4 \ 56 = 15$$

$$9-2$$

$$711 - 56 \ 23 - 15$$

$$= 131 = 1$$

$$9-4 \ 10 - 2$$

$$269 - 131$$

$$9 \ 711 = 23$$

$$10 - 4$$

$$980 - 711$$

$$= 269$$

$$10 - 9$$

$$10 \ 980$$

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6.14 Newton's Divided Difference Formula 6.75

example 4

Construct the divided difference for the data given below:

$$x \ -4 \ -1 \ 0 \ 2 \ 5$$

$$f(x) \ 1245 \ 33 \ 5 \ 9 \ 1335$$

Solution

Divided Difference Table

First Divided Second Divided Third Divided Fourth Divided

x f(x)

Difference Difference Difference Difference

$$-4 \ 1245$$

$$33 - 1245$$

$$= - 404$$

$$-1 + 4$$

$$-28 + 404$$

$$-1 \ 33 = 94$$

$$0+4$$

$$5 - 33 \ 10 - 94$$

$$= -28 = -14$$

$$0 +1 \ 2+4$$

$$2 + 28 \ 13 + 14$$

$$0 \ 5 = 10 = 13$$

$$2 +1 \ 5 +1$$

$$9-5 \ 88 - 10$$

$$=2 = 13$$

$$2-0 \ 5 +1$$

$$442 - 2$$

$$2 \ 9 = 88$$

$$5-0$$

$$1335 - 9$$

$$= 442$$

$$5-2$$

$$5 \ 1335$$

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6.76 Finite Differences and Interpolation

example 5

Complete f(9.2) from the following data by using Newton's divided difference interpolation formula.

x 8 9 9.5 11

f (x) 2.079442 2.197225 2.251292 2.397895

Solution

Divided Difference Table

First Divided Second Divided Third Divided

x f (x)

Difference Difference Difference

8 2.079442

0.117783

9 2.197225 - 0.006433

0.108134 0.000411

9.5 2.251292 - 0.005200

0.097735

11 2.397895

By Newton's divided difference formula,

$$f(x) = f(x_0) + (x - x_0) [x_0, x_1] + (x - x_0)(x - x_1) [x_0, x_1, x_2]$$

$$+ (x - x_0)(x - x_1)(x - x_2) [x_0, x_1, x_2, x_3]$$

$$f(9.2) = 2.079442 + (9.2 - 8)(0.117783) + (9.2 - 8)(9.2 - 9)(-0.006433)$$

$$+ (9.2 - 8)(9.2 - 9)(9.2 - 9.5)(0.00041)$$

$$= 2.079442 + 0.141340 - 0.001544 - 0.000030$$

$$= 2.219208$$

example 6

Using Newton's divided difference formula, compute $f(10.5)$ from the following data:

x 10 11 13 17

f (x) 2.3026 2.3979 2.5649 2.8332

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6.14 Newton's Divided Difference Formula 6.77

Solution

Divided Difference Table

First Divided Second Divided Third Divided

x f (x)

Difference Difference Difference

10 2.3026

0.0953

11 2.3979 - 0.0039

0.0835 0.0002

13 2.5649 - 0.0027

0.0671

17 2.8332

By Newton's divided difference formula,

$$\begin{aligned}
 f(x) &= f(x_0) + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] \\
 &+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] \\
 f(10.5) &= 2.3026 + (10.5 - 10)(0.0953) + (10.5 - 10)(10.5 - 11)(-0.0039) \\
 &+ (10.5 - 10)(10.5 - 11)(10.5 - 13)(0.0002) \\
 &= 2.3026 + 0.0477 + 0.00098 + 0.00013 \\
 &= 2.3514
 \end{aligned}$$

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6.80 Finite Differences and Interpolation

example 10

From the following table, find $f(x)$ using Newton's divided difference formula:

x 1 2 7 8

$f(x)$ 1 5 5 4

Solution

Divided Difference Table

First Divided Second Divided Third Divided

x $f(x)$

Difference Difference Difference

1 1

4

2

2 5 -

3

1

0

14

1

7 5 -

6

-1

8 4

By Newton's divided difference formula,

$$f(x) = f(x_0) + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

$$+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3]$$

$$\hat{E}^2 \hat{E}^1$$

$$= 1 + (x - 1)4 + (x - 1)(x - 2) \hat{A} - \sim + (x - 1)(x - 2)(x - 7) \hat{A} \hat{E} \sim -$$

$$\hat{E} - 3 \ 14$$

$$2 \ 2 \ 1$$

$$= 1 + 4x - 4 - (x - 3x + 2) + (x^3 - 10x^2 + 23x - 14)$$

$$3 \ 14$$

$$1 \ 3 \ 29 \ 2 \ 107 \ 16$$

$$= x - x + x -$$

$$14 \ 21 \ 14 \ 3$$

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example 12

Establish a cubic polynomial of the curve $y = f(x)$ passing through the points (1, -3), (3, 9), (4, 30), (6, 132). Hence, find $f(2)$.

Solution

Divided Difference Table

First Divided Second Divided Third Divided

$x \ f(x)$

Difference Difference Difference

1 -3

6

3 9 5

21 1

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4 30 10

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51

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6 132

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6.82 Finite Differences and Interpolation

By Newton's divided difference formula,

$$\begin{aligned}
 f(x) &= f(x_0) + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] \\
 &+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] \\
 &= -3 + (x - 1)(6) + (x - 1)(x - 3)(5) + (x - 1)(x - 3)(x - 4)(1) \\
 &= -3 + 6x - 6 + 5x^2 - 20x + 15 + x^3 - 8x^2 + 19x - 12 \\
 &= x^3 - 3x^2 + 5x - 6 \\
 f(2) &= (2)^3 - 3(2)^2 + 5(2) - 6 = 0
 \end{aligned}$$

example 13

The shear stress in kilopound per square foot (ksf) for 5 specimens in a clay stratum are as follows:

Depth (m) 1.9 3.1 4.2 5.1 5.8

Stress (ksf) 0.3 0.6 0.4 0.9 0.7

Use Newton's dividend difference interpolating polynomial to compute the stress at 4.5 m depth.

Solution

Divided Difference Table

Depth Stress First Divided Second Divided Third Divided Fourth Divided
x y Difference Difference Difference Difference

1.9 0.3

0.25

3.1 0.6 -0.1877

-0.1818 0.1739

4.2 0.4 0.3687 -0.1295

0.5556 -0.3313

5.1 0.9 -0.5258

-0.2857

5.8 0.7

By Newton's divided difference formula,

S_g S_{mi} by

D_g th

$$y(x) = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

gr ed

$$+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3]$$

En pil

+ (x - x₀) (x - x₁) (x - x₂) (x - x₃) [x₀, x₁, x₂, x₃, x₄]

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6.14 Newton's Divided Difference Formula 6.83

$$\begin{aligned} y(4.5) &= 0.3 + (4.5 - 1.9)(0.25) + (4.5 - 1.9)(4.5 - 3.1)(-0.1877) \\ &+ (4.5 - 1.9)(4.5 - 3.1)(4.5 - 4.2)(0.1739) \\ &+ (4.5 - 1.9)(4.5 - 3.1)(4.5 - 4.2)(4.5 - 5.1)(-0.1295) \\ &= 0.3 + 0.65 - 0.6832 + 0.1899 + 0.0848 \\ &= 0.5415 \text{ ksf} \end{aligned}$$

exercise 6.5

1

1. If $f(x) = 2$

, find the divided differences $f(a, b)$, $f(a, b, c)$, and $f(a, b, c, d)$.

$f(a, b, c, d)$.

$\frac{1}{(a+b)(ab+bc+ca)(abc+bcd+acd+abd)}$

Ans.: - 2

2

$\frac{1}{(a+b)(ab+bc+ca)(abc+bcd+acd+abd)}$

$\frac{1}{(a+b)(ab+bc+ca)(abc+bcd+acd+abd)}$

5. Find $y(x = 20)$ from the following table:

x 12 18 22 24 32

y(x) 146 836 19481 2796 9236

[ans.: 1305.36]

7. Using the divided difference table, find $f(x)$ which takes the values 1, 4, 40, 85 as $x = 0, 1, 3, 4$.

$\frac{1}{(x-0)(x-1)(x-3)(x-4)}$

[ans.: $x^4 + x^3 + x^2 + 1$]

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6.15 Inverse Interpolation

The process of evaluating the value of x for a value of y (which is not in the table) is called inverse interpolation. Lagrange's formula is a relation between two variables, either of which may be taken as the independent variable. On interchanging x and y in the Lagrange's interpolation formula,

$$\begin{aligned} & (y - y_1)(y - y_2) \dots (y - y_n)(y - y_0)(y - y_2) \dots (y - y_n) \\ x = & x_0 + x_1 + \dots \\ & (y_0 - y_1)(y_0 - y_2) \dots (y_0 - y_{n-1})(y_1 - y_0)(y_1 - y_2) \dots (y_1 - y_n) \\ & (y - y_0)(y - y_1) \dots (y - y_{n-1}) \\ & + x_n \dots (6.24) \\ & (y_n - y_0)(y_n - y_1) \dots (y_n - y_{n-1}) \end{aligned}$$

Equation (6.24) is used for inverse interpolation.

example 1

From the data given, find the value of x when $y = 13.5$.

x 93 96.2 100 104.2 108.7

y 11.38 12.80 14.70 17.07 19.91

Solution

By Lagrange's formula for inverse interpolation,

$$\begin{aligned} & (y - y_1)(y - y_2)(y - y_3)(y - y_4)(y - y_0)(y - y_2)(y - y_3)(y - y_4) \\ x = & x_0 + x_1 \\ & (y_0 - y_1)(y_0 - y_2)(y_0 - y_3)(y_0 - y_4)(y_1 - y_0)(y_1 - y_2)(y_1 - y_3)(y_1 - y_4) \\ & (y - y_0)(y - y_1)(y - y_3)(y - y_4)(y - y_0)(y - y_1)(y - y_2)(y - y_4) \\ & + x_2 + \\ & (y_2 - y_0)(y_2 - y_1)(y_2 - y_3)(y_2 - y_4)(y_3 - y_0)(y_3 - y_1)(y_3 - y_2)(y_3 - y_4) \\ & (y - y_0)(y - y_1)(y - y_2)(y - y_3) \\ & + x_4 \\ & (y_4 - y_0)(y_4 - y_1)(y_4 - y_2)(y_4 - y_3) \end{aligned}$$

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6.15 Inverse Interpolation 6.85

$$(13.5 - 12.80)(13.5 - 14.70)(13.5 - 17.07)(13.5 - 19.91)$$

$$\begin{aligned}
&= (93) \\
&(11.38 - 12.80)(11.38 - 14.70)(11.38 - 17.07)(11.38 - 19.91) \\
&(13.5 - 11.38)(13.5 - 14.70)(13.5 - 17.07)(13.5 - 19.91) \\
&+ (96.2) \\
&(12.80 - 11.38)(12.80 - 14.70)(12.80 - 17.07)(12.80 - 19.91) \\
&(13.5 - 11.38)(13.5 - 12.80)(13.5 - 17.07)(13.5 - 19.91) \\
&+ (100) \\
&(14.70 - 11.38)(14.70 - 12.80)(14.70 - 17.07)(14.70 - 19.91) \\
&(13.5 - 11.38)(13.5 - 12.80)(13.5 - 14.70)(13.5 - 19.91) \\
&+ (104.2) \\
&(17.07 - 11.38)(17.07 - 12.80)(17.07 - 14.70)(17.07 - 19.91) \\
&(13.5 - 11.38)(13.5 - 12.80)(13.5 - 14.70)(13.5 - 17.07) \\
&+ (108.7) \\
&(19.91 - 11.38)(19.91 - 12.80)(19.91 - 14.70)(19.91 - 17.07) \\
&x = -7.8137 + 68.4669 + 43.6076 - 7.2758 + 0.7711 \\
&= 97.7561
\end{aligned}$$

example 2

Find the root of the equation $f(x) = 0$, given that $f(30) = -30$, $f(34) = -13$, $f(38) = 3$, and $f(42) = 18$.

Solution

Let $x_0 = 30$, $x_1 = 34$, $x_2 = 38$, $x_3 = 42$

$y_0 = -30$, $y_1 = -13$, $y_2 = 3$, $y_3 = 18$

It is required to find x for $y = f(x) = 0$.

By Lagrange's formula for inverse interpolation,

$$\begin{aligned}
&(y - y_1)(y - y_2)(y - y_3)(y - y_0)(y - y_2)(y - y_3) \\
&x = x_0 + x_1 \\
&(y_0 - y_1)(y_0 - y_2)(y_0 - y_3)(y_1 - y_0)(y_1 - y_2)(y_1 - y_3) \\
&(y - y_0)(y - y_1)(y - y_3)(y - y_0)(y - y_1)(y - y_2) \\
&+ x_2 + x_3 \\
&(y_2 - y_0)(y_2 - y_1)(y_2 - y_3)(y_3 - y_0)(y_3 - y_1)(y_3 - y_2) \\
&(0 + 13)(0 - 3)(0 - 18)(0 + 30)(0 - 3)(0 - 18) \\
&= (30) + (34) \\
&(-30 + 13)(-30 - 3)(-30 - 18)(-13 + 30)(-13 - 3)(-13 - 18) \\
&(0 + 30)(0 + 13)(0 - 18)(0 + 30)(0 + 13)(0 - 3) \\
&+ (38) + (42) \\
&(3 + 30)(3 + 13)(3 - 18)(18 + 30)(18 + 13)(18 - 3) \\
&= -0.782 + 6.5323 + 33.6818 - 2.2016 \\
&= 37.2305
\end{aligned}$$

Hence, the root of $f(x) = 0$ is 37.2305.

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6.86 Finite Differences and Interpolation

exercIse 6.6

1. Find x given $y = 0.3887$ from the following data:

x 21 23 25

y 0.3706 0.4068 0.4433

5. Find the age corresponding to the annuity value 13.6 from the given

table:

Age (x) 30 35 40 45 50

Annuity

15.9 14.9 14.1 13.3 12.5

value (y)

[ans.: 43]

points to remember

Forward Differences

$\Delta y_n = y_n - y_{n-1}$

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Backward Differences

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$\nabla y_n = y_n - y_{n-1}$

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Points to Remember 6.87

Central Differences

$\Delta^2 y_1 = y_n - y_{n-1}$

n-

2

Newton's Forward Interpolation Formula

$r(r-1) \frac{\Delta^2 y_1}{2} + r(r-1)(r-2) \frac{\Delta^3 y_1}{6} + \dots$

$$y_r = y_0 + r D y_0 + \frac{r(r-1)}{2!} D^2 y_0 + \frac{r(r-1)(r-2)}{3!} D^3 y_0 + \dots + L$$

2! 3!

Newton's Backward Interpolation Formula

$$y_r = y_n + \frac{r}{h} (y_n - y_{n-1}) + \frac{r(r+1)}{2!} \frac{y_n - 2y_{n-1} + y_{n-2}}{h^2} + \dots + L$$

$$y_r = y_n + r \frac{y_n - y_{n-1}}{h} + \frac{r(r+1)}{2!} \frac{y_n - 2y_{n-1} + y_{n-2}}{h^2} + \dots + L$$

2! 3!

Gauss's Forward Interpolation Formula

$$y_r = y_0 + \frac{r}{h} D y_0 + \frac{r(r-1)}{2!} D^2 y_0 + \frac{r(r-1)(r-2)}{3!} D^3 y_0 + \dots + L$$

$$y_r = y_0 + r D y_0 + \frac{r(r-1)}{2!} D^2 y_0 + \frac{r(r-1)(r-2)}{3!} D^3 y_0 + \dots + L$$

2! 3!

$$(r+1) \frac{r(r-1)(r-2)}{4!} D^4 y_0 + \dots + L$$

$$+ D^4 y_0 + L$$

4!

Gauss's Backward Interpolation Formula

$$y_r = y_n + \frac{r}{h} D y_n + \frac{r(r+1)}{2!} D^2 y_n + \frac{r(r+1)(r+2)}{3!} D^3 y_n + \dots + L$$

$$y_r = y_n + r D y_n + \frac{r(r+1)}{2!} D^2 y_n + \frac{r(r+1)(r+2)}{3!} D^3 y_n + \dots + L$$

2! 3!

$$(r+2) \frac{r(r+1)(r+3)}{4!} D^4 y_n + \dots + L$$

$$+ D^4 y_n + L$$

4!

Stirling's Formula

$$y_r = y_0 + \frac{r}{h} D y_0 + \frac{r(r-1)}{2!} D^2 y_0 + \frac{r(r-1)(r-2)}{3!} D^3 y_0 + \dots + L$$

$$y_r = y_0 + r \frac{D y_0}{h} + \frac{r(r-1)}{2!} \frac{D^2 y_0}{h^2} + \frac{r(r-1)(r-2)}{3!} \frac{D^3 y_0}{h^3} + \dots + L$$

$$+ \frac{r(r-1)(r-2)(r-3)}{4!} \frac{D^4 y_0}{h^4} + \dots + L$$

$$+ \frac{r(r-1)(r-2)(r-3)(r-4)}{5!} \frac{D^5 y_0}{h^5} + \dots + L$$

$$+ \frac{r(r-1)(r-2)(r-3)(r-4)(r-5)}{6!} \frac{D^6 y_0}{h^6} + \dots + L$$

$$+ \frac{r(r-1)(r-2)(r-3)(r-4)(r-5)(r-6)}{7!} \frac{D^7 y_0}{h^7} + \dots + L$$

4!

Lagrange's Interpolation Formula

$$f(x) = y_0 \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x-x_0)(x-x_1)\dots(x-x_{n-1})} + y_1 \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x-x_1)(x-x_2)\dots(x-x_{n-1})(x-x_0)} + \dots + y_n \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} + \dots + y_n \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})}$$

$$f(x) = y_0 + y_1 + \dots + y_n$$

$$(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)$$

$$(x-x_0)(x-x_1)\dots(x-x_{n-1})$$

$$+ y_n$$

$$(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})$$

$$S_g S_{mi} b_y$$

$$D_g t_h$$

Newton's Divided Difference Formula

$$g_r e_d$$

$$y = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] + \dots + (x-x_0)\dots(x-x_{n-1})[x_0, x_1, \dots, x_n]$$

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+ (x - x0) (x - x1) (x - x2) [x0, x1, x2, x3] + ...

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+ (x - x0) (x - x1) ... (x - xn-1) [x, x0, x1, ..., xn]

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6.88 Finite Differences and Interpolation

Inverse Interpolation

(y - y1) (y - y2) ... (y - yn) (y - y0) (y - y2) ... (y - yn)

x = x0 + x1 + ...

(y0 - y1) (y0 - y2) ... (y0 - y1) (y1 - y0) (y1 - y2) ... (y1 - yn)

(y - y0) (y - y1) ... (y - yn - 1)

+ xn

(yn - y0) (yn - y1) ... (yn - yn - 1)

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Numerical Integration

chapter outline

7.1 Introduction

7.2 Newton-Cotes Quadrature Formula

7.3 Trapezoidal Rule

7.4 Simpson's 1/3 Rule

7.5 Simpson's 3/8 Rule

7.6 Gaussian Quadrature Formulae

7.1 Introduction

The process of evaluating a definite integral from a set of tabulated values of $f(x)$ is called numerical integration. This process when applied to a function of a single variable is known as quadrature. In numerical integration, $f(x)$ is represented by an interpolation formula and then it is integrated between the given limits. In this way, the quadrature formula is derived for approximate integration of a function defined by a set of numerical values only.

7.2 newton—cotes Quadrature Formula

Let the function $y = f(x)$ takes values $y_0, y_1, y_2, \dots, y_n$ for $x_0, x_1, x_2, \dots, x_n$ respectively (Fig. 7.1).

b

Let $I = \int_a^b f(x) dx$. Dividing the interval (a, b) by $y = f(x)$

a

into n sub-intervals of width h such that y_n

$x_0 = a, x_1 = x_0 + h, x_2 = x_0 + 2h, \dots, x_n = x_0 + nh = b$ $y_0, y_1, y_2, \dots$

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Dg th

b $x_0 + nh$

gr ed

$\int_a^b f(x) dx = \int_a^b f(x) dx$

En pil

o $x_0, x_1, x_2, \dots, x_n$

a x_0

m

Fig. 7.1

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7.2 Numerical Integration

Putting $x = x_0 + rh, dx = h dr$

When $x = x_0, r = 0$

When $x = x_0 + nh, r = n$

b n

$\int_a^b f(x) dx = h \int_0^n f(x_0 + rh) dr$

a 0

n

$\int_0^n r(r-1)(r-2) \dots (r-n) dr$

$= h \int_0^n \left[y_0 + r \Delta y_0 + \frac{r(r-1)}{2!} \Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!} \Delta^3 y_0 + \dots \right] dr$

0! °

2! 3!

[By Newton's forward interpolation formula]

n

$\int_0^n r^3 dr = \frac{r^4}{4} \Big|_0^n = \frac{n^4}{4}$

$\int_0^n r^2 dr = \frac{r^3}{3} \Big|_0^n = \frac{n^3}{3}$

3 2

$\int_0^n r^3 dr = \frac{r^4}{4} \Big|_0^n = \frac{n^4}{4}$

$$= \frac{h}{2} y_0 + \frac{h}{2} y_1 + \frac{h^2}{6} y''_0 + \frac{h^2}{6} y''_1 + \frac{h^3}{24} y'''_0 + \frac{h^3}{24} y'''_1 + \frac{h^4}{720} y^{(4)}_0 + \frac{h^4}{720} y^{(4)}_1 + \dots$$

$$x_0 + nh$$

$$\frac{h}{2} n(n-1) y''_0 + \frac{h}{2} n(n-1) y''_1 + \dots$$

$$\int_{x_0}^{x_0 + nh} f(x) dx = h \left[\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{h}{24} y''_0 + \frac{h}{24} y''_1 + \dots \right] \dots (7.1)$$

This equation is known as the Newton-Cotes quadrature formula.

7.3 trapezoidal rule

By the Newton-Cotes quadrature formula,

$$x_0 + nh$$

$$\frac{h}{2} n(n-1) y''_0 + \frac{h}{2} n(n-1) y''_1 + \dots$$

$$\int_{x_0}^{x_0 + nh} f(x) dx = h \left[\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{h}{24} y''_0 + \frac{h}{24} y''_1 + \dots \right] \dots (7.1)$$

Putting $n = 1$ in Eq. (7.1) and ignoring the differences of order higher than one,

$$x_0 + h$$

$$\frac{h}{2} y_0 + \frac{h}{2} y_1 + \frac{h^2}{6} y''_0 + \frac{h^2}{6} y''_1 + \dots$$

$$\int_{x_0}^{x_0 + h} f(x) dx = h \left[\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{h}{24} y''_0 + \frac{h}{24} y''_1 + \dots \right] \dots (7.1)$$

$$x_0$$

$$\frac{h}{2} y_0 + \frac{h}{2} y_1 + \frac{h^2}{6} y''_0 + \frac{h^2}{6} y''_1 + \dots$$

$$\int_{x_0}^{x_0 + h} f(x) dx = h \left[\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{h}{24} y''_0 + \frac{h}{24} y''_1 + \dots \right] \dots (7.1)$$

Similarly,

$$x_0 + 2h$$

$$h$$

$$\int_{x_0}^{x_0 + 2h} f(x) dx = h \left[\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{h}{24} y''_0 + \frac{h}{24} y''_1 + \dots \right] \dots (7.1)$$

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$$x_0 + h$$

Dg th

gr ed

$x_0 + nh$

En pil

h

$\int (y_{n-1} + y_n)$

$f(x) dx =$

m

2

Co

$x_0 + (n-1)h$

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7.3 Trapezoidal Rule 7.3

Adding all these integrals,

$x_0 + nh$

h

$\int f(x) dx = \frac{h}{2} (y_0 + y_n) + h (y_1 + y_2 + \dots + y_{n-1})$

x_0

$2 \int_0^\circ$

h

$[X + 2R]$

2

where X = extreme terms, R = remaining terms

This is known as the trapezoidal rule.

errors in the trapezoidal rule

Expanding $y = f(x)$ in the neighbourhood of $x = x_0$ by Taylor's series,

$(x - x_0)^2$

$y(x) = y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2!}y_0'' + \dots$ (7.2)

2!

where $y_0' = \left[\frac{dy}{dx} \right]_{x=x_0}$, and so on.

$x_1 x_1$

$\frac{h^2}{2} (x - x_0)^2$

$\int y dx = \int_0^1$

$\int y + (x - x_0)$

$0 \leq y \leq 1$

2!

$y_0' + \frac{h}{2} \frac{d^2 y}{dx^2}$

$x_0 \times 0$

x_1

$$(x - x_0)^2 (x - x_0)^3$$

$$= y_0 x + y_0 \frac{x^2}{2!} + y_0 \frac{x^3}{3!} + \dots$$

$$2! 3! x_0$$

$$(x_1 - x_0)^2 (x - x_0)^3$$

$$= y_0 (x_1 - x_0) + y_0 \frac{(x_1 - x_0)^2}{2!} + \frac{1}{3!} y_0 \frac{(x_1 - x_0)^3}{3!} + \dots$$

$$2! 3!$$

$$h^2 h^3$$

$$= h y_0 + y_0 \frac{h^2}{2!} + y_0 \frac{h^3}{3!} + \dots \quad (7.3)$$

$$2! 3!$$

where $x_1 - x_0 = h$

x_1

h

Also, $\int_{x_0}^{x_1} y \, dx \approx (y_0 + y_1) \frac{h}{2} = \text{Area of the first trapezium} = A_1 \dots (7.4)$

x_0

2

Putting $x = x_1$ in Eq. (7.2),

$$(x_1 - x_0)^2$$

$$y(x_1) = y_1 = y_0 + (x_1 - x_0) y_0' + y_0 \frac{(x_1 - x_0)^2}{2!} + \dots$$

$$2!$$

$$h^2$$

$$= y_0 + h y_0' + y_0 \frac{h^2}{2!} + \dots \quad (7.5)$$

$$2!$$

Substituting Eq. (7.5) in Eq. (7.4),

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Substituting Eq. (7.5) in Eq. (7.4),

Substituting Eq. (7.5) in Eq. (7.4),

$$A_1 \approx \int_{x_0}^{x_1} y \, dx \approx \int_{x_0}^{x_1} \left(y_0 + h y_0' + y_0 \frac{h^2}{2!} + \dots \right) dx$$

$$2! \int_{x_0}^{x_1} dx$$

Substituting Eq. (7.5) in Eq. (7.4),

$$2!$$

$$h^2 h^3$$

$$m$$

$$\approx h y_0 + y_0 \frac{h^2}{2!} + y_0 \frac{h^3}{3!} + \dots \quad (7.6)$$

Substituting Eq. (7.5) in Eq. (7.4),

$$2! 2(2!)$$

Substituting Eq. (7.5) in Eq. (7.4),

Substituting Eq. (7.5) in Eq. (7.4),

7.4 Numerical Integration

Subtracting Eq. (7.6) from Eq. (7.3),

$$x_1$$

$$E_1 = \frac{1}{3!} h^3 y'''(\xi_1)$$

$$\int_{x_0}^{x_1} y \, dx - A_1 = h \left[y_0' \frac{1}{2!} - y_0'' \frac{1}{3!} + \frac{1}{4!} y_0''' h^2 + \dots \right]$$

$$3$$

$$x_0$$

$$1/3$$

$$= -\frac{1}{6} h^3 y_0''' + \dots$$

$$1/2$$

$$1/3$$

Hence, the error in the first interval (x_0, x_1) , neglecting other terms, is $-\frac{1}{6} h^3 y_0'''$.

$$1/2$$

$$1$$

Similarly, the error in the interval (x_1, x_2) is $-\frac{1}{6} h^3 y_1'''$ and the error in the interval

$$1/2$$

(x_{n-1}, x_n) is $-\frac{1}{6} h^3 y_{n-1}'''$.

$$1/2$$

Hence, the total error is

$$1/3$$

$$E = -\frac{1}{6} h^3 (y_0''' + y_1''' + \dots + y_{n-1}''')$$

$$1/2$$

Let $y'''(x)$ be the largest value of y_0''' , y_1''' , ..., y_{n-1}''' where $x_0 < x < x_n$.

$$1/3$$

$$E < -\frac{1}{6} h^3 y'''(x)$$

$$1/2$$

$$(x - x_0)^2$$

$$< -\frac{1}{6} h^3 y'''(x) \quad [\because nh = x_n - x_0]$$

$$1/2$$

example 1

Find the area bounded by the curve and the x-axis from $x = 7.47$ to $x = 7.52$ from the following table, by using the trapezoidal rule.

x	7.47	7.48	7.49	7.50	7.51	7.52
f(x)	1.93	1.95	1.98	2.01	2.03	2.06

Solution

$a = 7.47$, $b = 7.52$, $h = 0.01$

$$7.52$$

$$\text{Area} = \int_{7.47}^{7.52} f(x) \, dx$$

$$7.47$$

By the trapezoidal rule,

7.52

h

$$\int_a^b f(x) dx = \frac{h}{2} [(y_0 + y_5) + 2(y_1 + y_2 + y_3 + y_4)]$$

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d

Dg th

7.47

gr ed

0.01

$$[(1.93 + 2.06) + 2(1.95 + 1.98 + 2.01 + 2.03)]$$

En pil

2

m

$$= 0.0996$$

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7.3 Trapezoidal Rule 7.5

example 2

Consider the following tabular values:

x 25.0 25.1 25.2 25.3 25.4 25.5 25.6

f (x) 3.205 3.217 3.232 3.245 3.256 3.268 3.280

Determine the area bounded by the given curve and the x-axis between

x = 25 and x = 25.6 by the trapezoidal rule.

Solution

$$a = 25, b = 25.6, h = 0.1$$

By the trapezoidal rule,

25.6

h

$$\int_a^b y dx = \frac{h}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

25

0.1

2

$$[(3.205 + 3.280) + 2(3.217 + 3.232 + 3.245 + 3.256 + 3.268)]$$

$$= 1.9461$$

example 3

Given the data below, find the isothermal work done on the gas if it is compressed from $v_1 = 22$ L to $v_2 = 2$ L.

v_2

Use $W = - \int p \, dv$

v_1

$v, L \quad 2 \quad 7 \quad 12 \quad 17 \quad 22$

$P, \text{ atm} \quad 12.20 \quad 3.49 \quad 2.049 \quad 1.44 \quad 1.11$

Solution

$v_1 = 22, v_2 = 2, h = 5$

By the trapezoidal rule,

v_2

$W = - \int p \, dv$

Sg Smi by

v_1

Dg th

2

gr ed

$= - \int p \, dv$

En pil

22

m

Co

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7.6 Numerical Integration

22

$= \int p \, dv$

2

h

$= \frac{h}{2} (y_1 + y_4) + 2 (y_2 + y_3)$

$2 \int_0^\circ$

5

$= [(12.20 + 1.11) + 2(3.49 + 2.049 + 1.44)]$

2

$= 68.17$

example 4

2

x

Use trapezoidal rule to evaluate $\int dx$, dividing the interval into

$0 \leq x \leq 2$

four equal parts.

Solution

$$a = 0, b = 2, n = 4$$

$$x_n - x_0 = 2 - 0$$

$$h = \frac{2 - 0}{4} = 0.5$$

$$n = 4$$

$$x$$

$$y = f(x) =$$

$$2 + x^2$$

$$x_0 = 0, x_1 = 0.5, x_2 = 1, x_3 = 1.5, x_4 = 2$$

$$y = f(x) \quad 0 \quad 0.3333 \quad 0.5774 \quad 0.7276 \quad 0.8165$$

$$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4$$

By the trapezoidal rule,

$$\int_a^b f(x) dx \approx \frac{h}{2} (y_0 + y_4) + h (y_1 + y_2 + y_3)$$

$$x = h$$

$$\int_0^2 (2 + x^2) dx \approx \frac{0.5}{2} (y_0 + y_4) + 0.5 (y_1 + y_2 + y_3)$$

$$= \frac{0.5}{2} (0 + 0.8165) + 0.5 (0.3333 + 0.5774 + 0.7276)$$

$$= 1.0233$$

$$2$$

$$[(0 + 0.8165) + 2(0.3333 + 0.5774 + 0.7276)]$$

$$= 1.0233$$

example 5

Sg Smi by

Dg th

gr ed

$$1$$

Evaluate $\int_0^1 e^x dx$, with $n = 10$ using the trapezoidal rule.

En pil

$$0$$

$$m$$

$$Co$$

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7.3 Trapezoidal Rule 7.7

Solution

$$a = 0, b = 1, n = 10$$

$$b - a = 1 - 0$$

$$h = \frac{1 - 0}{10} = 0.1$$

$$n = 10$$

$$y = f(x) = e^x$$

x 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.10

$f(x)$ 1 1.1052 1.2214 1.3499 1.4918 1.6487 1.8221 2.0138 2.2255 2.4596 2.7183

y_0 y_1 y_2 y_3 y_4 y_5 y_6 y_7 y_8 y_9 y_{10}

By the trapezoidal rule,

1

h

$$\int_0^1 e^x dx = \frac{1}{2} (y_0 + y_{10}) + (y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 + y_9)$$

x

0

0.1

2

$$[(1 + 2.7183) + 2(1.1052 + 1.2214 + 1.3499 + 1.4918 + 1.6487)$$

$$+ 1.8211 + 2.0138 + 2.2255 + 2.4596)]$$

$$= 1.7196$$

example 6

1

Calculate $\int_0^1 2e^x dx$ with $n = 10$ using the trapezoidal rule.

0

Solution

$$a = 0, b = 1, n = 10$$

$$b - a = 1 - 0$$

$$h = \frac{b - a}{n} = \frac{1 - 0}{10} = 0.1$$

$$n = 10$$

$$y = f(x) = 2e^x$$

x 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1

$f(x)$ 2 2.2103 2.4428 2.6997 2.9836 3.2974 3.6442 4.0275 4.4511 4.9192 5.4365

S_g S_{mi} by

D_g t_h

y_0 y_1 y_2 y_3 y_4 y_5 y_6 y_7 y_8 y_9 y_{10}

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m

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7.8 Numerical Integration

By the trapezoidal rule,

1

h

$$\int_0^2 e^x dx = \frac{1}{2} (y_0 + y_{10}) + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 + y_9)$$

$$= 3.4394$$

example 7

1

Compute the integral $\int_{-1}^1 e^x dx$ using the trapezoidal rule for $n = 4$.

x

-1

Solution

$$a = -1, b = 1, n = 4$$

$$x_0 = -1$$

$$h = \frac{b-a}{n} = 0.5$$

$$n = 4$$

$$y = f(x) = e^x$$

$$x: -1, -0.5, 0, 0.5, 1$$

$$f(x): 0.3679, 0.6065, 1, 1.6487, 2.7183$$

$$y_0, y_1, y_2, y_3, y_4$$

By the trapezoidal rule,

$$\int_{-1}^1 e^x dx$$

$$= \frac{1}{2} (y_0 + y_4) + 2(y_1 + y_2 + y_3)$$

$$= \frac{1}{2} (0.3679 + 2.7183) + 2(0.6065 + 1 + 1.6487)$$

$$= 2.39916$$

$$\int_{-1}^1 e^x dx = 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

$$= 2.39916$$

example 8

$$\int_1^2 e^{-x} dx$$

Evaluate $\int_1^2 e^{-x} dx$ with $n = 10$ using the trapezoidal rule.

Sg Smi by

Dg th

0

gr ed

Solution

En pil

$a = 0, b = 1, n = 10$

m

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7.4 Simpson's 1/3 Rule 7.9

$b - a = 1 - 0$

$h = \frac{b-a}{n} = 0.1$

$n = 10$

2

$y = f(x) = e^{-x}$

$x \quad 0 \quad 0.1 \quad 0.2 \quad 0.3 \quad 0.4 \quad 0.5 \quad 0.6 \quad 0.7 \quad 0.8 \quad 0.9 \quad 1.0$

$f(x) \quad 1 \quad 0.99 \quad 0.9608 \quad 0.9139 \quad 0.8521 \quad 0.7788 \quad 0.6977 \quad 0.6126 \quad 0.5273 \quad 0.4449 \quad 0.3679$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6 \quad y_7 \quad y_8 \quad y_9 \quad y_{10}$

By the trapezoidal rule,

1

h

$\int_a^b$

$-x^2$

$\Delta x = \frac{b-a}{n} = \frac{1-0}{10} = 0.1$

0

2

0.1

2

$[(1 + 0.3679) + 2(0.99 + 0.9608 + 0.9139 + 0.8521 + 0.7788$

$+ 0.6977 + 0.6126 + 0.5273 + 0.4449)]$

$= 0.7462$

7.4 Simpson's 1/3 rule

By the Newton-Cotes quadrature formula,

$x_0 + nh$

$\int_a^b f(x) dx = \frac{h}{n} \sum_{i=0}^{n-1} f(x_i)$

$\int_a^b f(x) dx = \frac{h}{n} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + f(x_n)]$

$\int_a^b f(x) dx = \frac{h}{n} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + f(x_n)]$

$\int_a^b f(x) dx = \frac{h}{n} [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + f(x_n)]$

24

$D y_0 + \dots$

...(7.7)

x_0

Putting $n = 2$ in Eq. (7.7) and ignoring the differences of order higher than 2,

$x_0 + 2h$

$\hat{E} 1 \sim$

$\int_{x_0}^{x_0+2h} f(x) dx = 2h \int y_0 + Dy_0 + D^2 y_0 \dots$

$\hat{I} 6^\circ$

x_0

$\hat{E} \hat{E} y - 2 y_1 + y_0 \hat{\sim}$

$= 2h \int y_0 + (y_1 - y_0) + \hat{A} 2 \hat{\sim} \dots$

$\hat{I} \hat{E} 6^\circ$

h

$= (y_0 + 4y_1 + y_2)$

3

Similarly,

$x_0 + 4h$

h

$\int_{x_0}^{x_0+4h} f(x) dx =$

3

$(y_2 + 4y_3 + y_4)$

$x_0 + 2h$

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$x_0 + nh$

En pil

h

$\int_{x_0}^{x_0+nh} f(x) dx = (y_{n-2} + 4y_{n-1} + y_n)$

3

m

$x_0 + (n-2)h$

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7.10 Numerical Integration

Adding all these integrals,

$x_0 + nh$

h

$$\int_a^b f(x) dx = \frac{h}{3} (y_0 + y_n) + \frac{4h}{3} (y_1 + y_3 + \dots + y_{n-1}) + \frac{2h}{3} (y_2 + y_4 + \dots + y_{n-2})$$

x_0

$3!^\circ$

h

3

$$[X + 4O + 2E]$$

where X = extreme terms, O = odd terms, E = even terms

This is known as Simpson's 1/3 rule.

note To apply this rule, the number of sub-intervals must be a multiple of 2.

errors in simpson's 1/3 rule

Expanding $y = f(x)$ in the neighbourhood of $x = x_0$ by Taylor's series,

$$(x - x_0)^2 (x - x_0)^3 (x - x_0)^4 \dots$$

$$y(x) = y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2!}y_0'' + \frac{(x - x_0)^3}{3!}y_0''' + \frac{(x - x_0)^4}{4!}y_0^{(4)} + \dots \quad (7.8)$$

$2! 3! 4!$

where $y_0' = \left[\frac{dy}{dx} \right]_{x=x_0}$, and so on.

$x^2 x^2$

$$\frac{h^2}{2!} (x - x_0)^2 (x - x_0)^3 (x - x_0)^4 \dots$$

$$\int_a^b y dx = \int_a^b \left[y_0 + \frac{(x - x_0)^2}{2!} y_0'' + \frac{(x - x_0)^3}{3!} y_0''' + \frac{(x - x_0)^4}{4!} y_0^{(4)} + \dots \right] dx$$

$$= \left[y_0 x + \frac{(x - x_0)^3}{3!} y_0'' + \frac{(x - x_0)^4}{4!} y_0''' + \frac{(x - x_0)^5}{5!} y_0^{(4)} + \dots \right]_a^b$$

$2!$

$$y_0'' + \frac{(x - x_0)^2}{2!} y_0'' + \frac{(x - x_0)^3}{3!} y_0''' + \frac{(x - x_0)^4}{4!} y_0^{(4)} + \dots$$

$3!$

$$y_0''' + \frac{(x - x_0)^2}{2!} y_0''' + \frac{(x - x_0)^3}{3!} y_0^{(4)} + \dots$$

$4!$

$$y_0^{(4)} + \frac{(x - x_0)^2}{2!} y_0^{(4)} + \frac{(x - x_0)^3}{3!} y_0^{(5)} + \dots$$

$$x_0 x_0$$

x^2

$$(x - x_0)^2 (x - x_0)^3 (x - x_0)^4 (x - x_0)^5 \dots$$

$$= y_0 x + \frac{(x - x_0)^3}{3!} y_0'' + \frac{(x - x_0)^4}{4!} y_0''' + \frac{(x - x_0)^5}{5!} y_0^{(4)} + \dots$$

$2! 3! 4! 5! x_0$

$$(x^2 - x_0)^2 (x - x_0)^3$$

$$= y_0 (x^2 - x_0) + \frac{(x - x_0)^2}{2!} y_0'' + \frac{(x - x_0)^3}{3!} y_0''' + \dots$$

$2! 3!$

$$(x^2 - x_0)^4 (x - x_0)^5 \dots$$

$$+ \frac{(x - x_0)^2}{2!} y_0'' + \frac{(x - x_0)^3}{3!} y_0''' + \dots$$

$4! 5!$

$$4h^2 y_0 + \frac{8h^3}{3!} y_0'' + \frac{16h^4}{4!} y_0''' + \frac{32h^5}{5!} y_0^{(4)} + \dots$$

$$= \frac{4h^2}{2!} y_0 + \frac{8h^3}{3!} y_0'' + \frac{16h^4}{4!} y_0''' + \frac{32h^5}{5!} y_0^{(4)} + \dots$$

$2! 3! 4! 5!$

$$4h^3 + 2h^4 + 4h^5 + \dots$$

$$= 2h y_0 + 2h^2 y_0' + y_0'' + y_0''' + y_0 + \dots \quad (7.9)$$

$$3 \cdot 3 \cdot 15!$$

$$\text{where } x_2 - x_0 = 2h$$

$$x_2$$

$$h$$

$$\text{Also, } \int y \, dx = 3 (y_0 + 4 y_1 + y_2) = \text{Area in the interval } (x_0, x_2) = A_1 \dots \quad (7.10)$$

Sg Smi by

Dg th

$$x_0$$

gr ed

En pil

m

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7.4 Simpson's 1/3 Rule 7.11

Putting $x = x_1$ in Eq. (7.8),

$$(x_1 - x_0)^2 (x - x_0)^3 (x - x_1)^4$$

$$y(x_1) = y_1 = y_0 + (x_1 - x_0) y_0' + y_0'' + \frac{1}{2} y_0''' + \frac{1}{24} y_0^{(4)} + \dots$$

$$2! \cdot 3! \cdot 4!$$

$$h^2 h^3 h^4 + \dots$$

$$= y_0 + h y_0' + y_0'' + y_0''' + y_0 + \dots \quad (7.11)$$

$$2! \cdot 3! \cdot 4!$$

Putting $x = x_2$ in Eq. (7.8),

$$4h^2 + 8h^3 + 16h^4 + \dots$$

$$y(x_2) = y_2 = y_0 + 2h y_0' + y_0'' + y_0''' + y_0 + \dots \quad (7.12)$$

$$2! \cdot 3! \cdot 4!$$

Substituting Eq. (7.11) and (7.12) in Eq. (7.10),

$$4h^3 + 2h^4 + 5h^5 + \dots$$

$$A_1 = 2h y_0 + 2h^2 y_0' + y_0'' + y_0''' + y_0 + \dots \quad (7.13)$$

$$3 \cdot 3 \cdot 18$$

Subtracting Eq. (7.13) from Eq. (7.9),

$$x_2$$

$$\frac{1}{5} h^5$$

$$\int y \, dx - A_1 = \frac{1}{5} h^5 - 18 h^5 y_0 + \dots$$

$$5 \text{ iv}$$

$$x_0$$

$$1 \cdot 5 \text{ iv}$$

$$= -h y_0 + \frac{1}{5} h^2 y''(x_0) + \frac{1}{90} h^3 y'''(x_0) + \frac{1}{180} h^4 y^{(4)}(x_0) + \dots$$

90

Hence, the error in the interval (x_0, x_2) , neglecting higher powers of h , is

1 5 iv

$$= -h y_0 + \frac{1}{5} h^2 y''(x_0) + \frac{1}{90} h^3 y'''(x_0) + \frac{1}{180} h^4 y^{(4)}(x_0) + \dots$$

90

1 5 iv

Similarly, the error in the interval (x_2, x_4) is $-\frac{1}{5} h^2 y''(x_2) + \frac{1}{90} h^3 y'''(x_2) + \frac{1}{180} h^4 y^{(4)}(x_2) + \dots$

90

Hence, the total error is

$$E = -\left(\frac{1}{5} h^2 y''(x_0) + \frac{1}{5} h^2 y''(x_2) + \dots \right) + \left(\frac{1}{90} h^3 y'''(x_0) + \frac{1}{90} h^3 y'''(x_2) + \dots \right) + \left(\frac{1}{180} h^4 y^{(4)}(x_0) + \frac{1}{180} h^4 y^{(4)}(x_2) + \dots \right)$$

1 5 iv

90

$$h y_0 + y_2 + \frac{1}{5} h^2 y''(x_0) + \frac{1}{5} h^2 y''(x_2) + \dots$$

iv iv iv

Let $y_{iv}(x)$ be the largest value of $y_0, y_2, \dots, y_{2n-2}$ where $x_0 < x < x_{2n}$.

1

$$E < -\frac{1}{5} h^2 y_{iv}(x) + \frac{1}{90} h^3 y'''(x) + \frac{1}{180} h^4 y^{(4)}(x) + \dots$$

90

$$= \frac{1}{5} h^2 y_{iv}(x) - \frac{1}{90} h^3 y'''(x) + \frac{1}{180} h^4 y^{(4)}(x) - \dots$$

$$< -\frac{1}{5} h^2 y_{iv}(x) \quad [\because 2nh = x_{2n} - x_0]$$

180

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Dg th

gr ed

En pil

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7.12 Numerical Integration

example 1

Consider the following values:

x 10 11 12 13 14 15 16

y 1.02 0.94 0.89 0.79 0.71 0.62 0.55

16

Find $\int_{10}^{16} y \, dx$ by Simpson's 1/3 rule.

10

Solution

$$a = 10, b = 16, h=1$$

By Simpson's 1/3 rule,

16

h

$$\int y \, dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

10

1

3

$$[(1.02 + 0.55) + 4(0.94 + 0.79 + 0.62) + 2(0.89 + 0.71)]$$

$$= 4.7233$$

example 2

A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is given as follows:

t (s) 0 10 20 30 40 50 60 70 80

2

a (m/s²) 30 31.63 33.34 35.47 37.75 40.33 43.25 46.69 50.67

By Simpson's 1/3 rule, find the velocity at t = 80 s.

Solution

$$a = 0, b = 80, h = 10$$

By Simpson's 1/3 rule,

80

$$\text{Velocity} = \int a \, dt$$

0

h

$$= [(y_0 + y_8) + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)]$$

Sg Smi by

3

Dg th

10

gr ed

$$= [(30 + 50.67) + 4(31.63 + 35.47 + 40.33 + 46.69)$$

3

En pil

$$+ 2(33.34 + 37.75 + 43.25)]$$

m

$$= 3086.1 \, \text{m/s}$$

Co

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7.4 Simpson's 1/3 Rule 7.13

example 3

A river is 80 metres wide. The depth 'd' in metres at a distance x metres from one bank is given by the following table. Calculate the area of cross section of the river using Simpson's 1/3 rule.

x 0 10 20 30 40 50 60 70 80

y 0 4 7 9 12 15 14 8 7

Solution

a = 0, b = 80, h = 10

80

$A = \int_0^{80} y \, dx$

0

By Simpson's 1/3 rule,

80

h

$\int_0^{80} y \, dx = \frac{h}{3} [(y_0 + y_8) + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)]$

0

10

$= \frac{10}{3} [(0 + 7) + 4(4 + 9 + 15 + 8) + 2(7 + 12 + 14)]$

3

$= 723.33 \, \text{m}^2$

example 4

6

1

Evaluate $\int_0^6 (1 + x) \, dx$ taking h = 1 using Simpson's 1/3 rule. Hence,

0

obtain an approximate value of log 7.

Solution

a = 0, b = 6, h=1

b-a 6-0

n= =6

h 1

1

$y = f(x) =$

$1+x$

x 0 1 2 3 4 5 6

Sg Smi by

Dg th

f(x) 1 0.5 0.3333 0.25 0.2 0.1667 0.1429

gr ed

$y_0 y_1 y_2 y_3 y_4 y_5 y_6$

En pil

m

Co

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7.14 Numerical Integration

By Simpson's 1/3 rule,

6

1 h

$$\int_0^1 (1+x) dx = \frac{1}{6} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

0

1

3

$$[(1 + 0.1429) + 4(0.5 + 0.25 + 0.1667) + 2(0.3333 + 0.2)]$$

$$= 1.9588 \dots (1)$$

By direct integration,

6

1 6

$$\int_0^1 (1+x) dx = \log(1+x) \Big|_0^1 = \log 2 \dots (2)$$

0

From Eqs (1) and (2),

$$\log 2 = 1.9588$$

example 5

5

x d

Evaluate $\int_0^5 (4x+5) dx$ by using Simpson's 1/3 rule, taking 10 equal parts.

$$\int_0^5 (4x+5) dx$$

Hence, find the approximate value of $\log_e 5$.

Solution

$$a = 0, b = 5, n = 10$$

$$b - a = 5 - 0$$

$$h = \frac{5 - 0}{10} = 0.5$$

$$n = 10$$

$$1$$

$$y = f(x) =$$

$$4x + 5$$

$$x \quad 0 \quad 0.5 \quad 1.0 \quad 1.5 \quad 2.0 \quad 2.5 \quad 3.0 \quad 3.5 \quad 4.0 \quad 4.5 \quad 5.0$$

$$f(x) \quad 0.2 \quad 0.1428 \quad 0.1111 \quad 0.0910 \quad 0.0769 \quad 0.0667 \quad 0.0588 \quad 0.0526 \quad 0.0476 \quad 0.0435 \quad 0.04$$

$$y_0 y_1 y_2 y_3 y_4 y_5 y_6 y_7 y_8 y_9 y_{10}$$

By Simpson's 1/3 rule,

5

$d \times h$

$$\int_0^5 (4x + 5) dx = \frac{1}{3} (y_0 + y_5) + \frac{4}{3} (y_1 + y_3 + y_5 + y_7 + y_9) + \frac{2}{3} (y_2 + y_4 + y_6 + y_8)$$

0

0.5

$$= [(0.2 + 0.04) + 4(0.1428 + 0.0910 + 0.0667 + 0.0526 + 0.0435)$$

Sg Smi by

Dg th

3

$$+ 2(0.1111 + 0.0769 + 0.0588 + 0.0476)]$$

gr ed

En pil

$$= 0.4026 \dots (1)$$

m

Co

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7.4 Simpson's 1/3 Rule 7.15

By the direct method,

5 5

$$dx \log_e (4x + 5)$$

$$\int_0^5 (4x + 5) dx = 4$$

0 0

1

$$= (\log_e 25 - \log_e 5)$$

4

1 25

$$= \log_e$$

4 5

1

$$= \log_e 5 \dots (2)$$

4

Equating Eqs (1) and (2),

1

$$\log_e 5 = 0.4026$$

4

$$\log_e 5 = 1.6104$$

example 6

6 3

Evaluate the integral $\int_{-2}^6 (1 + x^2) dx$

by Simpson's 1/3 rule with taking

-2

6 sub-intervals. Use four digits after the decimal point for calculations.

Solution

$a = -2, b = 6, n=6$

$b - a = 6 - (-2) = 8$

$h = \frac{b-a}{n} = \frac{8}{6} = \frac{4}{3}$

$n = 6$

3

$y = f(x) = (1 + x^2)^2$

$x_0 = -2, x_1 = -1.33, x_2 = -0.67, x_3 = 0, x_4 = 0.67, x_5 = 1.33, x_6 = 6$

$x_0 = -2, x_1 = -1.33, x_2 = -0.67, x_3 = 0, x_4 = 0.67, x_5 = 1.33, x_6 = 6$

$f(x)$

11.1803 1.7360 1.7360 11.1803 42.1479 108.7094 225.0622

Sg Smi by

$y_0, y_1, y_2, y_3, y_4, y_5, y_6$

Dg th

gr ed

En pil

m

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7.16 Numerical Integration

By Simpson's 1/3 rule,

$h = \frac{b-a}{n} = \frac{8}{6} = \frac{4}{3}$

$h = \frac{4}{3}$

$\int_{-2}^6 (1 + x^2)^2 dx =$

$\frac{1}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$

$= \frac{1}{3} [(11.1803 + 225.0622) + 4(1.7360 + 11.1803 + 108.7094) + 2(1.7360 + 42.1479)]$

$= 360.2280$

$= 360.2280$

$= 360.2280$

$= 360.2280$

$[(11.1803 + 225.0622) + 4(1.7360 + 11.1803 + 108.7094)$

$+ 2(1.7360 + 42.1479)]$

$= 360.2280$

example 7

0.6 2

Using Simpson's 1/3 rule, find $\int_0^{0.6} e^{-x} dx$ by taking $n = 6$.

0

Solution

$a = 0, b = 0.6, n=6$

$b - a = 0.6 - 0$

$h = \frac{b-a}{n} = 0.1$

$n = 6$

2

$y = f(x) = e^{-x}$

$x \quad 0 \quad 0.1 \quad 0.2 \quad 0.3 \quad 0.4 \quad 0.5 \quad 0.6$

$f(x) \quad 1 \quad 0.99 \quad 0.9608 \quad 0.9139 \quad 0.8521 \quad 0.7788 \quad 0.6977$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

By Simpson's 1/3 rule,

0.6

h

$\int_0^{0.6} e^{-x} dx$

$= \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$

$= \frac{0.1}{3} [(1 + 0.6977) + 4(0.99 + 0.9139 + 0.7788) + 2(0.9608 + 0.8521)]$

$= 0.5351$

0.1

3

3

$[(1 + 0.6977) + 4(0.99 + 0.9139 + 0.7788) + 2(0.9608 + 0.8521)]$

$= 0.5351$

example 8

$\int_0^{\pi/2} \cos x dx$ by

$\Delta x = \frac{\pi/2 - 0}{n}$

$\Delta x = \frac{\pi/2 - 0}{n}$

3

2

Estimate $\int_0^{\pi/2} \cos x dx$ by using Simpson's 1/3 rule with 6 intervals.

$\Delta x = \frac{\pi/2 - 0}{n}$

0

m

Co

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7.4 Simpson's 1/3 Rule 7.17

Solution

$$a = 0, b = 3, n = 6$$

$$b-a = 3-0$$

$$h = \frac{b-a}{n} = \frac{3-0}{6} = 0.5$$

$$n = 6$$

$$y = f(x) = \cos^2 x$$

$$x \quad 0 \quad 0.5 \quad 1.0 \quad 1.5 \quad 2.0 \quad 2.5 \quad 3.0$$

$$f(x) \quad 1 \quad 0.9999 \quad 0.9996 \quad 0.9993 \quad 0.9988 \quad 0.9981 \quad 0.9973$$

$$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$$

By Simpson's 1/3 rule,

$$\int_a^b f(x) dx \approx \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$h = 0.5$$

$$\int_0^3 \cos^2 x \, dx \approx \frac{0.5}{3} [(1 + 0.9973) + 4(0.9999 + 0.9993 + 0.9981) + 2(0.9996 + 0.9988)]$$

$$= 2.9978$$

$$= 2.9978$$

$$= 2.9978$$

$$= 2.9978$$

$$= 2.9978$$

$$= 2.9978$$

example 9

$$p = \frac{\pi}{2}$$

$$n = 6$$

Compute the integral $\int_0^{\pi/2} \sin x \, dx$ for $n = 6$ with an accuracy to 0.0001

$$0.0001$$

decimal places using Simpson's 1/3 rule.

Solution

$$p = \frac{\pi}{2}$$

$$a = 0, b = \frac{\pi}{2}, n = 6$$

$$h = \frac{b-a}{n} = \frac{\pi/2 - 0}{6} = \frac{\pi}{12}$$

$$h = \frac{\pi}{12}$$

$$h = \frac{\pi}{12}$$

$$b-a = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$$h = \frac{\pi}{12}$$

$$n = 6$$

$$y = f(x) = \sin x$$

$$x \quad 0 \quad \frac{\pi}{12} \quad \frac{2\pi}{12} \quad \frac{3\pi}{12} \quad \frac{4\pi}{12} \quad \frac{5\pi}{12} \quad \frac{6\pi}{12} = \frac{\pi}{2}$$

$$S_6 = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$x \quad 0$$

Dg th

12 6 4 3 12 2

gr ed

f(x) 0 0.5087 0.7071 0.8409 0.9306 0.9828 1.0

En pil

y0 y1 y2 y3 y4 y5 y6

m

Co

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7.18 Numerical Integration

By Simpson's 1/3 rule,

p

2

h

$\int_0^p \sin x \, dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$

0

p

36

$[(0 + 1) + 4(0.5087 + 0.8409 + 0.9828) + 2(0.7071 + 0.9306)]$

$= 1.1873$

example 10

The speed v metres per second, of a car, t seconds after it starts, is shown in the following table:

t 0 12 24 36 48 60 72 84 96 108 120

v 0 3.60 10.08 18.90 21.60 18.54 10.26 4.50 4.5 5.4 9.0

Using Simpson's 1/3 rule, find the distance travelled by the car in 2 minutes.

Solution

Let s (metres) distance be travelled in t (seconds).

ds

$=v$

dt

$\int ds = \int v \, dt$

$s = \int v \, dt$

The distance travelled in 2 minutes i.e., 120 seconds is

120

$s = \int v \, dt$

0

Also, $h = 12$ seconds

By Simpson's 1/3 rule,

120

h

$$\int_0^{12} v \, dt = \frac{h}{3} [(y_0 + y_{10}) + 4(y_1 + y_3 + y_5 + y_7 + y_9) + 2(y_2 + y_4 + y_6 + y_8)]$$

0

12

$$[(0 + 9.0) + 4(3.60 + 18.90 + 18.54 + 4.50 + 5.4)$$

Sg Smi by

Dg th

3

gr ed

$$+ 2(10.08 + 21.60 + 10.26 + 4.5)]$$

En pil

$$= 1222.56 \text{ metres}$$

m

Co

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7.5 Simpson's 3/8 Rule 7.19

7.5 Simpson's 3/8 rule

By the Newton-Cotes quadrature formula,

$x_0 + nh$

$$\int_{x_0}^{x_0 + nh} f(x) \, dx = \frac{h}{n} \left[\frac{1}{n} y_0 + \frac{3}{2n} D y_0 + \frac{3}{2n} D^2 y_0 + \frac{1}{n} D^3 y_0 \right]$$

$$\int_{x_0}^{x_0 + nh} f(x) \, dx = \frac{h}{n} \left[y_0 + D y_0 + \frac{3}{2} D^2 y_0 + \frac{1}{2} D^3 y_0 \right]$$

$\hat{I} \approx \frac{h}{24} [y_0 + 3y_1 + 3y_2 + y_3]$

$D y_0 +$

24

$D y_0 + \frac{3}{2} D^2 y_0 + \frac{1}{2} D^3 y_0$

...(7.14)

x_0

Putting $n = 3$ in Eq. (7.14) and ignoring the differences of order higher than 3,

$x_0 + 3h$

$$\int_{x_0}^{x_0 + 3h} f(x) \, dx = \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3]$$

$$\int_{x_0}^{x_0 + 3h} f(x) \, dx = \frac{3h}{8} [y_0 + 3D y_0 + \frac{3}{2} D^2 y_0 + \frac{1}{2} D^3 y_0]$$

$\hat{I} \approx \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3]$

x_0

3h

8

$$(y_0 + 3y_1 + 3y_2 + y_3)$$

$$x_0 + 6h$$

$$3h$$

Similarly,

$$\int_{x_0}^{x_0+8h} f(x) dx =$$

$$8$$

$$(y_3 + 3y_4 + 3y_5 + y_6)$$

$$x_0 + 3h$$

$$x_0 + nh$$

$$3h$$

$$\int_{x_0}^{x_0+8h} f(x) dx =$$

$$8$$

$$(y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

$$x_0 + (n-3)h$$

Adding all these integrals,

$$x_0 + nh$$

$$3h$$

$$\int_{x_0}^{x_0+8h} f(x) dx = \frac{h}{8} (y_0 + y_n) + 2h (y_3 + y_6 + \dots + y_{n-3}) + 3h (y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1})$$

$$x_0$$

$$8h$$

$$3h$$

$$8$$

$$[X + 2T + 3R]$$

where X = extreme terms, T = multiple of three terms, R = remaining terms

This is known as Simpson's 3/8 rule.

note To apply this rule, the number of sub-intervals must be a multiple of 3.

errors in simpson's 3/8 rule

Expanding $y = f(x)$ in the neighbourhood of $x = x_0$ by Taylor's series,

$$(x - x_0)^2 (x - x_0)^3 (x - x_0)^4 \dots$$

$$y(x) = y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2!}y_0'' + \frac{(x - x_0)^3}{3!}y_0''' + \frac{(x - x_0)^4}{4!}y_0^{(4)} + \dots \quad (7.15)$$

Sg Smi by

$$2! 3! 4!$$

Dg th

gr ed

where $y_0' = [y'(x)]_{x=x_0}$, and so on.

En pil

m

Co

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7.20 Numerical Integration

x^3

$\int_{x_0}^{x_3} (x - x_0)^2 (x - x_0)^3 (x - x_0)^4 \, dx$

$\int y \, dx = \int_0^1$

$\int y + (x - x_0) y_0 \, dx +$

$2!$

$y_0 \, dx +$

$3!$

$y_0 \, dx +$

$4!$

$y_0 + \frac{1}{2}$

$x_0 \times 0$

x^3

$(x - x_0)^2 (x - x_0)^3 (x - x_0)^4 (x - x_0)^5 \, dx$

$= y_0 x + y_0 \, dx + y_0 \, dx + y_0 \, dx + y_0 + \frac{1}{2}$

$2! 3! 4! 5! x_0$

$(x^3 - x_0)^2 (x - x_0)^3 (x - x_0)^4$

$= y_0 (x^3 - x_0) + y_0 \, dx + 3 y_0 \, dx + 3 y_0 \, dx$

$2! 3! 4!$

$(x^3 - x_0)^5 \, dx$

$+ y_0 + \frac{1}{2}$

$5!$

$9h^2 27h^3 81h^4 243h^5 \, dx$

$= 3h y_0 + y_0 \, dx + y_0 \, dx + y_0 \, dx + y_0 + \frac{1}{2} \dots (7.16)$

$2! 3! 4! 5!$

where $x^3 - x_0 = 3h$

x^3

$3h$

Also, $\int y \, dx = (y_0 + 3y_1 + 3y_2 + y_3) = \text{Area in the interval } (x_0, x_3) = A_1 \dots (7.17)$

x_0

8

Putting $x = x_1$ in Eq. (7.15),

$h^2 h^3 h^4 \, dx$

$y(x_1) = y_1 = y_0 + hy_0 \, dx + y_0 \, dx + y_0 \, dx + y_0 + \frac{1}{2} \dots (7.18)$

$2! 3! 4!$

Putting $x = x_2$ in Eq. (7.16),

$4h^2 8h^3 16h^4 \, dx$

$y(x_2) = y_2 = y_0 + 2hy_0 \, dx + y_0 \, dx + y_0 \, dx + y_0 + \frac{1}{2} \dots (7.19)$

$2! 3! 4!$

Putting $x = x_3$ in Eq. (7.17),

$9h^2/27 + h^3/81 + h^4/iv$

$$y(x_3) = y_3 = y_0 + 3hy_0' + y_0''h^2 + y_0'''h^3/6 + y_0^{(4)}h^4/24 + \dots \quad (7.20)$$

$2! 3! 4!$

Substituting Eqs (7.18), (7.19) and (7.20) in Eq. (7.17),

$9h^2/27 + h^3/81 + h^4/33 + h^5/iv$

$$A_1 = 3hy_0' + y_0''h + y_0'''h^2/2 + y_0^{(4)}h^3/6 + y_0^{(5)}h^4/24 + \dots \quad (7.21)$$

$2! 3! 4! 16$

Subtracting Eq. (7.21) from Eq. (7.16),

x_3

$\int_{x_0}^{x_3} y dx - A_1 = \frac{h^4}{40} - \frac{16}{16} h^5 y_0^{(5)} + \dots$

$$\int_{x_0}^{x_3} y dx - A_1 = \frac{h^4}{40} - \frac{16}{16} h^5 y_0^{(5)} + \dots$$

$5 iv$

x_0

$3/5 iv$

$$= -\frac{h^5}{80} y_0^{(5)} + \dots$$

So the error is

80

Δg_{th}

$3/5 iv$

graded

Hence, the error in the interval (x_0, x_3) , neglecting higher powers of h , is $-\frac{h^5}{80} y_0^{(5)}$.

En pil

80

$3/5 iv$

m

Similarly, the error in the interval (x_3, x_6) is $-\frac{h^5}{80} y_3^{(5)}$.

Co

80

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7.5 Simpson's 3/8 Rule 7.21

Hence, the total error is

$E = -$

$3/5 iv$

80

$$h^5 \left(y_0^{(5)} + y_3^{(5)} + \dots + y_{n-3}^{(5)} \right)$$

$iv iv iv$

Let $y_{iv}(x)$ be the largest value of $y_0, y_3, \dots, y_{n-3}$ where $x_0 < x < x_{3n}$.

3

$E \leftarrow nh \sum_{i=1}^n y_i(x)$

80

$(x - x_0)^4$

$\leftarrow -3nh \sum_{i=1}^n y_i(x) \hat{=} 3nh = x_3n - x_0 \sim^\circ$

80

example 1

3

1

Evaluate $\int_0^3 dx$ with $n = 6$ by using Simpson's 3/8 rule and, hence,

$\int_0^1 \frac{1}{1+x}$

calculate $\log 2$.

Solution

$a = 0, b = 3, n=6$

$b - a = 3 - 0$

$h = \frac{3-0}{6} = 0.5$

$n = 6$

1

$y = f(x) =$

$\frac{1}{1+x}$

$x \quad 0 \quad 0.5 \quad 1 \quad 1.5 \quad 2 \quad 2.5 \quad 3$

$f(x) \quad 1 \quad 0.6667 \quad 0.5 \quad 0.4 \quad 0.3333 \quad 0.2857 \quad 0.25$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

By Simpson's 3/8 rule,

3

$\frac{1}{3}h$

$\int_0^3 \frac{1}{1+x} dx = \frac{8}{3} \hat{=} (y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5) \sim^\circ$

0

$3(0.5)$

8

$[(1 + 0.25) + 2(0.4) + 3(0.6667 + 0.5 + 0.3333 + 0.2857)]$

$= 1.3888 \dots(1)$

By direct integration,

3

$\frac{1}{3}$

$\int_0^3 \frac{1}{1+x} dx = \log(1+x) \Big|_0^3$

$\sum_{i=1}^n S_i$ by

Dg th

0

$= \log 4$

gr ed

En pil

$$= \log(2)2$$

m

$$= 2 \log 2$$

Co

...(2)

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7.22 Numerical Integration

From Eqs (1) and (2),

$$2 \log 2 = 1.3888$$

$$\log 2 = 0.6944$$

example 2

p 2

sin

X

Evaluate $\int_0^1 x \, dx$ by using Simpson's 3/8 rule.

$$0.5 + 4 \cos x$$

Solution

$$a = 0, b = p$$

Dividing the interval into six equal parts, i.e., $n = 6$,

$$b-a \quad p \quad -0 \quad p$$

h= = =

n 6 6

$\sin 2x$

$$y = f(x) =$$

$$5 + 4 \cos x$$

p p p 2p 5p

x 0 p

6 3 2 3 6

f (x) 0 0.02954 0.10714 0.2 0.25 0.16277 0

y0 y1 y2 y3 y4 y5 y6

By Simpson's 3/8 rule,

p

$\sin 2 \times 3h$

$$\int (5 + 4 \cos x) dx = 8 \hat{E}(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)^\circ$$

0

$$3\hat{p}$$

$8 \text{ \AA} \sim 6 \text{ \AA}$

$$[(0 + 0) + 2(0.2) + 3(0.02954 + 0.10714 + 0.25 + 0.16277)]$$

p

$$= (2.04835)$$

16

$$= 0.40219$$

example 3

p

Sg Smi by

Dg th

2 1

$$\dot{U} = 1 - 2 \sin t$$

2

gr ed

Find d using one of the methods of numerical

En pil

0

integration.

m

Co

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7.5 Simpson's 3/8 Rule 7.23

Solution

È p̃

Dividing the interval $[0, \pi]$ into six equal parts and applying Simpson's 3/8 rule, $\hat{I} \approx 2^\circ$

p

$$a = 0, b = \pi, n=6$$

2

p

-0

$$b-a = 2\pi$$

$$h = \frac{b-a}{n} = \frac{2\pi}{6} = \frac{\pi}{3}$$

$$n = 6$$

1

$$y = f(t) = 1 - \sin^2 t$$

2

$$p \approx \frac{1}{5} \pi$$

t 0

12 6 4 3 12 2

$f(x)$ 1 0.9831 0.9354 0.8660 0.7906 0.7304 0.7071

$y_0 y_1 y_2 y_3 y_4 y_5 y_6$

By Simpson's 3/8 rule,

p

2

$1/3 h$

$\int_0^1 (1 - 2 \sin t) dt = \frac{8}{3} [(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)]$

2

d

0

$3 \hat{p}$

$8 \hat{p}^2$

$[(1 + 0.7071) + 2(0.8660)$

$+ 3(0.9831 + 0.9354 + 0.7906 + 0.7304)]$

$= 1.3496$

example 4

p

2

$\hat{p}$

Find $\int_0^{\pi/2} \sin q dq$ by Simpson's 3/8 rule, dividing the interval $[0, \pi/2]$ into

6 equal parts.

six equal parts.

Solution

p

$a = 0, b = \pi/2, n=6$

2

$S_g = \sum_{i=1}^n y_i$

$D_g = \frac{1}{n} \sum_{i=1}^n y_i$

p

-0

grad

$b-a = 2p$

$h = \frac{b-a}{n} = \frac{2p}{6} = \frac{p}{3}$

En pil

$n = 6$

m

$y = f(q) = \sin q$

Co

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7.24 Numerical Integration

p p p p 5p p

q 0

12 6 4 3 12 2

f (q) 1 1.2953 1.6487 2.0281 2.3773 2.6247 2.7182

y0 y1 y2 y3 y4 y5 y6

By Simpson's 3/8 rule,

p

2

3h

Úe

sin q

d q= È(y0 + y6) + 2 (y3) + 3 (y1 + y2 + y4 + y5)

0

8 Î °

3Ê p ^

8 ÁË 12 ~-

[(1 + 2.7182) + 2(2.0281)

+ 3(1.2953 + 1.6487 + 2.3773 + 2.6247)]

= 3.1012

example 5

1

sin x 1

By Simpson's 3/8 rule, evaluate Ú taking h = .

0 x 6

Solution

1

a = 0, b = 1, h =

6

b - a 1- 0

n= = 6

h 1

6

sin x

y = f (x) =

x

sin x

lim =1

$x \in \mathbb{R}$

1 1 1 2 5

$x \in \mathbb{R}$

6 3 2 3 6

$f(x) = 1, 0.9954, 0.9816, 0.9589, 0.9276, 0.8882, 0.8415$

$y_0, y_1, y_2, y_3, y_4, y_5, y_6$

Sg Smi by

Dg th

By Simpson's 3/8 rule,

gr ed

En pil

1

$\sin x$

$\int_a^b f(x) dx = \frac{h}{8} [(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)]$

m

Co

0

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7.5 Simpson's 3/8 Rule 7.25

3×1^6

8×6^{10}

$[(1 + 0.8415) + 2(0.9589) + 3(0.9954 + 0.9816 + 0.9276 + 0.8882)]$
 $= 0.9461$

example 6

The velocity of a train which starts from rest is given by the following table, the time being reckoned in minutes from the start and speed in km/h.

Time 3 6 9 12 15 18

Velocity 22 29 31 20 4 0

Estimate approximately the distance covered in 18 minutes by Simpson's 3/8 rule.

Solution

Let s km distance be covered in t minutes.

ds

$=v$

dt

$\int ds = \int v dt$

d

$$s = \int v \, dt$$

The distance covered in 18 minutes is

18

$$s = \int_0^{18} v \, dt$$

0

Since the train starts from rest, at $t = 0$, $v = 0 \Rightarrow y_0 = 0$

Time (t) 0 3 6 9 12 15 18

Velocity (v) 0 22 29 31 20 4 0

$y_0 \, y_1 \, y_2 \, y_3 \, y_4 \, y_5 \, y_6$

3 1

Also, $h = 3 \text{ minutes} = \frac{3}{60} \text{ hours}$

60 20

S_g S_{mi} by

D_g t_h

By Simpson's 3/8 rule,

gr ed

18

En pil

3h

$$\int_0^{18} v \, dt = \frac{h}{8} [(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)]$$

d

m

Co

0

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7.26 Numerical Integration

$$3 \hat{1}^{\wedge}$$

$$8 \hat{2}^{\wedge} 20^{\wedge}$$

$$[(0 + 0) + 2(31) + 3(22 + 29 + 20 + 4)]$$

$$= 5.38125$$

example 7

Find the volume of a solid of revolution formed by rotating about the x-axis the area bounded by the lines $x = 0$, $x = 1.5$, $y = 0$, and the curve passing through the following points:

x 0.00 0.25 0.50 0.75 1.00 1.25 1.50

y 1.00 0.9826 0.9589 0.9089 0.8415 0.7624 0.7589

Solution

Volume is given by

$$V = \int_0^1 \pi y^2 dx$$

x 0.00 0.25 0.50 0.75 1.00 1.25 1.50

y² 1.00 0.9655 0.9195 0.8261 0.7081 0.5812 0.5759

y₀ y₁ y₂ y₃ y₄ y₅ y₆

$$h = 0.25$$

By Simpson's 3/8 rule,

$$\frac{3h}{8} \int_0^1 y^2 dx = \frac{3(0.25)}{8} [(1.00 + 0.5759) + 2(0.8261) + 3(0.9655 + 0.9195 + 0.7081 + 0.5812)]$$

$$= 1.1954$$

$$V = \pi \int_0^1 y^2 dx = \pi (1.1954) = 3.7555$$

$$V = \pi \int_0^1 y^2 dx = \pi (1.1954) = 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

$$= 3.7555$$

example 8

Sg Smi by

Dg th

5.2

Evaluate $\int_4^{5.2} \log x$ using the trapezoidal rule and Simpson's 3/8 rule,

gr ed

En pil

4

take $h = 0.2$.

m

Co

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7.5 Simpson's 3/8 Rule 7.27

Solution

$$a = 4, b = 5.2, h = 0.2$$

$$b - a = 5.2 - 4$$

$$n = \frac{b - a}{h} = \frac{5.2 - 4}{0.2} = 6$$

$h = 0.2$

$y = f(x) = \log x$

$x \quad 4 \quad 4.2 \quad 4.4 \quad 4.6 \quad 4.8 \quad 5.0 \quad 5.2$

$f(x) \quad 1.3863 \quad 1.4351 \quad 1.4816 \quad 1.5261 \quad 1.5686 \quad 1.6094 \quad 1.6487$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

By the trapezoidal rule,

5.2

h

$\int \log x \, dx = \frac{h}{2} (y_0 + y_6) + h (y_1 + y_2 + y_3 + y_4 + y_5)$

4

0.2

2

$[(1.3863 + 1.6487)$

$+ 2(1.4351 + 1.4816 + 1.5261 + 1.5686 + 1.6094)]$

$= 1.8277$

By Simpson's 3/8 rule,

5.2

$3h$

$\int \log x \, dx = \frac{3h}{8} (y_0 + y_6) + \frac{3h}{4} (y_3) + \frac{3h}{8} (y_1 + y_2 + y_4 + y_5)$

4

$3(0.2)$

8

$[(1.3863 + 1.6487) + 2(1.5261)$

$+ 3(1.4351 + 1.4816 + 1.5686 + 1.6094)]$

$= 1.8278$

example 9

1

$dx = 1$

Evaluate $\int_0^1 (1 + x^2) \, dx$ taking $h = 0.2$ using Simpson's 3/8 rule and the

0

trapezoidal rule.

Solution

$S_g = \sum_{i=1}^n S_{mi}$ by

$D_g = \frac{1}{n} \sum_{i=1}^n D_{th}$

1

graded

$a = 0, b = 1, h =$

0.2

Enrollment

m

Co

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7.28 Numerical Integration

$b - a = 1 - 0$

$n = 6$

$h = \frac{1}{6}$

1

6

$y = f(x) =$

$1 + x^2$

$1 \quad 1 \quad 1 \quad 2 \quad 5$

$x \quad 0 \quad 1$

$6 \quad 3 \quad 2 \quad 3 \quad 6$

$f(x) \quad 36 \quad 9 \quad 4 \quad 9 \quad 36 \quad 1$

1

$37 \quad 10 \quad 5 \quad 13 \quad 61 \quad 2$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

By Simpson's 3/8 rule,

1

$dx = 3h$

$\int_0^1 (1 + x^2) dx = \frac{8}{6} (y_0 + y_6) + 2 (y_3) + 3 (y_1 + y_2 + y_4 + y_5)$

0

$= \frac{1}{6} (36 + 1) + 2 (9) + 3 (9 + 4 + 9 + 36)$

$= \frac{1}{6} (37 + 18 + 117) = \frac{172}{6} = 28.6667$

$\hat{=} 28.6667$

$= 0.7854$

By the trapezoidal rule,

1

$dx = h$

$\int_0^1 (1 + x^2) dx = \frac{2}{6} (y_0 + y_6) + 2 (y_1 + y_2 + y_3 + y_4 + y_5)$

0

$= \frac{1}{6} (36 + 1) + 2 (9 + 4 + 9 + 36)$

$= \frac{1}{6} (37 + 117) = \frac{154}{6} = 25.6667$

$\hat{=} 25.6667$

$= 0.7842$

example 10

6

dx

Evaluate $\int_0^1 (1+x^2) dx$ by using (i) trapezoidal rule, (ii) Simpson's 1/3 rule,

$0 \leq x \leq 1$

2

(iii) Simpson's 3/8 rule.

Solution

Sg Smi by

Dg th

$a = 0, b = 1$

gr ed

Dividing the interval into six equal parts, i.e., $n = 6$,

En pil

m

Co

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7.5 Simpson's 3/8 Rule 7.29

$b-a = 1-0 = 1$

$h = \frac{1}{6}$

$n = 6$

1

$y = f(x) =$

$1 + x^2$

$x \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$

$f(x) \quad 1 \quad 0.5 \quad 0.2 \quad 0.1 \quad 0.0588 \quad 0.0385 \quad 0.027$

$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

(i) By the trapezoidal rule,

6

$d \times h$

$\int_0^1 (1+x^2) dx = \frac{h}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$

0

1

$= \frac{1}{12} [(1 + 0.027) + 2(0.5 + 0.2 + 0.1 + 0.0588 + 0.0385)]$

21

$= 1.4108$

(ii) By Simpson's 1/3 rule,

6

$d \times h$

$\int_0^1 (1+x^2) dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$

0

1

3

$$[(1 + 0.027) + 4(0.5 + 0.1 + 0.0385) + 2(0.2 + 0.0588)]$$

$$= 1.3662$$

(iii) By Simpson's 3/8 rule,

6

$$d \times 3h$$

$$\int_1^6 x^2 = \frac{8}{3} [(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)]$$

0

3

8

$$[(1 + 0.027) + 2(0.1) + 3(0.5 + 0.2 + 0.0588 + 0.0385)]$$

$$= 1.3571$$

Sg Smi by

Dg th

gr ed

En pil

m

Co

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points to remember

Newton-Cotes Quadrature Formula

$$x_0 + nh$$

$$\int_{x_0}^{x_0 + nh} f(x) dx = \frac{h}{n} \left[\frac{n-1}{2} (y_0 + y_n) + \sum_{i=1}^{n-2} y_i \right]$$

$$\int_{x_0}^{x_0 + nh} f(x) dx = \frac{h}{n} \left[\frac{n-1}{2} (y_0 + y_n) + \sum_{i=1}^{n-2} y_i \right]$$

$$x_0 \text{ to } x_0 + nh$$

Trapezoidal Rule

$$x_0 + nh$$

h

$$\int_{x_0}^{x_0 + nh} f(x) dx = \frac{h}{2} (y_0 + y_n) + h \sum_{i=1}^{n-1} y_i$$

x0

$$2 \int_0^1$$

Simpson's 1/3 Rule

$$x_0 + nh$$

h

$$\int_{x_0}^{x_0 + nh} f(x) dx = \frac{h}{3} (y_0 + y_n) + \frac{4h}{3} \sum_{i=1}^{n-1} y_i$$

x0

$$3 \int_0^1$$

Simpson's 3/8 Rule

$$x_0 + nh$$

3h

$$\int f(x) dx = E(y_0 + y_n) + 2(y_3 + y_6 + \dots + y_{n-3})$$

x0

8 Ĩ

$$+3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1})^{\sim \circ}$$

Sg Smi by

Dg th

gr ed

En pil

m

Co

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2. ERROR ANALYSIS AND COMPUTER ARITHMETIC ERRORS IN COMPUTATION

When a calculator or digital computer is used to perform calculations, an unavoidable error is generated.

This error arises because the arithmetic performed in a machine involves numbers with only a finite number of digits hence many calculations are performed with approximate representations of the actual numbers. In addition to inaccurate representation of numbers, the arithmetic performed in a computer is not exact.

The arithmetic generally involves manipulating binary digits or bits by various shifting or logical operation.

Data Types

A data type defines the set of values that an expression can produce or a variable can contain. The data type of a variable or expression also defines the operations that can be performed on the variable or expression. The type of a variable is established by the variable's declaration, while the type of an expression is determined by the definitions of its operators and the types of their operands.

Amongst other data types, the integer types and floating-point types are considered arithmetic types, since arithmetic can be performed on them

Integer Representation.

Now that we have reviewed how base-10 numbers can be represented in binary form, it is simple to conceive of how integers are represented on a computer.

The most straightforward approach, called the signed magnitude method, employs the first bit of a word to indicate the sign, with a 0 for positive and a 1 for negative. The remaining bits are used to store the number.

1

The representation of the decimal integer -173 on a 16-bit computer using the signed magnitude method.

Note that the signed magnitude method described above is not used to represent integers on conventional computers. A preferred approach called the 2's complement technique directly incorporates the sign into the number's magnitude rather than providing a separate bit to represent plus or minus

Problem 1

Determine the range of integers in base-10 that can be represented on a 16-bit computer.

Floating-Point Representation

Fractional quantities are typically represented in computers using floating-point form. In this approach, the number is expressed as a fractional part, called a mantissa or significand, and an integer part, called an exponent or characteristic, as in

$m.be$

where m = the mantissa, b = the base of the number system being used, and e = the exponent.

2

For instance, the number 156.78 could be represented as 0.15678×10^3 in a floating-point base-10 system. The figure below shows one way that a floating-point number could be stored in a word. The first bit is reserved for the sign, the next series of bits for the signed exponent, and the last bits for the mantissa.

Note that the mantissa is usually normalized if it has leading zero digits. This is retain any additional significant figure when the number is stored.

1 bit 5 bit 10 bit

sign exponent integer, mantissa, significand 16 bit system

1 bit 8 bit 23 bits

sign exponent integer, mantissa, significand 32 bit system

1 bit 11 bits 52 bits

sign exponent integer, mantissa, significand 64 bit system

Binary Machine Numbers Representation

For example

Suppose the quantity $1/34 = 0.029411765 = 0.0294 \times 10^0$. But in normalized form is 0.2941×10^{-1} . The consequence of normalization is that the absolute value of m is limited. That is

1

$\leq m < 1$

m

Floating-point representation allows both fractions and very large numbers to be expressed on the computer. However, it has some disadvantages.

For example, floating-point numbers take up more room and take longer to process than integer numbers. More significantly, however, their use introduces a source of error because the mantissa holds only a finite number of significant figures. Thus, a round-off error is introduced.

3

Summary 1

Summary 2

There are two ways to do it; Chopping or Rounding

4

Numerical Errors

1) Round off error

The round off error of a number is the error introduced by rounding off the decimal representation of the number to a certain decimal place.

E.g 416.5678 → 416.6

2) Truncation or chopping errors

These are errors caused by chopping or truncating (prematurely breaking off) a finite or infinite sequence of computational steps necessary for producing an exact result. Chopping is not recommended because it introduces an error that is systematic and can be large.

E.g 416.5678

→ 416.5 (this is 4 digit chopping when it normalized)

Errors of all types are collectively called "BUGS ". The process of locating and removing bugs is called ' DEBUGGING'. Various compilers provide diagnostic which indicates all errors in a source programme except error in logic.

Definition of Terms

Let α = True value or exact value of a number and

α^* = The computed or approximated value of a number.

1. Then the absolute error ϵ is given by

$$\epsilon = \alpha - \alpha^*$$

2. Relative error ϵ_r = $\frac{\epsilon}{\alpha}$ = $\frac{\alpha - \alpha^*}{\alpha}$

$$\epsilon_r = 1 - \frac{\alpha^*}{\alpha}$$

3. Relative error ϵ_r , $\alpha = \alpha^* + \epsilon$

The absolute error (ϵ) in a measurement X is:

$$\epsilon = |\alpha - \alpha^*|$$

5

The relative error in a measurement x ($x \neq 0$) is the ratio of the absolute error to the true value.

$$\text{Absolute error} = x - x^*$$

$$|x - x^*| = |x| \left| \frac{x - x^*}{x} \right|$$

$$|x| \left| \frac{x - x^*}{x} \right| = \text{Absolute error}$$

$$\left| \frac{x - x^*}{x} \right| = \text{Relative error}$$

$$\text{Percentage relative error, } \%100 = \left| \frac{x - x^*}{x} \right| \times 100$$

$$\text{Percentage error}$$

Problem 2

Given that $x = 0.4356789 \times 10^4$

Let x be rounded off to 0.4357×10^4 by (rounding off)

And x be chopped off to 0.4356×10^4 by (chopping off)

Thus,

Calculate the error, absolute error, relative error and the true percent relative error

Solution

$$i. \text{ Absolute error} = x - x^*$$

$$= (0.4357 - 0.4356)10^4$$

$$= 10^{-4} \times 10^4$$

$$ii. \text{ Absolute error} = |x - x^*|$$

$$= 10^{-4} \times 10^4$$

$$= 10^{-4} \times 10^4$$

$$iii. \text{ Relative error} = \frac{|x - x^*|}{x} = \frac{10^{-4}}{0.4357 \times 10^4}$$

$$= \frac{10^{-4}}{0.4357 \times 10^4}$$

$$= \frac{10^{-4}}{0.4357 \times 10^4}$$

$$iv. \%100 = \left| \frac{x - x^*}{x} \right| \times 100$$

$$= \frac{10^{-4}}{0.4357 \times 10^4} \times 100$$

$$0.23 \times 10^{-3} \times 100 = 0.023\%$$

6

Finite-Digit Arithmetic & Errors in Computer Arithmetic

Assume the following point representation as $f(x)$ and $f(y)$ even for real numbers x and y and the symbols $\oplus$, $\ominus$, $\otimes$, $\oslash$ represents a machine addition, subtraction, multiplication and division respectively.

Assume a K - digit operation

$$x \oplus y = \text{round}\{f(x) \oplus f(y)\}$$

$$x \ominus y = \text{round}\{f(x) \ominus f(y)\}$$

$$x \otimes y = \text{round}\{f(x) \otimes f(y)\}$$

$$x \oslash y = \text{round}\{f(x) \oslash f(y)\}$$

Problem 3

Given that $x = 1/3$ and $y = 5/7$ and that five-digit chopping is used for the arithmetic calculation involving x and y . Compute the absolute and relative error in the arithmetic and, normalized and round up the mantissa to 3 or 4 dp.

Solution

Operation True value 5 digit Absolute Relative error

Chopping error

$$x + y = 22/21 = 1.0476 \quad 9.05 \times 10^{-6} \quad 8.64 \times 10^{-6}$$

21

$$x - y = -8/21 = -0.38095 \quad 2.381 \times 10^{-6} \quad 6.25 \times 10^{-6}$$

21

$$x \otimes y = 5/0.23809 = 5.238 \times 10^{-6} \quad 22.0 \times 10^{-6}$$

21

$$x \oslash y = 7/0.46666 = 6.666 \times 10^{-6} \quad 14.29 \times 10^{-6}$$

15

7

Problem 4

Numerical values x and y are stored in the computer as approximations

$\tilde{x}$ and $\tilde{y}$ which are multiplied together. Neglecting any further truncation or round-off error. Show that the product of relative errors (R.E) is the sum of the R.E of the factors

Solution

$$x - \tilde{x}$$

$$\tilde{x} =$$

$$x$$

$$\tilde{x} - x = x - \tilde{x}$$

$$\tilde{x} - x = x - \tilde{x}$$

$$\tilde{x} = x - (\tilde{x} - x)$$

$$\tilde{x} = x \{1 - (\tilde{x} - x)/x\} \dots \text{equ1}$$

$$y - \tilde{y}$$

$$\tilde{y} =$$

$$y$$

$$\tilde{y} - y = y - \tilde{y}$$

$$\tilde{y} - y = y - \tilde{y}$$

$$\tilde{y} = y - (\tilde{y} - y)$$

$$\tilde{y} = y \{1 - (\tilde{y} - y)/y\} \dots \text{equ2}$$

Multiply Equ1 & Equ2

$$\tilde{x} \cdot \tilde{y} = x \{1 - (\tilde{x} - x)/x\} \cdot y \{1 - (\tilde{y} - y)/y\}$$

$$\tilde{x} \cdot \tilde{y} = xy \{1 - (\tilde{x} - x)/x\} \{1 - (\tilde{y} - y)/y\}$$

$$x * y = (1 - \{x + y\} + x \cdot y)$$

$$x * y = (1 - \{x + y\} + x \cdot y)$$

$$x * y$$

$$= 1 - \{x + y\} + x \cdot y$$

$$x$$

$$x * y$$

$$x = 1$$

$$x$$

$$1 = 1 - \{x + y\} + x \cdot y$$

$$0 = -\{x + y\} + x \cdot y$$

$$x \cdot y = x + y$$

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Exercise 1

Numerical values x and y are stored in the computer as approximations

$x * y$ where e_x and e_y are the error respectively. Neglecting any further truncation or round-off error. Show that;

The error in the sum is equal to the sum of the errors

The error in the difference is equal to sum of difference of the error

Problem 5

Using a 3-digit chopping and 3-digit rounding off. Estimate the relative error in evaluating a function $f(x) = x^3 - 6x^2 + 3x - 0.149$ at $x = 4.71$

Solution

$$x^3 - 6x^2 + 3x$$

$$\text{Exact } 4.71 \quad 104.487111 \quad 22.1841 \quad 133.1046 \quad 14.13$$

$$\text{3-digit } 4.71 \quad 104 \quad 22.1 \quad 133 \quad 14.1$$

chopping

$$\text{3-digit } 4.71 \quad 105 \quad 22.2 \quad 133 \quad 14.1$$

rounding

off

Exact

$$\begin{aligned} f(4.71) &= 104.487111 - 133.1046 + 14.13 - 0.149 \\ &= -14.636489 \end{aligned}$$

3-digit chopping

$$\begin{aligned} f(4.71) &= 104 - 133 + 14.1 - 0.149 \\ &= -15.0 \end{aligned}$$

3-digit rounding off

$$\begin{aligned} f(4.71) &= 105 - 133 + 14.1 - 0.149 \\ &= -14.0 \end{aligned}$$

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Relative Error for 3-digit chopping

$$-14.636489 - (-15)$$

$$\square\square = ||$$

$$-14.636489$$

$$= 0.0248$$

Relative Error for 3-digit Rounding-off

$$-14.636489 - (-14.0)$$

$$\square\square = ||$$

$$-14.636489$$

$$= 0.0434$$

Remark

Polynomials should always be expressed in nested form before performing an evaluation because this form minimizes the number of arithmetic calculations.

$$\square(\square) = \square^3 - 6\square^2 + 3\square - 0.149 - - - 1$$

$$\square\square \square\square\square\square\square\square\square\square \square\square \square\square\square\square\square \square\square\square\square \square\square$$

$$\square(\square) = ((\square - 6)\square + 3)\square - 0.149 - - - 2$$

The former (1) have a lose in accuracy (relative error) while the latter (2) have an improved accuracy.

Miscellaneous

The errors associated with both calculations and measurements can be characterized with regard to their accuracy and precision. Accuracy refers to how closely a computed or measured value agrees with the true value. Precision refers to how closely individual computed or measured values agree with each other.

General formular for error propagation for function:

The foregoing approach can be generalized to functions that are dependent on more than one independent variable. This is accomplished with a multivariable version of the Taylor series.

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For example, if we have a function of two independent variables x and z , the Taylor series can be written as follows while dropping all second-order and higher terms.

If $\square = \square(\square, \dots, \square)$ is any function of $x, \dots, z$

$$\square\square \square\square$$

$$\square(\square\square+1, \dots, \square\square+1) = \square(\square, \dots, \square) + (\square\square+1 - \square) + (\square - \square) + \dots$$

$$\square\square \square\square \square\square+1$$

$$\square\square \square\square$$

$$\square(\square\square+1, \dots, \square\square+1) - \square(\square, \dots, \square) = (\square\square+1 - \square) + (\square\square+1 - \square)$$

$$\square\square \square\square$$

equ 2

where all partial derivatives are evaluated at the base point i. If all second-order and

higher terms are dropped, Eq 2. above can be written as

Eq 3

$$\Delta f \approx \left| \frac{\partial f}{\partial x} \right| \Delta x + \left| \frac{\partial f}{\partial z} \right| \Delta z + \left| \frac{\partial f}{\partial t} \right| \Delta t \quad (3)$$

Eq 3

Where Δx is the error in x and Δz is the error in z and Δf is the approximate error in f

Problem 6

The deflection y of the top of a sailboat mast is

$$y = \frac{FL^3}{8EI} \quad (4)$$

$$y =$$

$$8EI$$

Where F = a uniform side loading, L = height (m) , E = the modulus of elasticity (N/m²), and I = the moment of inertial (m⁴). Estimate the error in y given the following data:

$$F = 750 \quad \Delta F = 30$$

$$L = 9 \quad \Delta L = 0.03$$

$$E = 7.5 \times 10^9 \quad \Delta E = 5 \times 10^7$$

$$I = 0.0005 \quad \Delta I = 0.000005$$

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Solution

Employing equation 3 above;

$$\Delta y = \left| \frac{\partial y}{\partial F} \right| \Delta F + \left| \frac{\partial y}{\partial L} \right| \Delta L + \left| \frac{\partial y}{\partial E} \right| \Delta E + \left| \frac{\partial y}{\partial I} \right| \Delta I$$

$$\Delta y = \left| \frac{\partial y}{\partial F} \right| \Delta F + \left| \frac{\partial y}{\partial L} \right| \Delta L + \left| \frac{\partial y}{\partial E} \right| \Delta E + \left| \frac{\partial y}{\partial I} \right| \Delta I$$

$$\Delta y = \left| \frac{\partial y}{\partial F} \right| \Delta F + \left| \frac{\partial y}{\partial L} \right| \Delta L + \left| \frac{\partial y}{\partial E} \right| \Delta E + \left| \frac{\partial y}{\partial I} \right| \Delta I$$

$$\Delta y = \left| \frac{\partial y}{\partial F} \right| \Delta F + \left| \frac{\partial y}{\partial L} \right| \Delta L + \left| \frac{\partial y}{\partial E} \right| \Delta E + \left| \frac{\partial y}{\partial I} \right| \Delta I$$

$$\Delta y = \Delta y + \Delta y + 2 \Delta y + \Delta y$$

$$8 \times 10^9 \times 0.000005 + 8 \times 10^9 \times 0.000005$$

Substituting the appropriate values gives

$$\Delta y = 0.006561 + 0.002187 + 0.001094 + 0.00164 = 0.011482$$

Therefore, $y = 0.164025 \pm 0.011482$. In other words, y is between 0.152543 and 0.175507 m. The validity of these estimates can be verified by substituting the extreme values for the variables into the equation to generate an exact minimum of

$$720 \times 8.974$$

$$8(7.55 \times 10^9)0.000505$$

$$8(7.55 \times 10^9)0.000505$$

$$780 \times 9.034$$

$$\square\square\square\square = 0.175790$$

$$8(7.45 \times 109)0.000495$$

Thus, the first-order estimates from Taylor series are reasonably close to the exact values.

EXERCISES

1: Evaluate the polynomial

$$\square = \square^3 - 5\square^2 + 6\square - 0.55$$

at $x = 1.37$. Use 3-digit arithmetic with chopping. Evaluate the percent relative error.

2: Repeat (1) but express y as

$$y = ((x - 5)x + 6)x + 0.55$$

Evaluate the error and compare with part (1) stating what could have led to this discrepancy

3: State the Taylor's theorem and express mathematically its series

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3ii Recall that the velocity of the falling parachutist can be computed by

$$\square\square\square$$

$$-(\)\square$$

$$\square(\square) = (1 - \square\square)$$

$$\square$$

Using a first-order error analysis to estimate the error of v at $t = 6$, if $g = 9.81$ and $m = 50$ but $c = 12.5 \pm 1.5$

3iii: A missile leaves the ground with an initial velocity u forming an angle θ_0 with the vertical as shown in below. The laws of mechanics can be used to show that

$$\square\square$$

$$\sin 2\square =$$

$$\square$$

Where g is earth gravitation acceleration at 9.81. It is desired to fire the missiles and reach the design maximum range $R = 90\text{km}$ with an accuracy of $\pm 2\%$. Determine the range of values for θ_0 if $u = 1000\text{m/s}$.

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NUMERICAL METHODS

A numerical method is an approximate computer method for solving a mathematical problem which often has no analytical solution. In analytical solution, the problem is framed in a well understood form, calculating the exact

solution. A numerical method makes guesses at the solution and the test whether the problem is solved well enough to stop.

The problems themselves can come from many fields of application, e.g., biology, chemistry, physics, engineering, medicine, education, entertainment, the internet, forensics, financial markets, etc

The stakes are often high, so the results must be accurate, reliable, and robust.

Disasters Caused By Poor Numerical Computing

Here are some examples of real life disasters that occurred because of poor computing practices (and not because of programming errors per se).

❑ CSI: Round-off error (aka Salami attacks)

Hackers diverted round-off errors from normal bank calculations to their own accounts, netting them substantial sums of money over time.

❑ Patriot missile failure, February 25, 1991:

A Scud missile killed 28 soldiers in Dhahran, Saudi Arabia, when the Patriot missile meant to intercept it miscalculated the Scud's position. The error came from an integer to floating-point conversion of 9.5×10^{-8} per tenth of a second multiplied by the product of the up-time of the Patriot (100 hours) and the velocity of the Scud (1676 metres per second); i.e., 573 metres.

Trade - off in choosing a Numerical Method

1. Accuracy vs. computational cost: Higher order numerical methods often provide greater accuracy but require more computational resources and time.

1

2. Stability vs. convergence: Some numerical methods may be more stable but converge slower, while others may converge quickly but be highly sensitive to initial conditions.

3. Robustness vs. complexity: More complex numerical methods may provide better results in a wider range of situations, but they may also be more difficult to implement and require a deeper understanding of the underlying mathematical principles.

4. Applicability vs. simplicity: Some numerical methods may be more applicable to a wider range of problems but may also be more complex and difficult to implement, while simpler methods may be limited in their applicability.

5. Model complexity vs. computational efficiency: More complex numerical methods may be able to handle more complex models, but may require significantly more computational time and resources.

Other Includes;

- Type of Mathematical Problem
- Type, Availability, Precision, Cost, and Speed of Computer
- Program Development Cost versus Software Cost versus Run-Time Cost
- Characteristics of the Numerical Method
- Mathematical Behavior of the Function, Equation, or Data.
- Ease of Application
- Maintenance.

2

1. BINARY SYSTEMS

Modern computers of high-speed deal with real number in the binary system in contrast to the decimal systems that humans prefer to use. The binary 'sys' uses two digits; 0's and 1's only to represent numbers. It is called "bits". E.g., 11100101 has 8 bits. 8 bits make one byte.

In computing, ASCII (American Standard Code for Information Interchange) is an 8-bit code for characters. That is, it uses 8-bits to represent a letter. Example, the word 'CAT' in a word processor can be 01000011₂, 01000001₂, and 01010100₂.

Computers carry out their operation in binary number systems. Computers communicate with its human users in decimal system but work internally in binary system. Hence, there must be a conversion process that is executed internally by the computer.

Conversion of Binary to Decimal

Examples: a) Convert 10011011₂ to base 10

b) Convert 1001.11101₂ to base 10.

Solution

$$\begin{aligned} \text{a) } 10011011_2 &= 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + \\ &1 \times 2^1 + 1 \times 2^0 \\ &= 128 + 0 + 0 + 16 + 8 + 0 + 2 + 1 \\ &= 155_{10} \end{aligned}$$

$$\begin{aligned} \text{b) } 1001.11101_2 &= 1 \times 2^0 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times \\ &2^{-2} + 1 \times 2^{-3} + 0 \times 2^{-4} + 1 \times 2^{-5} \\ &= 8 + 1 + 1/2 + 1/4 + 1/8 + 1/16 + 1/32 \\ &= 9.96910 \end{aligned}$$

3

Conversion from Decimal (Base 10) to Binary Base

a) Convert 1510 to binary number

2 15

2 7r1

2 3r1

2 1r1

0r1

$15_{10} = 1111_2$

b) Convert 25_{10} to base 2

2 25

2 12 r 1

2 6r0

2 3r0

1r1

0r1

$15_{10} = 11001_2$

BINARY CODED DECIMAL SYSTEM

BCDS is the code used by microprocessors. The computer system expresses a decimal number by four binary numbers.

How does the computer process 155?

2 1 2 5 2 5

2 0r1 2 2r1 2 2r1

2 1r0 2 1r0

0r1 0r1

So 155 is;

1 = 0001

5 = 0101

5 = 0101

NOTE: You will add zero to the front so as to make each 4 digits

4

Conversion from Binary to Octal System

STEP 1 - Convert to base 10

STEP 2 - Convert to base 8

a) Convert 10110_2 to octal

Solution

STEP 1 - Convert to base 10

$$10110_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 22_{10}$$

STEP 2 - Convert to base 8

8 22

8 2r6

8 0r2

$$= 26$$

$$101102 = 268$$

b) Convert 10000101_2 to octal

Solution

STEP 1 - Convert to base 10

$$10000101 = 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 133$$

STEP 2 - Convert to base 8

$$8 \overline{) 133}$$

$$8 \overline{) 16} \text{ r } 5$$

$$8 \overline{) 2} \text{ r } 0$$

$$0 \text{ r } 2$$

$$10000101_2 = 2058$$

5

Conversion from Octal to Binary

a) Convert 3078 to binary

Solution

STEP 1 - Convert to base 10

$$3078 = 3 \times 8^2 + 0 \times 8^1 + 7 \times 8^0 \\ = 192 + 0 + 7 \\ = 19910$$

STEP 2 - Convert to base 2

$$2 \overline{) 199}$$

$$2 \overline{) 99} \text{ r } 1$$

$$2 \overline{) 49} \text{ r } 1$$

$$2 \overline{) 24} \text{ r } 1$$

$$2 \overline{) 12} \text{ r } 0$$

$$2 \overline{) 6} \text{ r } 0$$

$$2 \overline{) 3} \text{ r } 0$$

$$2 \overline{) 1} \text{ r } 1$$

$$0 \text{ r } 1$$

$$19910 = 110001112$$

EXCERCISE

1) Convert 47628 to binary.

Converting a binary fraction to octal

a) Convert 11101.01112

Solution

STEP 1 - Convert to base 10

$$11101.01112$$

$$\begin{aligned}
 &= 1 \times 24 + 1 \times 24 + 1 \times 22 + 0 \times 21 + 1 \times 20 + 0 \times 2^{-1} \\
 &+ 1 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4} \\
 &= 16 + 8 + 4 + 1 + 1/2 + 1/4 + 1/8 + 1/16 \\
 &= 29.4375210
 \end{aligned}$$

6

STEP 2 - Convert to base 8

8 29

8 3r5

0r3

29 = 358

the fraction part is as follow

$$0.43752 \times 8 = 3.5 \text{ } \square\square\square\square 3$$

$$0.5 \times 8 = 4 \text{ } \square\square\square\square 4$$

Continue until no fractional part or till you are okay with the precision

$$0.43752 = .348 \text{ Thus } 11101.01112 = 35.348$$

Convert a decimal fraction to binary

a) Convert 5.8725 to its binary equivalent

2 5

2 2r1

2 1r0

0r1

$$0.8725 \times 2 = 1.745 \text{ } \square\square\square\square 1$$

$$0.745 \times 2 = 1.49 \text{ } \square\square\square\square 1$$

$$0.49 \times 2 = 0.98 \text{ } \square\square\square\square 0$$

$$0.98 \times 2 = 1.96 \text{ } \square\square\square\square 1$$

Assume there was no remainder at this point, our answer becomes;

$$= 101.11012$$

NOTE: Solve till you have no remainder.

Exercise

1) Convert 11011011.110112 to octal.

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BINARY OPERATIONS (Arithmetic)

A microprocessor uses the following four binary arithmetic to carry out its operations. They include addition, subtraction, multiplication, division.

1) Addition

a. Add 10102 $\square\square\square$ 10012

1 0 1 0

+ 1 0 0 1

1 0 0 1 1

b. Add 110010012 □□□ 10011102

```

1 1 0 0 1 0 0 1
+ 1 0 0 1 1 1 0
1 0 0 0 1 0 1 1 1

```

2) Subtraction

a. Subtract 10012 □□□□ 10102

```

1 0 1 0
- 1 0 0 1
0 0 1 0

```

b. Subtract 0011002 □□□□ 00110102

```

0 0 1 1 0 1 0
- 0 0 1 1 0 0
0 0 0 1 1 1 0

```

c. Subtract 10101112 □□□□ 11101102

```

1 1 1 0 1 1 0
- 1 0 1 0 1 1 1
0 0 1 1 1 1 1

```

8

3) Multiplication

a. Evaluate 11001012 × 100112

```

1 1 0 0 1 0 1
X 1 0 0 1 1
1 1 0 0 1 0 1
1 1 0 0 1 0 1
0 0 0 0 0 0 0
0 0 0 0 0 0 0
1 1 0 0 1 0 1
1 1 1 0 1 1 1 1 1 1 1

```

4) Division

To divide two numbers, which result is an exact division, we basically need to follow four steps: division, multiplication, subtraction, and next digit.

To perform a binary division, we need to follow the same process as we do for dividing regular numbers but, in this case, we only need to decide if it's going to be a 1 or a 0.

a. Divide 110012 □□ 101

```

1 0 1
101 1 1 0 0 1
- 101

```

00101 add 0 to quotient,

101

000

ANS = 101

b. Divide 10110012 by 111

To divide two numbers, which result is an exact division, we basically need to follow four steps: division, multiplication, subtraction, and next digit.

First, we need to identify digit by digit.

Is 111 less or equal than 101? No, so we get another digit.

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Is 111 less or equal than 101? No, so we repeat and get another digit.

Is 111 less or equal than 1011? Yes, so in this case we don't need to think how many times can 1011 be divided by 111, instead, we just add a 1 to the product.

We keep on the usual until nothing to bring down and the remainder is read or attached.

1100

111 1 0 1 1 0 0 1

111

1000

111

00101

Exercise

A. Add 111101111 and 101010111

B. What is the binary subtraction of 111101111 and 101010111

C. Multiply 111101111 and 101010111

D. Divide 111101111 by 101

HEXADECIMAL NUMBER SYSTEM

Hexadecimal is the name of the numbering system that is base 16. This system, therefore, has numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, and 15. That means that two-digit decimal numbers 10, 11, 12, 13, 14, and 15 must be represented by a single numeral to exist in this numbering system. To address the two-digit decimal values, the alphabetic characters A, B, C, D, E, and F are used to represent these values in hexadecimal and are treated as valid numerals.

These digits are represented as follows:

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

0 1 2 3 4 5 6 7 8 9 A B C D E F

Convert 20710 to hex

16 207

16 12 R 15 -> F

0 R 12 -> C

$$10110110_2 = 20710_{10} = 12F_{16}$$

10

Convert $10F_{16}$ to decimal number

$$= 1 \times 16^2 + F \times 16^1 + 0 \times 16^0$$

$$= 256 + 10 \times 16$$

$$= 256 + 160$$

$$= 416$$

Exercise

a. Convert 1110011011_2 to Hexadecimal

b. Convert $12F_{16}$ to decimal.

Miscellaneous

Two's Complement

Two's complement is an alternative way of representing negative binary numbers. This alternative coding system also has the unique property that subtraction (or the addition of a negative number) can be performed using addition hardware. Architects of early computers were thus able to build arithmetic and logic units that performed operations of addition and subtraction using only adder hardware. (As it turns out since multiplication is just successive addition and division is just successive subtraction it was possible to use simple adder hardware to perform all of these operations.)

Example

Find the two's complement form of -6

Step 1

Decide how many bits you are going to use for all your operations. For our purposes we will always use 8 bits.

If we were using 8-bits the left-most bit will contain the sign. This would leave 7 bits to hold the number.

XXXX XXXX

This is the sign bit

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This sign bit is reserved and is no longer one of the digits that make up the binary number. Using two's complement, the computer recognizes the presence of a one (1) in the leftmost bit which tells the machine that before it does mathematics it needs to convert negative numbers into their two's complement equivalent.

0000 11102 the sign bit is 0 so the number is positive

The binary number is 7-digits long,

1000 01102 the sign bit is 1 so the number is negative

The binary number is only 7-digits long,

Step 2

Strip the sign bits off the numbers.

Step 3

Convert the negative number into it's two's complement form

So therefore

$-610 = 1000\ 01102$

Write down the number

without the sign bit $000\ 01102$

a) Flip all the digits

The 1 $\rightarrow$ 0, the 0 $\rightarrow$ 1 $111\ 10012$

b) Add 1 to this number + 1

c) This is now - 6 in the $111\ 10102$

two's complement format

Exercise

a) Find the 2s complement of -1410

b) Using the 2s complement form perform the following arithmetic operation

$1410 - 610$ and $610 - 1410$

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IOMoARcPSD|36554936

Curve Fitting

Chapter Outline

5.1 Introduction

5.2 Least Square Method

5.3 Fitting of Linear Curves

5.4 Fitting of Quadratic Curves

5.5 Fitting of Exponential and Logarithmic Curves

5.1 introduction

Curve fitting is the process of finding the 'best-fit' curve for a given set of data. It is the representation of the relationship between two variables by means of an algebraic equation. On the basis of this mathematical equation, predictions can be made in many statistical problems.

Suppose a set of n points of values $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of the two variables x and y are given. These values are plotted on a rectangular coordinate system, i.e., the xy -plane. The resulting set of points is known as a scatter diagram (Fig. 5.1).

The scatter diagram exhibits the trend and it is possible to visualize a smooth curve approximating the data. Such a curve is known as an approximating curve.

y y

o x o x

Fig. 5.1

Dg mi y

Sg S d b

th

. e

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5.2 Curved Fitting

5.2 LEAST SQUARE METHOD

From a scatter diagram, generally, more than one curve y may be seen to be appropriate to the given set of data. The

$P(x_i, y_i)$

method of least squares is used to find a curve which passes through the maximum number of points.

$Q(x_i, y)$

Let $P(x_i, y_i)$ be a point on the scatter diagram (Fig. 5.2).

Let the ordinate at P meet the curve $y = f(x)$ at Q and the o x

M

x-axis at M.

Fig. 5.2

Distance $QP = MP - MQ$

$= y_i - y$

$= y_i - f(x_i)$

The distance QP is known as deviation, error, or residual and is denoted by d_i . It may be positive, negative, or zero depending upon whether P lies above, below, or on the curve. Similar residuals or errors corresponding to the remaining $(n - 1)$ points may be obtained. The sum of squares of residuals, denoted by E, is given as

n

$$E = \sum_{i=1}^n d_i^2 = \sum_{i=1}^n [y_i - f(x_i)]^2$$

$i=1$

If $E = 0$ then all the n points will lie on $y = f(x)$. If $E \neq 0$, $f(x)$ is chosen such that E is minimum, i.e., the best fitting curve to the set of points is that for which E is minimum. This method is known as the least square method. This method does not attempt to determine the form of the curve $y = f(x)$ but it determines the values of the parameters of the equation of the curve.

5.3 FITTING OF LINEAR CURVES

Let (x_i, y_i) , $i = 1, 2, \dots, n$ be the set of n values and let the relation between x and y be $y = a + bx$. The constants a and b are selected such that the straight line is the best fit to the data.

The residual at $x = x_i$ is

$$d_i = y_i - f(x_i)$$

$$= y_i - (a + bx_i) \quad i = 1, 2, \dots, n$$

n

$$E = \sum_{i=1}^n d_i^2$$

$$i=1$$

n

$$= \sum_{i=1}^n (y_i - (a + bx_i))^2$$

2

$$i=1$$

n

$$= \sum_{i=1}^n (y_i - a - bx_i)^2$$

Differentiate

with respect to

a and

$$i=1$$

$\frac{\partial E}{\partial a}$

and

$\frac{\partial E}{\partial b}$

and set

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5.3 Fitting of Linear Curves 5.3

For E to be minimum,

$$(i) \frac{\partial E}{\partial a} = 0$$

$\frac{\partial E}{\partial a}$

n

$$= \sum_{i=1}^n 2(y_i - a - bx_i)(-1) = 0$$

$$i=1$$

n

$$= \sum_{i=1}^n (y_i - a - bx_i) = 0$$

$$i=1$$

n

$$\sum_{i=1}^n y_i = n\hat{a} + b\sum_{i=1}^n x_i$$

$$i=1 \quad i=1 \quad i=1$$

$$\hat{y} = n\hat{a} + b\hat{x}$$

∂E
 $(ii) = 0$
 ∂b
 n

$$\sum_{i=1}^n (y_i - a - bx_i)(-x_i) = 0$$

 $i = 1$
 n

$$\sum_{i=1}^n (x_i y_i - ax_i - bx_i^2) = 0$$

 $i = 1$
 $n \ n \ n$

$$\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2$$

 $i = 1 \ i = 1 \ i = 1$

$$\sum x_i y_i = a \sum x_i + b \sum x_i^2$$

These two equations are known as normal equations. These equations can be solved simultaneously to give the best values of a and b . The best fitting straight line is obtained by substituting the values of a and b in the equation $y = a + bx$.

Example 1

Fit a straight line to the following data:

x 1 2 3 4 6 8

y 2.4 3 3.6 4 5 6

Solution

Let the straight line to be fitted to the data be

$$y = a + bx$$

The normal equations are

$$\sum y = na + b \sum x \dots(1)$$

$$\sum xy = a \sum x + b \sum x^2$$

Dg mi y

Sg S d b

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5.4 Chapter 5 Curved Fitting

Here, $n = 6$

$x \ y \ x^2 \ xy$

1 2.4 1 2.4

2 3 4 6

3 3.6 9 10.8

4 4 16 16

6 5 36 30

8 6 64 48

$$\hat{A}x = 24 \quad \hat{A}y = 24 \quad \hat{A}x^2 = 130 \quad \hat{A}xy = 113.2$$

Substituting these values in Eqs (1) and (2),

$$24 = 6a + 24b \dots(3)$$

$$113.2 = 24a + 130b \dots(4)$$

Solving Eqs (3) and (4),

$$a = 1.9764$$

$$b = 0.5059$$

Hence, the required equation of the straight line is

$$y = 1.9764 + 0.5059x$$

Example 2

Fit a straight line to the following data. Also, estimate the value of y at

x = 2.5.

x 0 1 2 3 4

y 1 1.8 3.3 4.5 6.3

Solution

Let the straight line to be fitted to the data be

$$y = a + bx$$

The normal equations are

$$\hat{A}y = n\hat{A}a + b\hat{A}x \dots(1)$$

$$\hat{A}xy = a\hat{A}x + b\hat{A}x^2 \dots(2)$$

Here, n = 5

Dg mi y

Sg S d b

th

. e

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5.3 Fitting of Linear Curves 5.5

x y x² xy

0 1 0 0

1 1.8 1 1.8

2 3.3 4 6.6

3 4.5 9 13.5

4 6.3 16 25.2

$\hat{A}x = 10$ $\hat{A}y = 16.9$ $\hat{A}x^2 = 30$ $\hat{A}xy = 47.1$

Substituting these values in Eqs (1) and (2),

$16.9 = 5a + 10b \dots(3)$

$47.1 = 10a + 30b \dots(4)$

Solving Eqs (3) and (4),

$a = 0.72$

$b = 1.33$

Hence, the required equation of the straight line is

$y = 0.72 + 1.33x$

At $x = 2.5$,

$y(2.5) = 0.72 + 1.33(2.5) = 4.045$

Example 3

A simply supported beam carries a concentrated load $P(\text{lb})$ at its midpoint. Corresponding to various values of P , the maximum deflection $Y(\text{in})$ is measured. The data is given below:

P 100 120 140 160 180 200

Y 0.45 0.55 0.60 0.70 0.80 0.85

Find a law of the form $Y = a + bP$ using the least square method.

Solution

Let the straight line to be fitted to the data be

$Y = a + bP$

The normal equations are

$\hat{A}Y = na + b\hat{A}P$

Dg mi y

...(1)

Sg S d b

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5.6 Chapter 5 Curved Fitting

$$\hat{Y} = a + bP^2 \dots(2)$$

Here, $n = 6$

$\sum Y = 1000$

$\sum P = 900$

$\sum P^2 = 10000$

$\sum YP = 14400$

$\sum P^3 = 19600$

$\sum YP^2 = 25600$

$\sum YP^3 = 32400$

$\sum YP^4 = 40000$

$\sum YP^5 = 170$

$\sum YP^6 = 621$

Substituting these values in Eqs (1) and (2),

$$3.95 = 6a + 900b \dots(3)$$

$$621 = 900a + 142000b \dots(4)$$

Solving Eqs (3) and (4),

$$a = 0.0476$$

$$b = 0.0041$$

Hence, the required equation of the straight line is

$$Y = 0.0476 + 0.0041P$$

Example 4

Fit a straight line to the following data. Also, estimate the value of y at

$x = 70$.

x 71 68 73 69 67 65 66 67

y 69 72 70 70 68 67 68 64

Solution

Since the values of x and y are larger, we choose the origin for x and y at 69 and 67 respectively,

Let $X = x - 69$ and $Y = y - 67$

Let the straight line to be fitted to the data be

$$Y = a + bX$$

The normal equations are

$$\sum Y = na + b\sum X \dots(1)$$

$$\sum XY = a\sum X + b\sum X^2 \dots(2)$$

Dg mi y

Sg S d b

th

. e

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5.3 Fitting of Linear Curves 5.7

Here, $n = 8$

x y X Y X² XY

71 69 2 2 4 4

68 72 - 1 5 1 - 5

73 70 4 3 16 12

69 70 0 3 0 0

67 68 - 2 1 4 - 2

65 67 - 4 0 16 0

66 68 - 3 1 9 - 3

67 64 - 2 3

- 4 6

$\hat{A}X = -6 \hat{A}Y = 12 \hat{A}X = 54$

2

$\hat{A}XY = 12$

Substituting these values in Eqs (1) and (2),

$12 = 8a - 6b \dots(3)$

$12 = -6a + 54b \dots(4)$

Solving Eqs (3) and (4),

$a = 1.8182$

$b = 0.4242$

Hence, the required equation of the straight line is

$Y = 1.8182 + 0.4242 X$

$y - 67 = 1.8182 + 0.4242(x - 69)$

$y = 0.4242x + 39.5484$

$y(x = 70) = 0.4242(70) + 39.5484 = 69.2424$

Example 5

Fit a straight line to the following data taking x as the dependent variable.

x 1 3 4 6 8 9 11 14

y 1 2 4 4 5 7 8 9

Solution

If x is considered the dependent variable and y the independent variable, the equation of the straight line to be fitted to the data is

$x = a + by$

Dg mi y

Sg S d b

th

. e

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5.8 Curved Fitting

The normal equations are

$$\hat{A}x = na + b\hat{A}y \dots(1)$$

$$\hat{A}xy = a\hat{A}y + b\hat{A}y^2 \dots(2)$$

Here, $n = 8$

$x \ y \ y^2 \ xy$

1 1 1 1

3 2 4 6

4 4 16 16

6 4 16 24

8 5 25 40

9 7 49 63

11 8 64 88

14 9 81 126

$$\hat{A}x = 56 \ \hat{A}y = 40 \ \hat{A}y^2 = 256$$

2

$$\hat{A}xy = 364$$

Substituting these values in Eqs (1) and (2),

$$56 = 8a + 40b \dots(3)$$

$$364 = 40a + 256b \dots(4)$$

Solving Eqs (3) and (4),

$$a = -0.5$$

$$b = 1.5$$

Hence, the required equation of the straight line is

$$x = -0.5 + 1.5y$$

Example 6

If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form $P = mW + c$ connecting P and W using the following data:

$P \ 12 \ 15 \ 21 \ 25$

$W \ 50 \ 70 \ 100 \ 120$

where P and W are taken in kg-wt. Compute P when $W = 150$ kg.

Solution

Let the linear curve (straight line) fitted to the data be

$Dg \text{ mi } y$

$Sg \text{ S d b}$

th

$$P = mW + c = c + mW$$

. e

gr il

En mp

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5.3 Fitting of Linear Curves 5.9

The normal equations are

$$\hat{A}P = nc + mW \dots(1)$$

$$\hat{A}PW = c\hat{A}W + m\hat{A}W^2 \dots(2)$$

Here, $n = 4$

$P \ W \ W^2 \ PW$

12 50 2500 600

15 70 4900 1050

21 100 10000 2100

25 120 14400 3000

$$\hat{A}P = 73 \ \hat{A}W = 340 \ \hat{A}W^2 = 31800 \ 2$$

$$\hat{A}PW = 6750$$

Substituting these values in Eqs (1) and (2),

$$73 = 4c + 340m \dots(3)$$

$$6750 = 340c + 31800m \dots(4)$$

Solving Eqs (3) and (4),

$$c = 2.2759$$

$$m = 0.1879$$

Hence, the required equation of the straight line is

$$P = 0.1879W + 2.2759$$

When $W = 150$ kg,

$$P = 0.1879(150) + 2.2759 = 30.4609$$

ExERCiSE 5.1

1. Fit the line of best fit to the following data:

$x \ 0 \ 5 \ 10 \ 15 \ 20 \ 25$

$y \ 12 \ 15 \ 17 \ 22 \ 24 \ 30$

Ans. : $y = 0.7x + 11.28$

2. The results of a measurement of electric resistance R of a copper bar at various temperatures $t^{\circ}\text{C}$ are listed below:

$t^{\circ}\text{C}$ 19 25 30 36 40 45 50

R 76 77 79 80 82 83 85

Find a relation $R = a + bt$ where a and b are constants to be determined.

Ans. : $R = 70.0534 + 0.2924 t^{\circ}$

Dg mi y

Sg S d b

th

. e

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5.10 Chapter 5 Curved Fitting

3. Fit a straight line to the following data:

x 1.53 1.78 2.60 2.95 3.42

y 33.50 36.30 40.00 45.85 53.40

Ans. : $y = 19 + 9.7 x^{\circ}$

4. Fit a straight line to the following data:

x 100 120 140 160 180 200

y 0.45 0.55 0.60 0.70 0.80 0.85

Ans. : $y = 0.0475 + 0.00407 x^{\circ}$

5. Find the relation of the type $R = aV + b$, when some values of R and V obtained from an experiment are

V 60 65 70 75 80 85 90

R 109 114 118 123 127 130 133

Ans. : $R = 0.8071V + 61.4675^{\circ}$

5.4 FITTING OF QUADRATIC CURVES

Let (x_i, y_i) , $i = 1, 2, \dots, n$ be the set of n values and let the relation between x and y be $y = a + bx + cx^2$. The constants a , b , and c are selected such that the parabola is the best fit to the data. The residual at $x = x_i$ is

$$d_i = y_i - f(x_i)$$

$$= y_i - a - b x_i - c x_i^2$$

n

$$E = \sum d_i^2$$

$$i = 1$$

$$2$$

$$()$$

$$n$$

$$= \hat{E} \sum_{i=1}^n y_i - a + b x_i + c x_i^2$$

$$i = 1$$

$$\hat{I}^\circ$$

$$2$$

$$()$$

$$n$$

$$= \hat{A} y_i - a - b x_i - c x_i^2$$

$$i = 1$$

For E to be minimum,

$$\partial E$$

$$(i) = 0$$

$$\partial a$$

$$Dg_{mi} y$$

$$n$$

$$Sg \ S \ d \ b$$

$$th$$

$$\hat{A}^2 (y_i - a - b x_i - c x_i^2) (-1) = 0$$

$$. e$$

$$gr \ il$$

$$i = 1$$

$$En \ mp$$

$$Co$$

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5.4 Fitting of Quadratic Curves 5.11

$$n$$

$$\hat{A} (y_i - a - b x_i - c x_i^2) = 0$$

$$i = 1$$

$$n \ n \ n \ n$$

$$\hat{A} y_i = a \hat{A} + b \hat{A} x_i + c \hat{A} x_i^2$$

$$i = 1 \ i = 1 \ i = 1 \ i = 1$$

$$\hat{A} y_i = n a + b \hat{A} x + c \hat{A} x^2$$

$$(ii) \partial E = 0$$

$$\partial b$$

$$n$$

$$\sum_{i=1}^n (y_i - a - bx_i - cx_i^2)(-x_i) = 0$$

$$\sum_{i=1}^n (x_i y_i - ax_i - bx_i^2 - cx_i^3) = 0$$

$$\sum_{i=1}^n (x_i^2 y_i - ax_i^2 - bx_i^3 - cx_i^4) = 0$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

$$\sum_{i=1}^n x_i^2 y_i = a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4$$

These equations are known as normal equations. These equations can be solved simultaneously to give the best values of a, b, and c. The best fitting parabola is obtained by substituting the values of a, b, and c in the equation $y = a + bx + cx^2$.

Example 1

Fit a least squares quadratic curve to the following data:

x 1 2 3 4

y 1.7 1.8 2.3 3.2

Dg mi y

Estimate y(2.4).

Sg S d b

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5.12 Chapter 5 Curved Fitting

Solution

Let the equation of the least squares quadratic curve (parabola) be $y = a + bx + cx^2$.

The normal equations are

$$\hat{y} = na + b\hat{x} + c\hat{x}^2 \dots(1)$$

$$\hat{xy} = a\hat{x} + b\hat{x}^2 + c\hat{x}^3 \dots(2)$$

$$\hat{x}^2 y = a\hat{x}^2 + b\hat{x}^3 + c\hat{x}^4 \dots(3)$$

Here, $n = 4$

$$\begin{array}{ccccccc} x & y & x^2 & x^3 & x^4 & xy & x^2y \\ 1 & 1.7 & 1 & 1 & 1 & 1.7 & 1.7 \\ 2 & 1.8 & 4 & 8 & 16 & 3.6 & 7.2 \\ 3 & 2.3 & 9 & 27 & 81 & 6.9 & 20.7 \\ 4 & 3.2 & 16 & 64 & 256 & 12.8 & 51.2 \end{array}$$

$$S_x = 10 \quad S_y = 9.2$$

$$S_x = 30.3$$

$$S_x = 100$$

$$4$$

$$S_x = 354 \quad S_{xy} = 25.2$$

$$S_{xy} = 80.8$$

$$4$$

$$S_x = 354 \quad S_{xy} = 25.2$$

$$S_{xy} = 80.8$$

Substituting these values in Eqs (1), (2), and (3),

$$9 = 4a + 10b + 30c \dots(4)$$

$$25 = 10a + 30b + 100c \dots(5)$$

$$80.8 = 30a + 100b + 354c \dots(6)$$

Solving Eqs (4), (5), and (6),

$$a = 2$$

$$b = -0.5$$

$$c = 0.2$$

Hence, the required equation of least squares quadratic curve is

$$y = 2 - 0.5x + 0.2x^2$$

$$y(2.4) = 2 - 0.5(2.4) + 0.2(2.4)^2 = 1.952$$

Example 2

Fit a second-degree polynomial using least square method to the following data:

$$\begin{array}{ccccccc} x & 0 & 1 & 2 & 3 & 4 \\ y & 1 & 1.8 & 1.3 & 2.5 & 6.3 \end{array}$$

$$Dg \quad mi \quad y$$

$$Sg \quad S \quad d \quad b$$

$$th$$

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5.4 Fitting of Quadratic Curves 5.13

Solution

Let the equation of the least squares quadratic curve be $y = a + bx + cx^2$. The normal equations are

$$\hat{y} = na + b\hat{x} + c\hat{x}^2 \dots(1)$$

$$\hat{xy} = a\hat{x} + b\hat{x}^2 + c\hat{x}^3 \dots(2)$$

$$\hat{x^2y} = a\hat{x}^2 + b\hat{x}^3 + c\hat{x}^4 \dots(3)$$

Here, $n = 5$

$x \ y \ x^2 \ x^3 \ x^4 \ xy \ x^2y$

0 1 0 0 0 0 0

1 1.8 1 1 1 1.8 1.8

2 1.3 4 8 16 2.6 5.2

3 2.5 9 27 81 7.5 22.5

4 6.3 16 64 256 25.2 100.8

$\hat{x} = 10 \ \hat{y} = 12.9 \ \hat{x} = 30$

2

$\hat{x} = 100 \ \hat{x} = 354 \ \hat{xy} = 37.1 \ \hat{x}^2y = 130.3$

3 4 2

Substituting these values in Eqs (1), (2), and (3),

$$12.9 = 5a + 10b + 30c \dots(4)$$

$$37.1 = 10a + 30b + 100c \dots(5)$$

$$130.3 = 30a + 100b + 354c \dots(6)$$

Solving Eqs (4), (5), and (6),

$$a = 1.42$$

$$b = -1.07$$

$$c = 0.55$$

Hence, the required equation of the least squares quadratic curve is

$$y = 1.42 - 1.07x + 0.55x^2$$

Example 3

By the method of least squares, fit a parabola to the following data:

$x \ 1 \ 2 \ 3 \ 4 \ 5$

$y \ 5 \ 12 \ 26 \ 60 \ 97$

Also, estimate y at $x = 6$.

Solution

Let the equation of the parabola be $y = a + bx + cx^2$. The normal equations are

Dg mi y

Sg S d b

th

$$\hat{y} = na + b\hat{x} + c\hat{x}^2 \dots(1)$$

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5.14 Curved Fitting

$$\hat{xy} = a\hat{x} + b\hat{x}^2 + c\hat{x}^3 \dots(2)$$

$$\hat{x}^2y = a\hat{x}^2 + b\hat{x}^3 + c\hat{x}^4 \dots(3)$$

Here, $n = 5$

x y x² x³ x⁴ xy x²y

1 5 1 1 1 5 5

2 12 4 8 16 24 48

3 26 9 27 81 78 234

4 60 16 64 256 240 960

5 97 25 125 625 485 2425

$$\hat{x} = 15 \quad \hat{y} = 200 \quad \hat{x} = 55$$

2

$$\hat{x} = 225 \quad \hat{x} = 979 \quad \hat{xy} = 832$$

3 4

$$\hat{x}y = 3672$$

2

Substituting these values in Eqs (1), (2), and (3),

$$200 = 5a + 15b + 55c \dots(4)$$

$$832 = 15a + 55b + 225c \dots(5)$$

$$3672 = 55a + 225b + 979c \dots(6)$$

Solving Eqs (4), (5), and (6),

$$a = 10.4$$

$$b = -11.0857$$

$$c = 5.7143$$

Hence, the required equation of the parabola is

$$y = 10.4 - 11.0857x + 5.7143x^2$$

$$y(6) = 10.4 - 11.0857(6) + 5.7143(6)^2 = 149.6006$$

Example 4

Fit a second-degree parabolic curve to the following data.

x 1 2 3 4 5 6 7 8 9

y 2 6 7 8 10 11 11 10 9

Solution

Let $X = x - 5$

$Y = y - 10$

Let the equation of the parabola be $Y = a + bX + cX^2$.

The normal equations are

$$\hat{Y} = na + b\hat{X} + c\hat{X}^2$$

Dg mi y

Sg S d b

th

...(1)

. e

gr il

En mp

Co

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5.4 Fitting of Quadratic Curves 5.15

$$\hat{X} = a \hat{X} + b\hat{X}^2 + c \hat{X}^3$$

Y ...(2)

$$\hat{X}^2 Y = a \hat{X}^2 + b \hat{X}^3 + c \hat{X}^4 \dots (3)$$

Here, $n = 9$

x y X Y X² X³ X⁴ XY X²Y

1 2 -4 -8 16 -64 256 32 -128

2 6 -3 -4 9 -27 81 12 -36

3 7 -2 -3 4 -8 16 6 -12

4 8 -1 -2 1 -1 1 2 -2

5 10 0 0 0 0 0 0

6 11 1 1 1 1 1 1

7 11 2 1 4 8 16 2 4

8 10 3 0 9 27 81 0 0

9 9 4 -1 16 64 256 -4 -16

$$SX = 0 \quad SY = -16 \quad SX^2 = 60 \quad SX^3 = 0 \quad SX^4 = 708 \quad SXY = 51 \quad SX^2Y = -189$$

Substituting these values in Eqs (1), (2), and (3),

$$-16 = 9a + 60c \dots (4)$$

$$51 = 60b \dots (5)$$

$$-189 = 60a + 708c \dots (6)$$

Solving Eqs (4), (5), and (6),

$$a = 0.0043$$

$$b = 0.85$$

$$c = -0.2673$$

Hence, the required equation of the parabola is

$$Y = 0.0043 + 0.85X - 0.2673X^2$$

$$y - 10 = 0.0043 + 0.85(x - 5) - 0.2673(x - 5)^2$$

$$y = 10 + 0.0043 + 0.85(x - 5) - 0.2673(x^2 - 10x + 25)$$

$$= 10 + 0.0043 + 0.85x - 4.25 - 0.2673x^2 + 2.673x - 6.6825$$

$$= -0.9282 + 3.523x - 0.2673x^2$$

Example 5

Fit a second-degree parabola $y = a + bx^2$ to the following data:

x 1 2 3 4 5

Dg mi y

Sg S d b

1.8 5.1 8.9 14.1 19.8

th

y

. e

gr il

En mp

Co

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5.16 Curved Fitting

Solution

Let the curve to be fitted to the data be

$$y = a + bx^2$$

The normal equations are

$$\hat{A}y = na + b\hat{A}x^2 \dots(1)$$

$$\hat{A}x^2y = a\hat{A}x^2 + b\hat{A}x^4 \dots(2)$$

Here, $n = 5$

x y x² x⁴ x²y

1 1.8 1 1 1.8

2 5.1 4 16 20.4

3 8.9 9 81 80.1

4 14.1 16 256 225.6

5 19.8 25 625 495

$$\hat{A}y = 49.7 \quad \hat{A}x = 55$$

2

$$\hat{A}x = 979$$

4

$$\hat{A}x y = 822.9$$

2

Substituting these values in Eqs (1) and (2),

$$49.7 = 5a + 55b \dots(3)$$

$$822.9 = 55a + 979b \dots(4)$$

Solving Eqs (3) and (4),

$$a = 1.8165$$

$$b = 0.7385$$

Hence, the required equation of the curve is

$$y = 1.8165 + 0.7385x^2$$

Example 6

Fit a curve $y = ax + bx^2$ for the following data:

x 1 2 3 4 5 6

y 2.51 5.82 9.93 14.84 20.55 27.06

Solution

Let the curve to be fitted to the data be

$$y = ax + bx^2$$

The normal equations are

$$\sum y = \sum ax + \sum bx^2$$

$$\sum xy = \sum ax^2 + \sum bx^3$$

that is

$$\sum xy = a \sum x^2 + b \sum x^3$$

or

$$\dots(1)$$

Similarly

$$\sum y = a \sum x + b \sum x^2 \dots(2)$$

or

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5.4 Fitting of Quadratic Curves 5.17

$$\hat{A}x^2 y = a \hat{A}x^3 + b \hat{A}x^4 \dots(2)$$

$$\sum y \sum x^2 \sum x^3 \sum x^4 \sum xy \sum x^2 y$$

$$1 \ 2.51 \ 1 \ 1 \ 1 \ 2.51 \ 2.51$$

$$2 \ 5.82 \ 4 \ 8 \ 16 \ 11.64 \ 23.28$$

$$3 \ 9.93 \ 9 \ 27 \ 81 \ 29.79 \ 89.37$$

$$4 \ 14.84 \ 16 \ 64 \ 256 \ 59.36 \ 237.44$$

$$5 \ 20.55 \ 25 \ 125 \ 625 \ 102.75 \ 513.75$$

$$6 \ 27.06 \ 36 \ 216 \ 1296 \ 162.36 \ 974.16$$

$$S_x 2 = 91 \ S_x 3 = 441 \ S_x 4 = 2275 \ S_{xy} = 368.41 \ S_x 2 y = 1840.51$$

Substituting these values in Eqs (1) and (2),

$$368 \diamond 41 = 91a + 441b \dots(3)$$

$$1840 \diamond 51 = 441 a + 2275 b \dots(4)$$

Solving Eqs (3) and (4),

$$a = 2.11$$

$$b = 0.4$$

Hence, the required equation of the curve is

$$y = 2 \diamond 11x + 0 \diamond 4 x^2$$

ExERCiSE 5.2

1. Fit a parabola to the following data:

$$x \ -2 \ -1 \ 0 \ 1 \ 2$$

$$y \ 1.0 \ 1.8 \ 1.3 \ 2.5 \ 6.3$$

$$[\text{Ans. : } y = 1.48 + 1.13x + 0.55x^2]$$

2

2. Fit a curve $y = ax + bx^2$ to the following data:

$$x \ -2 \ -1 \ 0 \ 1 \ 2$$

$$y \ -72 \ -46 \ -12 \ 35 \ 93$$

$$[\text{Ans. : } y = 41.1x + 2.147 x^2]$$

Dg mi y

Sg S d b

th

. e

gr il

En mp

Co

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5.18 Curved Fitting

2

3. Fit a parabola $y = a + bx + cx^2$ to the following data:

$$x \ 0 \ 2 \ 5 \ 10$$

$$y \ 4 \ 7 \ 6.4 \ -6$$

$$[\text{Ans. : } y = 4.1 + 1.979x - 0.299x^2]$$

2

4. Fit a curve $y = a + ax + a x^2$

0 1 2

for the given data:

x 3 5 7 9 11 13

y 2 3 4 6 5 8

[Ans. : $y = 0.7897 + 0.4004x + 0.0089x^2$]

5.5 FITTING OF EXPONENTIAL AND LOGARITHMIC CURVES

Let (x_i, y_i) , $i = 1, 2, \dots, n$ be the set of n values and let the relation between x and y be $y = ab^x$.

Taking logarithm on both the sides of the equation $y = ab^x$,

$$\log_e y = \log_e a + x \log_e b$$

Putting $\log_e y = Y$, $\log_e a = A$, $x = X$, and $\log_e b = B$,

$$Y = A + BX$$

This is a linear equation in X and Y . The normal equations are

$$\sum Y = nA + B \sum X$$

$$\sum XY = A \sum X + B \sum X^2$$

$$\sum XY = A \sum X + B \sum X^2$$

Solving these equations, A and B , and, hence, a and b can be found. The best fitting exponential curve is obtained by substituting the values of a and b in the equation $y = ab^x$.

Similarly, the best fitting exponential curves for the relation $y = axb$ and $y = aeb^x$ can be obtained.

Example 1

Find the law of the form $y = ab^x$ to the following data:

x 1 2 3 4 5 6 7 8

y 1 1.2 1.8 2.5 3.6 4.7 6.6 9.1

Dg mi y

Sg S d b

th

. e

gr il

En mp

Co

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5.5 Fitting of Exponential and Logarithmic Curves 5.19

Solution

$$y = ab^x$$

Taking logarithm on both the sides,

$$\log_e y = \log_e a + x \log_e b$$

Putting $\log_e y = Y$, $\log_e a = A$, $x = X$ and $\log_e b = B$,

$$Y = A + BX$$

The normal equations are

$$\hat{A} Y = nA + B \hat{A} X \dots(1)$$

$$\hat{A} XY = A \hat{A} X + B \hat{A} X^2 \dots(2)$$

Here, $n = 8$

$x \ y \ X \ Y \ X^2 \ XY$

1 1 1 0.0000 1 0.0000

2 1.2 2 0.1823 4 0.3646

3 1.8 3 0.5878 9 1.7634

4 2.5 4 0.9163 16 3.6652

5 3.6 5 1.2809 25 6.4045

6 4.7 6 1.5476 36 9.2856

7 6.6 7 1.8871 49 13.2097

8 9.1 8 2.2083 64 17.6664

$$\hat{A} X = 36 \ \hat{A} Y = 8.6103 \ \hat{A} X^2 = 204 \ \hat{A} XY = 52.3594$$

Substituting these values in Eqs (1) and (2),

$$8.6103 = 8A + 36B \dots(3)$$

$$52.3594 = 36A + 204B \dots(4)$$

Solving Eqs (3) and (4),

$$A = -0.3823$$

$$B = 0.3241$$

$$\log_e a = A$$

$$\log_e a = -0.3823$$

$$a = 0.6823$$

$$\log_e b = B$$

$$\log_e b = 0.3241$$

$$\text{Dg mi y}$$

$$\text{Sg S d b}$$

$$b = 1.3828$$

th

. e

gr il

En mp

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5.20 Curved Fitting

Hence, the required law is

$$y = 0.6823 (1.3828)^x$$

Example 2

Fit a curve of the form $y = abx$ to the following data by the method of least squares:

x 1 2 3 4 5 6 7

y 87 97 113 129 202 195 193

Solution

$$y = abx$$

Taking logarithm on both the sides,

$$\log y = \log a + x \log b$$

Putting $\log y = Y$, $\log a = A$, $x = X$ and $\log b = B$,

$$Y = A + BX$$

The normal equations are

$$\hat{A}Y = nA + B\hat{A}X \dots(1)$$

$$\hat{A}XY = A\hat{A}X + B\hat{A}X^2 \dots(2)$$

Here, $n = 7$

x y X Y X² XY

1 87 1 4.4659 1 4.4659

2 97 2 4.5747 4 9.1494

3 113 3 4.7274 9 14.1822

4 129 4 4.8598 16 19.4392

5 202 5 5.3083 25 26.5415

6 195 6 5.2730 36 31.6380

7 193 7 5.2627 49 36.8389

$$\hat{A}X = 28 \quad \hat{A}Y = 34.4718 \quad \hat{A}X^2 = 140 \quad 2$$

$$\hat{A}XY = 142.2551$$

Substituting these values in Eqs (1) and (2),

$$34.4718 = 7A + 28B \dots(3)$$

$$142.2551 = 28A + 140B \dots(4)$$

Solving Eqs (3) and (4),

$$A = 4.3006$$

Dg mi y

Sg S d b

th

$$B = 0.156$$

. e

gr il

En mp

Co

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5.5 Fitting of Exponential and Logarithmic Curves 5.21

$$\log_e a = A$$

$$\log_e a = 4.3006$$

$$a = 73.744$$

$$\log_e b = B$$

$$\log_e b = 0.156$$

$$b = 1.1688$$

Hence, the required curve is

$$y = 73.744 (1.1688)^x$$

Example 3

Fit a curve of the form $y = ax^b$ to the following data:

x 20 16 10 11 14

y 22 41 120 89 56

Solution

$$y = ax^b$$

Taking logarithm on both the sides,

$$\log_e y = \log_e a + b \log_e x$$

Putting $\log_e y = Y$, $\log_e a = A$, $b = B$ and $\log_e x = X$,

$$Y = A + BX$$

The normal equations are

$$\hat{A} Y = nA + B \hat{X} \dots(1)$$

$$\hat{A} XY = A \hat{X} + B \hat{X}^2 \dots(2)$$

Here, $n = 5$

x y X Y X² XY

20 22 2.9957 3.0910 8.9742 9.2597

16 41 2.7726 3.7136 7.6873 10.2963

10 120 2.3026 4.7875 5.3019 11.0237

11 89 2.3979 4.4886 5.7499 10.7632

14 56 2.6391 4.0254 6.9648 10.6234

$$\hat{A} X = 13.1079 \quad \hat{A} = 20.1061$$

Y

$$\hat{A} X = 34.6781$$

2

$$\hat{A} XY = 51.9663$$

Substituting these values in Eqs (1) and (2),

$$20.1061 = 5A + 13.1079 B$$

Dg mi y

...(3)

Sg S d b

th

$$51.9663 = 13.1079 A + 34.6781 B \dots(4)$$

. e

gr il

En mp

Co

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5.22 Curved Fitting

Solving Eqs (3) and (4),

$$A = 10.2146$$

$$B = -2.3624$$

$$\log_e a = A$$

$$\log_e a = 10.2146$$

$$a = 27298 \diamond 8539$$

$$\text{and } b = B = -2.3624$$

Hence, the required equation of the curve is

$$y = 27298.8539 x^{-2.3624}$$

Example 4

Fit a curve of the form $y = ae^{bx}$ to the following data:

x 1 3 5 7 9

y 115 105 95 85 80

Solution

$$y = ae^{bx}$$

Taking logarithm on both the sides,

$$\log_e y = \log_e a + bx \log_e e$$

$$= \log_e a + bx$$

Putting $\log_e y = Y$, $\log_e a = A$, $b = B$ and $x = X$,

$$Y = A + BX$$

The normal equations are

$$\hat{A} = nA + B\hat{A} X$$

Y

...(1)

$$\hat{A} X = A\hat{A} X + B\hat{A} X^2$$

Y

...(2)

Here, $n = 5$

x y X Y X² XY

1 115 1 4.7449 1 4.7449

3 105 3 4.6539 9 13.9617

5 95 5 4.5539 25 22.7695

7 85 7 4.4427 49 31.0989

9 80 9 4.3820 81 39.438

$\hat{A} X = 25$ $\hat{A} Y = 22.7774$ $\hat{A} X^2 = 165$ $\hat{A} XY = 112.013$

Dg mi y

Sg S d b

th

. e

gr il

En mp

Co

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5.5 Fitting of Exponential and Logarithmic Curves 5.23

Substituting these values in Eqs (1) and (2),

$22.7774 = 5 A + 25 B \dots(3)$

$112.013 = 25A + 165B \dots(4)$

Solving Eqs (3) and (4),

$A = 4.7897$

$B = - 0.0469$

$\log_e a = A$

$\log_e a = 4.7897$

$a = 120.2653$

$b = B = - 0.0469$

and

Hence, the required equation of the curve is

$y = 120.2653 e^{-0.0469 x}$

Example 5

bx

Fit the exponential curve $y = ae^{bx}$ to the following data:

x 0 2 4 6 8

y 150 63 28 12 5.6

Solution

$y = ae^{bx}$

Taking logarithm on both the sides,

$\log_e y = \log_e a + bx \log_e e$

$= \log_e a + bx$

Putting $\log_e y = Y$, $\log_e a = A$, $b = B$ and $x = X$,

$Y = A + BX$

The normal equations are

$$\hat{Y} = nA + b\hat{X} \dots(1)$$

$$\hat{XY} = A\hat{X} + B\hat{X}^2 \dots(2)$$

Dg mi y

Sg S d b

th

. e

gr il

En mp

Co

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5.24 Curved Fitting

Here, $n = 5$

x y X Y X² XY

0 150 0 5.0106 0 0

2 63 2 4.1431 4 8.2862

4 28 4 3.3322 16 13.3288

6 12 6 2.4849 36 14.9094

8 5.6 8 1.7228 64 13.7824

$$\hat{X} = 20 \quad \hat{Y} = 16.6936 \quad \hat{X}^2 = 120 \quad \hat{XY} = 50.3068$$

Substituting these values in Eqs (1) and (2),

$$16.6936 = 5A + 20B \dots(3)$$

$$50.3068 = 20A + 120B \dots(4)$$

Solving Eqs (3) and (4),

$$A = 4.9855$$

$$B = -0.4117$$

$$\log_e a = A$$

$$\log_e a = 4.9855$$

$$a = 146.28$$

$$\text{and } b = B = -0.4117$$

Hence, the required equation of the curve is

$$y = 146.28 e^{-0.4117 x}$$

Example 6

The pressure and volume of a gas are related by the equation $PV^g = c$.

Fit this curve to the following data:

P 0.5 1.0 1.5 2.0 2.5 3.0

V 1.62 1.00 0.75 0.62 0.52 0.46

Solution

$$PV^g = c$$

Taking logarithm on both the sides,

$$\log_e P + g \log_e V = \log_e c$$

$$\frac{1}{g} \log_e V = \log_e c - \log_e P$$

$$\log_e V = \log_e c - \log_e P$$

$$\frac{1}{g} \log_e V = \log_e c - \log_e P$$

$$\log_e V = \log_e c - \log_e P$$

Putting $\log_e V = y$, $\log_e c = a$, $\log_e P = x$, $- = b$,

$$Dg \text{ mi } y$$

$$g \text{ g}$$

$$Sg \text{ S d b}$$

$$th$$

$$. e$$

$$y = a + bx$$

$$gr \text{ il}$$

$$En \text{ mp}$$

$$Co$$

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5.5 Fitting of Exponential and Logarithmic Curves 5.25

The normal equations are

$$\hat{A}y = na + b\hat{A}x$$

$$\hat{A}xy = a\hat{A}x + b\hat{A}x^2$$

Here, $n = 6$

$$P \text{ V x y } x^2 \text{ xy}$$

$$0.5 \ 1.62 \ -0.6931 \ 0.4824 \ 0.4804 \ -0.3343$$

$$1.0 \ 1.00 \ 0 \ 0 \ 0 \ 0$$

$$1.5 \ 0.75 \ 0.4055 \ -0.2877 \ 0.1644 \ -0.1166$$

$$2.0 \ 0.62 \ 0.6931 \ -0.4780 \ 0.4804 \ -0.3313$$

$$2.5 \ 0.52 \ 0.9163 \ -0.6539 \ 0.8396 \ -0.5992$$

$$3.0 \ 0.46 \ 1.0986 \ -0.7765 \ 1.2069 \ -0.8531$$

$$\hat{A}x = 2.4204 \ \hat{A}y = -1.7137 \ \hat{A}x = 3.1717$$

$$2$$

$$\hat{A}xy = -2.2345$$

Substituting these values in Eqs (1) and (2),

$$-1.7137 = 6a + 2.4204 b \dots(3)$$

$$-2.2345 = 2.4204a + 3.1717 b \dots(4)$$

Solving Eqs (3) and (4),

$$a = -0.002$$

$$b = -0.7029$$

1

$$- = b$$

g

$$g = 1.4227$$

1

$$\log_e c = a$$

g

1

$$\log_e c = -0.002$$

$$1.4227$$

$$c = 0.9972$$

Hence, the required equation of the curve is

$$PV(1.4227) = 0.9972$$

ExERCiSE 5.3

1. Fit the curve $y = ab^x$

to the following data:

x 2 3 4 5 6

y 144 172.3 207.4 248.8 298.5

Dg mi y

Sg S d b

th

x

[Ans.: $y = 100(1.2)^x$]

. e

gr il

En mp

Co

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5.26 Curved Fitting

2. Fit the curve $y = ae^{bx}$

to the following data:

x 0 2 4

y 5.012 10 31.62

$$0.46x$$

[Ans.: $y = 4.642e^{0.46x}$]

3. Fit the curve $y = ax^b$

to the following data:

x 1 2 3 4

y 2.50 8.00 19.00 50.00

2.09

[Ans.: $y = 2.227x$]

g

4. Estimate g by fitting the ideal gas law $PV = c$ to the following data:

P 16.6 39.7 78.5 115.5 195.3 546.1

V 50 30 20 15 10 5

[Ans.: $g = 1.504$]

points to Remember

Fitting of Linear Curves

(i) The normal equations for the straight line $y = a + bx$ are

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

(ii) The normal equations for the straight line $x = a + by$ are

$$\sum x = na + b \sum y$$

$$\sum xy = a \sum y + b \sum y^2$$

Fitting of Quadratic Curves

(i) The normal equations for the least squares quadratic curve (parabola)

$y = a + bx + cx^2$ are

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

(ii) The normal equations for the curve $y = a + bx^2$ are

$$\sum y = na + b \sum x^2$$

Dg mi y

$$\sum x^2 y = a \sum x^2 + b \sum x^4$$

Sg S d b

th

. e

gr il

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Points to Remember 5.27

(iii) The normal equations for the curve $y = ax + bx^2$ are

$$\sum xy = a \sum x + b \sum x^3$$

$$\sum x^2 y = a \sum x^3 + b \sum x^5$$

Fitting of Exponential and Logarithmic Curves

For the curve $y = ab^x$,

Taking logarithm on both the sides of the equation $y = ab^x$,

$$\log_e y = \log_e a + x \log_e b$$

Putting $\log_e y = Y$, $\log_e a = A$, $x = X$, and $\log_e b = B$,

$$Y = A + BX$$

This is a linear equation in X and Y . The normal equations are

$$\sum Y = nA + B \sum X$$

$$\sum XY = A \sum X + B \sum X^2$$

$$\sum XY = A \sum X + B \sum X^2$$

Similarly, the best fitting exponential curves for the relation $y = ax^b$ and $y = ae^{bx}$ can be obtained.

Dg mi y

Sg S d b

th

. e

gr il

En mp

Co

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Paper 8 of 8 — GET 411 HANDOUTS

Level 400 • First Semester • Handout

Computer Arithmetic and errors

MTHBD/CMPBD 423

- Truncation Error

$x^2 \times 3$

$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

$\frac{2}{6}$

- Round-Off Error

When the true value of a number is not stored exactly by a computers representation, (either by conversion to a nonterminating fraction, in the computers base, or input beyond precision) the error

associated with this imperfection is called Round-Off Error.

- Error in Original Data

In mathematical models, you may have coefficients that are obtained by imperfect real world measurements. Or perhaps the model does not perfectly reflect the real world. Nothing we can do about this but is is worth doing a sensitivity analysis on the parameters.

- Propagated Errors:

Suppose a real number x is represented by the machine as x^* where $x^* = x(1 + \delta)$ and δ (the initial

relative error) is small.

Suppose I want to calculate x^2 . In the machine I'll get $(x^*)^2 = x^2 (1 + \delta)^2$. Now I get an error

$E = (x^*)^2 - x^2 = x^2 (1 + \delta)^2 - x^2 \approx x^2 (2\delta)$ which can be very big, especially if x is big. If I look

$\frac{E}{x^2}$

at relative error $= \frac{E}{x^2}$, I still get a relative error of 2δ . Notice, the relative error doubled. This is an example of an error being propagated.

- Computer Arithmetic

– machine numbers: Three parts: the sign, the fraction part called the mantissa or significand, and the exponent called the characteristic.

$\pm .d_1 d_2 d_3 \dots d_p \times B^e$

* B = the base that is used generally: 2, 16, 10

* d_i 's digits or bits $0 < d_1 < B$ and $0 \leq d_i < B$ for $i \geq 1$.

* p = the precision the number of significand bits.

* e = the characteristic maybe 7 bits long.

– MATLAB's max and min values (default is double precision):

* The maximum positive number is $\approx 10^{308}$.

* The smallest positive number is $\approx 10^{-308}$.

* The machine eps is $\approx 10^{-16}$. Machine eps is the largest number such that: $1 + \text{machine eps} \neq 1$.

– errors in conversion (round-off error)

1

In base 2, 10 is the repeating decimal .11002 . It will never be stored exactly in a base 2 machine.

100,000

1

In MATLAB: $|10,000 - \sum| \approx 2 \cdot 10^{-8}$

$k=1$ 10

- relative and absolute error

If p^* is the machine's approximation to p :

* absolute error = $|p - p^*|$

$|p - p^*|$

* relative error = provide $p \neq 0$.

$|p|$

1

- Arithmetic Accuracy (loss of significance)

Suppose we are using base 10, five digit (chopping) arithmetic.

Actual computer

variable Value Value

x 1/3 0.33333 x 100

y 5/7 0.71428 x 100

u 0.714251 0.71425 x 100

v 98765.9 0.98765 x 10⁵

w 0.111111 x 10⁻⁴ 0.111111 x 10⁻⁴

computer computer Actual Absolute Relative

operation Result Value Error Error

$x+y$ 0.10476 x 10¹ 22/21 0.190 x 10⁻⁴ 0.182 x 10⁻⁴

$y-x$ 0.38095 x 100 8/21 0.238 x 10⁻⁵ 0.625 x 10⁻⁵

$x \cdot y$ 0.23809 x 100 5/21 0.524 x 10⁻⁵ 0.220 x 10⁻⁴

$y \div x$ 0.21428 x 10¹ 15/7 0.571 x 10⁻⁴ 0.267 x 10⁻⁴

$y-u$ 0.30000 x 10⁻⁴ 0.34714 x 10⁻⁴ 0.471 x 10⁻⁵ 0.136

$(y-u) \div w$ 0.27000 x 10¹ 0.31243 x 10¹ 0.424 0.136

$(y-u) \cdot v$ 0.29629 x 10¹ 0.34825 x 10¹ 0.465 0.136

$u+v$ 0.98765 x 10⁵ 0.98766 x 10⁵ 0.161 x 10¹ 0.163 x 10⁻⁴

lesson: Subtracting similar values creates a large relative error. This is called a loss of significance. Also, adding two numbers of disparate size causes large absolute error. For example in MATLAB: $1,000,000 + 10^{-11} = 1,000,000$.

- Avoiding loss of significance due to subtracting similar numbers in the quadratic formula.

Recall the roots of $ax^2 + bx + c$ can be found by

$\sqrt{}$ $\sqrt{}$

$-b \pm \sqrt{b^2 - 4ac}$ $-b \pm \sqrt{b^2 - 4ac}$

$x_1 =$ and $x_2 = (1)$

2a 2a

Suppose $b^2 - 4ac > 0$, $b > 0$ and $b \approx b^2 - 4ac$. There is thus a good chance of x_1 containing a large error, especially if a is relatively small. We resolve this by changing the form of x_1 by rationalizing the numerator

$\sqrt{\sqrt{}}$!

$-b + b^2 - 4ac \quad -b - b^2 - 4ac \quad -2c$

$x_1 = \sqrt{} = \sqrt{(2)}$

$2a \quad -b - b^2 - 4ac \quad b + b^2 - 4ac$

With the above form for x_1 the subtraction of nearly equal numbers is eliminated. Therefore, it would be wise to calculate x_1 by equation (2) and x_2 by equation (1). A similar revision is needed if $b < 0$ in which case, x_2 from equation (1) is at risk of losing precision due to subtracting similar numbers.

2

- Avoiding round off error with nested multiplication

Even in the absence of subtracting similar numbers, there will be round-off error and the best way to minimize this is to reduce the number of computer operations. A good example of this is nested multiplication.

Consider the function $f(x) = x^3 - 6x^2 + 3x - 0.149$ evaluated at $x = 4.71$. We will assume three digit base 10 arithmetic for the machine computations.

* traditional evaluation: 5 multiplies and 3 adds/subtracts.

The relative error with either chopping or rounding is ≈ 0.04

* nested multiplication: 2 multiplies and 3 adds/subtracts.

$f(x) = ((x - 6)x + 3)x - 0.149$ (3)

The relative error with chopping is ≈ 0.0093

The relative error with rounding is ≈ 0.0025

• Error analysis of algorithms generally assumes perfect precision, ie. no round-off error. However, it is there and is worth keeping it in mind. Especially if you are doing many sequential calculations where the output from one is input into another. In this way, errors can be propagated and your final answer can be garbage.

3

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10.2 ITERATIVE METHODS FOR SOLVING LINEAR SYSTEMS

As a numerical technique, Gaussian elimination is rather unusual because it is direct. That is, a solution is obtained after a single application of Gaussian elimination. Once a “solution” has been obtained, Gaussian elimination offers no method of refinement. The lack of refinements can be a problem because, as the previous section shows, Gaussian elimination is sensitive to rounding error.

Numerical techniques more commonly involve an iterative method. For example, in

calculus you probably studied Newton's iterative method for approximating the zeros of a differentiable function. In this section you will look at two iterative methods for approximating the solution of a system of n linear equations in n variables.

The Jacobi Method

The first iterative technique is called the Jacobi method, after Carl Gustav Jacob Jacobi (1804–1851). This method makes two assumptions: (1) that the system given by

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

....

....

....

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

has a unique solution and (2) that the coefficient matrix A has no zeros on its main diagonal. If any of the diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$ are zero, then rows or columns must be interchanged to obtain a coefficient matrix that has nonzero entries on the main diagonal.

To begin the Jacobi method, solve the first equation for x_1 , the second equation for x_2 , and so on, as follows.

1

$$x_1 = (b - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n)$$

$$a_{11} \quad 1$$

1

$$x_2 = (b - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n)$$

$$a_{22} \quad 2$$

1

$$x_n = (b - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1})$$

$$a_{nn} \quad n$$

Then make an initial approximation of the solution,

$(x_1, x_2, x_3, \dots, x_n)$, Initial approximation

and substitute these values of x_i into the right-hand side of the rewritten equations to obtain the first approximation. After this procedure has been completed, one iteration has been

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performed. In the same way, the second approximation is formed by substituting the first approximation's x -values into the right-hand side of the rewritten equations. By repeated iterations, you will form a sequence of approximations that often converges to the actual solution. This procedure is illustrated in Example 1.

EXAMPLE 1 Applying the Jacobi Method

Use the Jacobi method to approximate the solution of the following system of linear equations.

$$5x_1 + 2x_2 + 3x_3 = 1$$

$$3x_1 - 9x_2 + x_3 = 2$$

$$2x_1 + x_2 - 7x_3 = 3$$

Continue the iterations until two successive approximations are identical when rounded to three significant digits.

Solution To begin, write the system in the form

$$x_1 = \frac{1}{3}(2 + 9x_2 - x_3)$$

$$x_2 = \frac{1}{2}(3 + 7x_3 - 2x_1)$$

$$x_3 = \frac{1}{7}(2 - 2x_1 + x_2)$$

Because you do not know the actual solution, choose

$$x_1 = 0, x_2 = 0, x_3 = 0 \text{ Initial approximation}$$

as a convenient initial approximation. So, the first approximation is

$$x_1 = \frac{1}{3}(2 + 9(0) - (0)) = 0.667$$

$$x_2 = \frac{1}{2}(3 + 7(0) - 2(0.667)) = 0.333$$

$$x_3 = \frac{1}{7}(2 - 2(0.667) + 0.333) = 0.143$$

Continuing this procedure, you obtain the sequence of approximations shown in Table 10.1.

TABLE 10.1

n 0 1 2 3 4 5 6 7

x_1 0.000 0.667 0.146 0.192 0.181 0.185 0.186 0.186

x_2 0.000 0.333 0.203 0.328 0.332 0.329 0.331 0.331

x_3 0.000 0.143 0.517 0.416 0.421 0.424 0.423 0.423

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Because the last two columns in Table 10.1 are identical, you can conclude that to three significant digits the solution is

$$x_1 = 0.186, x_2 = 0.331, x_3 = 0.423.$$

For the system of linear equations given in Example 1, the Jacobi method is said to converge. That is, repeated iterations succeed in producing an approximation that is correct to three significant digits. As is generally true for iterative methods, greater accuracy would require more iterations.

The Gauss-Seidel Method

You will now look at a modification of the Jacobi method called the Gauss-Seidel method, named after Carl Friedrich Gauss (1777–1855) and Philipp L. Seidel (1821–1896). This modification is no more difficult to use than the Jacobi method, and it often requires fewer iterations to produce the same degree of accuracy.

With the Jacobi method, the values of x_i obtained in the n th approximation remain unchanged until the entire $(n + 1)$ th approximation has been calculated. With the Gauss-Seidel method, on the other hand, you use the new values of each x_i as soon as they are known. That is, once you have determined x_1 from the first equation, its value is then used in the second equation to obtain the new x_2 . Similarly, the new x_1 and x_2 are used in the third equation to obtain the new x_3 , and so on. This procedure is demonstrated in

Example 2.

EXAMPLE 2 Applying the Gauss-Seidel Method

Use the Gauss-Seidel iteration method to approximate the solution to the system of equations given in Example 1.

Solution The first computation is identical to that given in Example 1. That is, using $(x_1, x_2, x_3) = (0, 0, 0)$ as the initial approximation, you obtain the following new value for x_1 .

$$x_1 = 15 - 25(0) - 35(0) = 0.200$$

Now that you have a new value for x_1 , however, use it to compute a new value for x_2 . That is,

$$x_2 = 29 - 39(0.200) - 19(0) = 0.156.$$

Similarly, use $x_1 = 0.200$ and $x_2 = 0.156$ to compute a new value for x_3 . That is,

$$x_3 = 37 - 27(0.200) - 17(0.156) = 0.508.$$

So the first approximation is $x_1 = 0.200$, $x_2 = 0.156$, and $x_3 = 0.508$. Continued iterations produce the sequence of approximations shown in Table 10.2.

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TABLE 10.2

n	0	1	2	3	4	5
x_1	0.000	0.200	0.167	0.191	0.186	0.186
x_2	0.000	0.156	0.334	0.333	0.331	0.331
x_3	0.000	0.508	0.429	0.422	0.423	0.423

Note that after only five iterations of the Gauss-Seidel method, you achieved the same accuracy as was obtained with seven iterations of the Jacobi method in Example 1.

Neither of the iterative methods presented in this section always converges. That is, it is possible to apply the Jacobi method or the Gauss-Seidel method to a system of linear equations and obtain a divergent sequence of approximations. In such cases, it is said that the method diverges.

EXAMPLE 3 An Example of Divergence

Apply the Jacobi method to the system

$$x_1 - 5x_2 = 4$$

$$7x_1 - x_2 = 6,$$

using the initial approximation $(x_1, x_2) = (0, 0)$, and show that the method diverges.

Solution As usual, begin by rewriting the given system in the form

$$x_1 = 4 + 5x_2$$

$$x_2 = 6 - 7x_1.$$

Then the initial approximation $(0, 0)$ produces

$$x_1 = 4 + 5(0) = 4$$

$$x_2 = 6 - 7(0) = 6$$

as the first approximation. Repeated iterations produce the sequence of approximations shown in Table 10.3.

TABLE 10.3

n	0	1	2	3	4	5	6	7
x1	0	4	34	174	1244	6124	42,874	214,374
x2	0	6	34	244	1244	8574	42,874	300,124

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For this particular system of linear equations you can determine that the actual solution is $x_1 = 1$ and $x_2 = 1$. So you can see from Table 10.3 that the approximations given by the Jacobi method become progressively worse instead of better, and you can conclude that the method diverges.

The problem of divergence in Example 3 is not resolved by using the Gauss-Seidel method rather than the Jacobi method. In fact, for this particular system the Gauss-Seidel method diverges more rapidly, as shown in Table 10.4.

TABLE 10.4

n	0	1	2	3	4	5
x1	0	4	174	6124	214,374	7,503,124
x2	0	34	1224	42,874	1,500,624	52,521,874

With an initial approximation of $(x_1, x_2) = (0, 0)$, neither the Jacobi method nor the Gauss-Seidel method converges to the solution of the system of linear equations given in Example 3. You will now look at a special type of coefficient matrix A , called a strictly diagonally dominant matrix, for which it is guaranteed that both methods will converge.

Definition of Strictly An $n \times n$ matrix A is strictly diagonally dominant if the absolute value of each entry

Diagonally Dominant on the main diagonal is greater than the sum of the absolute values of the other entries

in the same row. That is,

Matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$|a_{11}| > |a_{12}| + |a_{13}| + \dots + |a_{1n}|$$

$$|a_{22}| > |a_{21}| + |a_{23}| + \dots + |a_{2n}|$$

$$|a_{nn}| > |a_{n1}| + |a_{n2}| + \dots + |a_{n,n-1}|$$

EXAMPLE 4 Strictly Diagonally Dominant Matrices

Which of the following systems of linear equations has a strictly diagonally dominant coefficient matrix?

(a) $3x_1 + x_2 = 4$

$2x_1 + 5x_2 = 2$

(b) $4x_1 + 2x_2 + x_3 = 1$

$x_1 + 2x_3 = 4$

$3 \times 1 \quad 5 \times 2 \quad x_3 \quad 3$

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Solution (a) The coefficient matrix

$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

3

$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

5

$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

is strictly diagonally dominant because $3 > 0 + 0$ and $5 > 0 + 0$.

(b) The coefficient matrix

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

4 2

$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

3 5 1

is not strictly diagonally dominant because the entries in the second and third rows do not conform to the definition. For instance, in the second row $a_{21} = 1$, $a_{22} = 0$,

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$a_{23} = 2$, and it is not true that $a_{22} > a_{21} + a_{23}$. Interchanging the second and third rows in the original system of linear equations, however, produces the coefficient matrix

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

4 2

$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

1 0 2

and this matrix is strictly diagonally dominant.

The following theorem, which is listed without proof, states that strict diagonal dominance is sufficient for the convergence of either the Jacobi method or the Gauss-Seidel method.

Theorem 10.1 If A is strictly diagonally dominant, then the system of linear equations given by $Ax = b$

has a unique solution to which the Jacobi method and the Gauss-Seidel method will converge for any initial approximation.

the Jacobi and

Gauss-Seidel Methods

In Example 3 you looked at a system of linear equations for which the Jacobi and Gauss-Seidel methods diverged. In the following example you can see that by interchanging the rows of the system given in Example 3, you can obtain a coefficient matrix that is strictly diagonally dominant. After this interchange, convergence is assured.

EXAMPLE 5 Interchanging Rows to Obtain Convergence

Interchange the rows of the system

$$x_1 - 5x_2 = -4$$

$$7x_1 - x_2 = 6$$

to obtain one with a strictly diagonally dominant coefficient matrix. Then apply the Gauss-Seidel method to approximate the solution to four significant digits.

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Solution Begin by interchanging the two rows of the given system to obtain

$$7x_1 - x_2 = 6$$

$$x_1 - 5x_2 = -4.$$

Note that the coefficient matrix of this system is strictly diagonally dominant. Then solve for x_1 and x_2 as follows.

$$x_1 = \frac{6}{7} + \frac{1}{7}x_2$$

$$x_2 = 4.5 - 1.5x_1$$

Using the initial approximation $(x_1, x_2) = (0, 0)$, you can obtain the sequence of approximations shown in Table 10.5.

TABLE 10.5

n	0	1	2	3	4	5
x_1	0.0000	0.8571	0.9959	0.9999	1.000	1.000
x_2	0.0000	0.9714	0.9992	1.000	1.000	1.000

So you can conclude that the solution is $x_1 = 1$ and $x_2 = 1$.

Do not conclude from Theorem 10.1 that strict diagonal dominance is a necessary condition for convergence of the Jacobi or Gauss-Seidel methods. For instance, the coefficient matrix of the system

$$4x_1 - 5x_2 = 1$$

$$x_1 - 2x_2 = 3$$

is not a strictly diagonally dominant matrix, and yet both methods converge to the solution $x_1 = 1$ and $x_2 = 1$ when you use an initial approximation of $(x_1, x_2) = (0, 0)$. (See Exercises 21–22.)

SECTION 10.2 EXERCISES 585**SECTION 10.2 □ EXERCISES**

In Exercises 1–4, apply the Jacobi method to the given system of 19. Interchange the rows of the system of linear equations in

linear equations, using the initial approximation $(x_1, x_2, \dots, x_n) = (0, 0, \dots, 0)$. Exercise 11 to obtain a system with a strictly diagonally dom-

$(0, 0, \dots, 0)$. Continue performing iterations until two successive invariant coefficient matrix. Then apply the Gauss-Seidel method

approximations are identical when rounded to three significant digits. to approximate the solution to two significant digits.

20. Interchange the rows of the system of linear equations in

1. $3x_1 + x_2 = 2$ 2. $4x_1 + 2x_2 = 6$

Exercise 12 to obtain a system with a strictly diagonally dom-

$x_1 + 4x_2 = 5$ 3. $3x_1 + 5x_2 = 1$ inant coefficient matrix. Then apply the Gauss-Seidel method

3. $2x_1 + x_2 = 2$ 4. $4x_1 + x_2 + x_3 = 7$ to approximate the solution to two significant digits.

$x_1 + 3x_2 + x_3 = 2$ $x_1 + 7x_2 + 2x_3 = 2$ In Exercises 21 and 22, the coefficient matrix of the system of linear

$x_1 + x_2 + 3x_3 = 6$ 3. $3x_1 + 4x_3 = 11$ equations is not strictly diagonally dominant. Show that the Jacobi

5. Apply the Gauss-Seidel method to Exercise 1. and Gauss-Seidel methods converge using an initial approximation

of $x_1, x_2, \dots, x_n = 0, 0, \dots, 0$.

6. Apply the Gauss-Seidel method to Exercise 2.

7. Apply the Gauss-Seidel method to Exercise 3. 21. $4x_1 + 5x_2 = 1$ 22. $4x_1 + 2x_2 + 2x_3 = 0$

8. Apply the Gauss-Seidel method to Exercise 4. $x_1 + 2x_2 + 3x_3 = 7$

In Exercises 9–12, show that the Gauss-Seidel method diverges for $3x_1 + x_2 + 4x_3 = 5$

the given system using the initial approximation $x_1, x_2, \dots, x_n = 0$

In Exercises 23 and 24, write a computer program that applies the

$0, 0, \dots, 0$.

Gauss-Siedel method to solve the system of linear equations.

9. $x_1 + 2x_2 = 1$ 10. $x_1 + 4x_2 = 1$

23. $4x_1 + x_2 + x_3 = 3$

$2x_1 + x_2 + 3x_3 + 2x_2 = 2$

$x_1 + 6x_2 + 2x_3 + x_4 + x_5 = 6$

11. $2x_1 + 3x_2 = 7$ 12. $x_1 + 3x_2 + x_3 = 5$

$x_2 + 5x_3 + x_5 + x_6 = 5$

$x_1 + 3x_2 + 10x_3 = 9$ 3. $3x_1 + x_2 + 5x_4 + x_5 + x_7 + x_8 = 0$

$3x_1 + x_3 + 13x_2 + 2x_3 = 1$ $x_3 + x_4 + 6x_5 + x_6 + x_8 = 12$

In Exercises 13–16, determine whether the matrix is strictly diago- $x_3 + x_5 + 5x_6 = 12$

nally dominant. $x_4 + 4x_7 + x_8 = 2$

$x_1 + 2x_4 + x_5 + x_7 + 5x_8 = 2$

$x_3 = 0$

2 1

13. 14. 24. $4x_1 + x_2 + x_3 = 18$

5 1

$x_1 + 4x_2 + x_3 + x_4 = 18$

x_1

$x_1 + x_2 + x_3$

12 6 0 7 5

$x_2 + 4x_3 + x_4 + x_5 = 4$

15. 2 3 2 16. 1 4 1

$x_3 + 4x_4 + x_5 + x_6 = 4$

0 6 13 0 2 3

$x_4 = 4x_5 = x_6 = x_7 = 26$

17. Interchange the rows of the system of linear equations in $x_5 = 4x_6 = x_7 = x_8 = 16$

Exercise 9 to obtain a system with a strictly diagonally dominant coefficient matrix. Then apply the Gauss-Seidel method $x_6 = 4x_7 = x_8 = 10$

to approximate the solution to two significant digits.

18. Interchange the rows of the system of linear equations in

Exercise 10 to obtain a system with a strictly diagonally dominant coefficient matrix. Then apply the Gauss-Seidel method

to approximate the solution to two significant digits.